For this problem: $F = 1.00$ *in*, $D_P = 1.500$. $F/D_P = 0.667$. Then $C_{pf} = 0.042$.

Specify open gearing for the wood chipper, mounted to the frame. $C_{ma} = 0.264$.

Compute: $K_m = 1.0 + C_{pf} + C_{ma} + 0.042 + 0.264 = 1.31$

Step 13. Specify the size factor, K_s, from Table 9–2.

For this problem, $K_s = 1.00$ for $P_d = 12$.

Step 14. Specify the rim thickness factor, K_B, from Figure 9–14.

For this problem, specify a solid gear blank. $K_B = 1.00$.

Step 15. Specify a service factor, SF, typically from 1.00 to 1.50, based on uncertainty of data.

For this problem, there is no unusual uncertainty. Let $SF = 1.00$.

Step 16. Specify a reliability factor using Table 9–11 as a guideline.

For this problem, specify a reliability of 0.99. $K_R = 1.00$.

Step 17. Specify a design life. Compute the number of loading cycles for the pinion and the gear. Determine the stress cycle factors for bending (Y_N) and pitting (Z_N) for the pinion and the gear.

For this problem, intermittent use is expected. Specify the design life to be 3000 hours, similar to agricultural machinery. The numbers of loading cycles are:

$$N_{cP} = (60)(3000\text{ hr})(1750\text{ rpm})(1) = 3.15 \times 10^8\text{ cycles}$$
$$N_{cG} = (60)(3000\text{ hr})(463.2\text{ rpm})(1) = 8.34 \times 10^7\text{ cycles}$$

Then, from Figure 9–21, $Y_{NP} = 0.96$, $Y_{NG} = 0.98$. From Figure 9–22, $Z_{NP} = 0.92$, $Z_{NG} = 0.95$.

Step 18. Compute the expected bending stresses in the pinion and the gear using Equation (9–16).

$$s_{tP} = \frac{W_t P_d}{F J_P} K_o\, K_s\, K_m\, K_B\, K_v = \frac{(144)(12)}{(1.00)(0.325)}(1.75)(1.0)(1.31)(1.0)(1.35) = 16\,455\text{ psi}$$
$$s_{tG} = s_{tP}(J_P/J_G) = (16\,455)(0.325/0.410) = 13\,044\text{ psi}$$

Step 19. Adjust the bending stresses using Equation 9–30.

For this problem, for the pinion:

$$s_{atP} > s_{tP}\frac{K_R(SF)}{Y_{(NP)}} = (16\,455)\frac{(1.00)(1.00)}{0.96} = 17\,141\text{ psi}$$

For the gear:

$$s_{atG} > s_{tG}\frac{K_R(SF)}{Y_{(NG)}} = (13\,044)\frac{(1.00)(1.00)}{0.98} = 13\,310\text{ psi}$$

Step 20. Compute the expected contact stress in the pinion and the gear from Equation (9–23). Note that this value will be the same for both the pinion and the gear.

$$s_c = C_P\sqrt{\frac{W_t\, K_o\, K_s\, K_m\, K_v}{F D_P I}} = 2300\sqrt{\frac{(144)(1.75)(1.0)(1.31)(1.35)}{(1.00)(1.50)(0.104)}} = 122\,933\text{ psi}$$

Step 21. Adjust the contact stresses for the pinion and the gear using Equation (9–31).

$$s_{acP} > s_{cP}\frac{K_R(SF)}{Z_{NP}} = (122\,933)\frac{(1.00)(1.00)}{(0.92)} = 133\,623\text{ psi}$$

For the gear:

$$s_{acG} > s_{cG}\frac{K_R(SF)}{Z_{NG}} = (122\,933)\frac{(1.00)(1.00)}{(0.95)} = 129\,403\text{ psi}$$

Step 22. Specify materials for the pinion and the gear that will have suitable through hardening or case hardening to provide allowable bending and contact stresses greater than those required from Steps 19 and 21. Typically the contact stress in the pinion is the controlling factor. Refer to Figures 9–18 and 9–19 and Tables 9–9 and 9–10 for data on required hardness. Refer to Appendices 3 to 5 for properties of steel to specify a particular alloy and heat treatment.

For this problem, the contact stress for the pinion is the controlling factor, as is often the case. A steel must be specified that is rated to handle approximately $s_{ac} = 133.6$ ksi. First check Figure 9–19 to explore whether or not through-hardened steel is practical. We can use the equation for Grade 1 steel in U.S. units to determine the required Brinell hardness number, *HB.*

$$\text{Reqd. } HB = (s_{ac} - 29.10)/0.322 = (133.6 \text{ ksi} - 29.10)/0.322 = 324$$

This value is well within the recommended hardness for through-hardened steels. Using Appendix 4, we can specify SAE 4140 OQT 1000 steel having $HB = 341$ and 18% elongation indicating good ductility. We can also check the required hardness for the gear that has a required $s_{ac} = 129.4$ ksi.

$$\text{Reqd. } HB = (s_{ac} - 29.10)/0.322 = (129.4 \text{ ksi} - 29.10)/0.322 = 311$$

This value can be met by SAE 4140 OQT 1100 steel having $HB = 311$ and 20% elongation. However, because both the pinion and the gear experience nearly the same contact stress, it may be prudent to specify the same heat treatment for both to permit them to be produced by the same process.

9–10 GEAR DESIGN FOR THE METRIC MODULE SYSTEM

Here we take the principles, guidelines, and design methodology that were initially developed for the U.S. Customary unit system based on the diametral pitch, P_d, and adapt them to the SI Metric unit system based on the metric module, m. The primary variables involved are listed below with units in both systems.

Variables	U.S. units	SI units
Length (D, C, pitch)	in	mm
Force (W_t, W_r)	lb	N
Power, P	hp	watts or kW
Pitch line speed, v_t	ft/min	m/s
Stresses	psi or ksi	MPa

Some of the required adjustments to calculations and equations are listed below, to help you to mentally relate the two systems to each other. In general, we recommend that designs be completed in one system or the other with minimal conversions.

- Refer to Table 8–1 gear teeth and gear pair features.
- Charts, tables, and graphs in this chapter contain both sets of units.
- Face width, F, recommended limits:

  U.S. units (in): $8/P_d < F < 16/P_d$ Nominal: $12/P_d$
  SI units (mm): $8m < F < 16m$ Nominal: $12m$

- Pitch line speed, v_t:

  U.S. units (ft/min): $v_t = \pi Dn/12$ ft/min
  [D in inches, n in rpm]

  SI units (m/s): $v_t = \pi Dn/60\,000$ m/s
  [D in mm, n in rpm]

- Transmitted load, W_t:

  U.S. units (lb): $W_t = 33\,000(P)/v_t$ [P in hp, v_t in ft/min]
  SI units (m/s): $W_t = 1000(P)/v_t$ [P in kW, v_t in m/s]

The following example problem uses SI units. The procedure will be virtually the same as that used to design with U.S. Customary units in Example Problem 9–7.

Example Problem 9–8 A gear pair is to be designed to transmit 15.0 kilowatts (kW) of power to a large meat grinder in a commercial meat processing plant. The pinion is attached to the shaft of an electric motor rotating at 575 rpm. The gear must operate at 270 to 280 rpm. The gear unit will be enclosed and of commercial quality. Commercially hobbed (quality number A11), 20°, full-depth, involute gears are to be used in the metric module system. The maximum center distance is to be 200 mm. Specify the design of the gears.

Solution The nominal velocity ratio is

$$VR = 575/275 = 2.09$$

Specify an overload factor of $K_o = 1.50$ from Table 9–1 for a uniform power source and moderate shock for the meat grinder. Then compute design power,

$$P_{des} = K_o P = (1.50)(15 \text{ kW}) = 22.5 \text{ kW}$$

From Figure 9–11, $m = 5$ is a reasonable trial module. Then

$$N_P = 18 \quad \text{(design decision)}$$
$$D_P = N_P m = (18)(5) = 90 \text{ mm}$$
$$N_G = N_P(VR) = (18)(2.09) = 37.6 \quad \text{(Use 38)}$$
$$D_G = N_G m = (38)(5) = 190 \text{ mm}$$
$$\text{Final output speed} = n_G = n_P(N_P/N_G)$$
$$n_G = 575 \text{ rpm} \times (18/38) = 272 \text{ rpm} \quad \text{(OK)}$$

Center distance $= C = (N_P + N_G)m/2$[Table 8–1]

$$C = (18 + 38)(5)/2 = 140 \text{ mm} \quad \text{(OK)}$$

In SI units, the pitch line speed in meters per second (m/s) is

$$v_t = \pi D_P n_P/(60\,000) = [(\pi)(90)(575)]/(60\,000) = 2.71 \text{ m/s}$$

In SI units, the transmitted load, W_t, is in newtons (N). If the power, P, is in kW, and v_t is in m/s,

$$W_t = 1000(P)/v_t = (1000)(15)/(2.71) = 5536 \text{ N}$$

Face width, F: Let's specify the nominal $F = 12m = 12(5) = 60$ mm.

Factors in stress analysis:

$K_o = 1.50$ (found earlier)

$K_s = 1.00$ (Table 9–2; $m = 5$) $K_B = 1.00$ (Use solid gear blanks)

$K_R = 1.00$ (Table 9–11; 0.99 reliability) $SF = 1.00$ (No unusual conditions)

$K_v = 1.31$ (Figure 9–16; $A_v = 11$)

$K_m = 1.21$ (Figures 9–12 and 9–13; $F = 60$ mm; $F/D_P = 60/90 = 0.67$)

$J_P = 0.315$; $J_G = 0.380$ (Figure 9–10; $N_P = 18$, $N_G = 38$)

$C_p = 191$ (Table 9–7) $I = 0.092$ (Figure 9–17)

Pinion contact stress:

(Equation 9–23)

$$S_c = C_P\sqrt{\frac{W_t\, K_o\, K_s\, K_m\, K_v}{FD_P I}} = 191\sqrt{\frac{(5536)(1.50)(1.0)(1.21)(1.31)}{(60)(90)(0.092)}} = 983 \text{ MPa}$$

Adjustments for number of cycles, from Figures 9–21 and 9–22:

$$Y_{Np} = 0.94 \quad Z_{NP} = 0.91 \quad Y_{NG} = 0.96 \quad Z_{NG} = 0.92$$

Required $s_{acP} = s_c(SF)(K_R)/Z_{NP} = 983$ MPa $(1.0)(1.0)/0.91 = 1080$ MPa

Using $s_{acP} = 1080$ MPa, Figure 9–19 shows the required hardness = HB 396 for through-hardened Grade 1 steel. This is acceptable but near the upper end of recommended range.

Material specification:

From Figure A4–5 (other possibilities exist),

SAE 4340 OQT 800; HB 415; $s_y = 1324$ MPa; $s_u = 1448$ MPa; 12% elongation.

Check other stresses:

The contact stress for the gear and the bending stress for the pinion and the gear are expected to require less material hardness and strength.

Required $s_{acG} = s_c(SF)(K_R)/Z_{NG} = 983$ MPa$(1.0)(1.0)/0.92 = 1068$ MPa

This is slightly lower than for the pinion (OK)

$$S_{tP} = \frac{W_t\, K_o\, K_s\, K_B\, K_m\, K_v}{FmJ_P} = \frac{(5536)(1.50)(1)(1)(1.21)(1.31)}{(60)(5)(0.315)} = 139 \text{ MPa}$$

Required $s_{atP} = s_{tP}(SF)(K_R)/Y_{NP} = 139$ MPa$(1.0)(1.0)/0.94 = 148$ MPa

Referring to Figure 9–18, it is obvious that bending stress requires far lower hardness for the gear teeth, less than HB 180. The stress in the gear is always less than that in the pinion so it will obviously be safe as well.

Summary of the Design:

$P = 15.0$ kW from an electric motor to a large meat grinder

Pinion speed: $n_P = 575$ rpm Gear speed: $n_G = 272$ rpm

Number of teeth: $N_P = 18$; $N_G = 38$ Center distance: $C = 140.00$ mm

Module: $m = 5$ mm Diameters: $D_P = 90$ mm; $D_G = 190$ mm

Material: Steel-SAE 4340 OQT 800

Comment:

A redesign may be considered with several possible approaches:

1. Increase the face width, F, to lower the stresses and permit the choice of a material with more moderate required hardness and better ductility. The recommended upper limit of face width is $16m = 16(5) = 80$ mm.
2. Increase the size of pinion and its number of teeth (same module) to lower stresses.

   Possible trial: Module: $m = 5$ mm Number of teeth: $N_P = 22$; $N_G = 46$

   Center distance: $C = 170.00$ mm Diameters: $D_P = 110$ mm; $D_G = 230$ mm
3. Consider case-hardened steel for the initial design, rather than through-hardened steel. A smaller design is possible.

9–11 COMPUTER-AIDED SPUR GEAR DESIGN AND ANALYSIS

This section presents one approach to assisting the gear designer with the many calculations and judgments that must be made to produce an acceptable design. The spreadsheet shown in Figure 9–23 facilitates the completion of a prospective design for a pair of gears in a few minutes by an experienced designer. You must have studied all of the material here and in Chapter 8 in order to understand the data needed in the spreadsheet and to use it effectively.

The recommended use of the spreadsheet is to create a series of design iterations that allow you to progress toward an optimum design in a short amount of time. It follows the process outlined in Section 9–9 up to the point of computing the required allowable bending stress number and the allowable contact stress number for both the pinion and the gear. The designer must use those data to specify suitable materials for the gears and their heat treatments.

Given below is a discussion of the essential features of the spreadsheet. In general, it first calls for the input of basic performance data, allowing a proposed geometry to be specified. The final result is the completion of the stress analyses for bending and pitting resistance for both the pinion and the gear. Equations (9–16) and (9–30) are combined for the bending analysis. The analysis of pitting resistance uses Equations (9–23) and (9–31). The designer must provide data for the several factors in those equations taken from appropriate figures and charts or based on design decisions. Virtually all computations are performed by the spreadsheet, allowing the designer to exercise judgment based on the intermediate results.

The format used for the spreadsheet helps the designer follow the process. After defining the problem at the top of the sheet, the first column at the left calls for several pieces of input data. Any value in italics within a gray-shaded area must be entered by the designer. White areas offer the results of calculations and provide guidance. The upper part of the second column also guides the designer in determining values for the several factors needed to complete the stress analyses for bending and pitting resistance. The area at the lower right of the spreadsheet gives the primary output data for stresses on which the design decisions for materials and heat treatments are based.

The data in Figure 9–23 are taken from Example Problem 9–7 which was completed in the traditional manner in Section 9–9.

Discussion of the Use of the Spur Gear Design Spreadsheet

1. **Describing the application:** In the heading of the sheet, the designer is asked to describe the application for identification purposes and to focus on the basic uses for the gears. Use the nature of the prime mover and the driven machine to specify the overload factor, K_o, using Table 9–1 as a guide.
2. **Initial input data:** It is assumed that designers begin with a knowledge of the power transmission requirement, the rotational speed of the pinion of the gear pair, and the desired output speed. Using the nature of the application and the overload factor, K_o, compute the *design power* from

$$P_{\text{des}} = K_o P$$

   Then use Figure 9–11 to determine a trial value of the diametral pitch using the design power and the rotational speed of the pinion. The number of teeth in the pinion is a critical design decision because the size of the system depends on this value. Ensure against interference. An initial trial value of $N_P = 17$ to 20 is often a good choice.
3. **Number of gear teeth:** The spreadsheet computes the approximate number of gear teeth to produce the desired output speed from $N_G = N_P(n_G/n_P)$. But, of course, the number of teeth in any gear must be an integer, and the actual value of N_G is entered by the designer.
4. **Computed data:** The seven values reported in the middle of the first column are all determined from the input data, and they allow the designer to evaluate the suitability of the geometry of the proposed design at this point. Changes to the input data can be made at this time if any value is out of the desired range in the judgment of the designer.
5. **Secondary input data:** When a suitable geometry for the gears is obtained, the designer enters the data called for at the lower part of the first column of the spreadsheet. The locations of data in pertinent tables and figures are listed.
6. **Factors in design analysis:** The stress analysis requires many factors to account for the unique situation of the design being pursued. Again, guidance is offered, but the designer must enter the values of the required factors. Many of the factors can have a value of 1.00 for normal conditions.
7. **Alignment factor:** The alignment factor depends on two other factors: the pinion proportion factor and the mesh alignment factor as shown in Figures 9–12 and 9–13. The suggested values in the white areas are computed from the equations given in the figures. Note the listed value of F/D_P. If $F/D_P < 0.50$, use $F/D_P = 0.50$ to find C_{pf}. The designer must decide on the type of gearing to be used (open or closed) and the degree of precision to be designed into the system. The final result is computed from the input data.
8. **Size, and rim thickness factors:** Consult Table 9–2 along with Figure 9–14. Note that the rim thickness factor can be different for the pinion and the gear. Sometimes the smaller pinion is made from a solid blank while the larger gear can use a rim-and-spoke design.

DESIGN OF SPUR GEARS

APPLICATION:	*Example Problem 9–7*		
Wood Chipper driven by an electric motor			
Initial Input Data:			
Input Power:	$P =$	*3.00* hp	
Input Speed:	$n_P =$	*1750* rpm	
Diametral Pitch:	$P_d =$	*12*	
Number of Pinion Teeth:	$N_P =$	*18*	
Desired Output Speed:	$n_G =$	*462.5* rpm	
Computed number of gear teeth:		68.1	
Enter: Chosen No. of Gear Teeth:	$N_G =$	*68*	
Computed data:			
Actual Output Speed:	$n_G =$	463.2 rpm	
Gear Ratio:	$m_G =$	3.78	
Pitch Diameter—Pinion:	$D_P =$	1.500 in	
Pitch Diameter—Gear:	$D_G =$	5.667 in	
Center Distance:	$C =$	3.583 in	
Pitch Line Speed:	$v_t =$	687 ft/min	
Transmitted Load:	$W_t =$	144 lb	
Secondary Input Data:			
	Min	Nom	Max
Face Width Guidelines (in):	0.667	1.000	1.333
Enter: Face Width:	$F =$	*1.000* in	
Ratio: Face width/pinion diameter:	$F/D_P =$	0.67	
Recommended ratio	$F/D_P <$	2.00	
Enter: Elastic Coefficient:	$Cp =$	*2300*	Table 9–7
Enter: Quality Number:	$A_v =$	*11*	Table 9–5
Enter: Bending Geometry Factors:			
Pinion:	$J_P =$	*0.325*	Figure 9–10
Gear:	$J_G =$	*0.410*	Figure 9–17
Enter: Pitting Geometry Factor:	$I =$	*0.104*	Figure 9–17
REF:	$m_G =$	3.78	

Factors in Design Analysis:				
Alignment Factor, $K_m = 1.0 + C_{pf} + C_{ma}$	If $F < 1.0$	If $F > 1.0$	$F/D_P =$	0.67
Pinion Proportion Factor, $C_{pf} =$	0.042	0.042	$[0.50 < F/D_P < 2.00]$	
Enter: $C_{pf} =$	*0.042*	Figure 9–12		
Type of gearing:	Open	Commer.	Precision	Ex. Prec.
Mesh Alignment Factor, $C_{ma} =$	0.264	0.143	0.080	0.048
Enter: $C_{ma} =$	*0.264*	Figure 9–13		
Alignment Factor: $K_m =$	1.31	[Computed]		
Overload Factor: $K_o =$	*1.75*	Table 9–1		
Size Factor: $K_s =$	*1.00*	Table 9–2: Use 1.00 if $P_d >= 5$		
Pinion Rim Thickness Factor: $K_{BP} =$	*1.00*	Figure 9–14: Use 1.00 if solid blank		
Gear Rim Thickness Factor: $K_{BG} =$	*1.00*	Figure 9–14: Use 1.00 if solid blank		
Dynamic Factor: $K_v =$	1.35	[Computed: See Table 9–16]		
Service Factor: $SF =$	*1.00*	Use 1.00 if no unusual conditions		
Reliability Factor: $K_R =$	*1.00*	Table 9–11 Use 1.00 for $R = .99$		
Enter: Design Life:	*3000*	hours	See Table 9–12	
Pinion - Number of load cycles: $N_P =$	3.15E + 08	Guidelines: Y_N, Z_N		
Gear - Number of load cycles: $N_G =$	8.34E + 07	10^7	$>10^7$	$<10^7$
Bending Stress Cycle Factor: $Y_{NP} =$	*0.96*	1.00	0.96	Figure 9–21
Bending Stress Cycle Factor: $Y_{NG} =$	*0.98*	1.00	0.98	Figure 9–21
Pitting Stress Cycle Factor: $Z_{NP} =$	*0.92*	1.00	0.92	Figure 9–22
Pitting Stress Cycle Factor: $Z_{NG} =$	*0.95*	1.00	0.95	Figure 9–22
Stress Analysis: Bending				
Pinion: Required $s_{at} =$	17,102	psi	See Figure 9–18 or	
Gear: Required $s_{at} =$	13,280	psi	Table 9–9 or 9–10	
Stress Analysis: Pitting				
Pinion: Required $s_{ac} =$	133,471	psi	See Figure 9–19 or	
Gear: Required $s_{ac} =$	129,256	psi	Table 9–9 or 9–10	
Specify materials, alloy and heat treatment, for most severe requirement.				
One possible material specification:				
Pinion: Requires HB > 324; AISI 4140 OQT 1000, HB = 341, s_{ac} = 140,000 psi				
Gear: Requires HB > 311; AISI 4140 OQT 1000, HB = 341, s_{ac} = 140,000 psi				

FIGURE 9–23 Spreadsheet solution for Example Problem 9–9; alternate spur gear design using data for Example Problem 9–7

9. **Dynamic factor:** The spreadsheet uses the equations included in Table 9–6 to compute the dynamic factor using the quality number and pitch line speed found from data in the first column.
10. **Safety factor:** This is a design decision as discussed in Section 9–9. Often a value of 1.00 is used if no unusual conditions are expected that are not already accounted for in other factors. Larger safety factors allow for a higher degree of safety or to account for uncertainties. For extra safety, use *SF* up to 1.50.
11. **Reliability factor:** The designer must select a value from Table 9–11 according to the desired level of reliability.
12. **Stress cycle factors:** Here the designer must specify the design life in hours of operation for the gear pair being designed. Table 9–12 provides suggestions according to the use of the system. The number of cycles of stress is then computed for both the pinion and the gear, assuming the normal case of one cycle of one-direction stress per revolution. If the gears operate in a reversing mode, as idlers, or in planetary gear trains, this calculation must be adjusted to account for the multiple cycles of stress experienced in each revolution. Guidelines recommend factors of 1.00 for 10^7 cycles for which the allowable stress numbers are computed. For a larger number of cycles, equations given in Figures 9–21 and 9–22 are used to compute the recommended factors. Because a variety of data are given for the case of fewer than 10^7 cycles, the designer is referred to the figures to determine the factors. In any case, the user of the spreadsheet must enter the selected values.
13. **Stress analyses for bending and pitting resistance:** Finally, the required allowable bending stress number and the required allowable contact stress number are computed using Equations (9–30) and (9–31), adjusted for the special values of factors for the pinion and the gear.
14. **Specification of the materials and their heat treatment:** The final step is left to the designer to use the computed values from the stress analyses and to specify materials that will provide an adequate strength and surface hardness of the gear teeth. Pertinent data are listed in Figures 9–18 and 9–19 and Tables 9–9 and 9–10. The appendices tables for material properties may also be consulted once the required hardnesses of the materials are determined.

9–12 USE OF THE SPUR GEAR DESIGN SPREADSHEET

The spreadsheet developed in Section 9–11 is a useful tool that aids the designer in the process of completing a design for a pair of gears to be safe with regard to bending stresses in the teeth of the gears and for pitting resistance. The use of the spreadsheet was demonstrated for the data in Example Problem 9–7 as shown in Figure 9–23.

An important use for the spreadsheet is to propose and analyze several design alternatives and to work toward a goal of optimizing the design with regard to size, cost, or other parameters important to a particular design objective.

Designers for typical machine and vehicle drives would plan to use Grade 1 steels and standard quenching and tempering heat treatments. Where small size is critical or where cost is not a major concern, case hardening by carburizing, induction or flame hardening, or nitriding can be used. Use of more high-capacity Grade 2 or Grade 3 steels may also be specified if data are available. Therefore, it is usually desirable to produce several design alternatives that can be analyzed for cost and manufacturability. Then the final selection can be made with assurance that a reasonably optimum design has been identified.

Successive Iterations. We now continue the design process by making carefully selected changes in design decisions using the *Guidelines for Adjustments in Successive Iterations* from Section 9–9, just before Example Problem 9–7.

The design for the wood chipper drive from Example Problem 9–7 was very satisfactory for the goal of having an efficient design using through-hardened steel. Use of SAE 4140 OQT 1000 steel with a hardness of HB 341 is capable of resisting potential pitting caused by the pinion contact stress of approximately 134 ksi.

Now, how can we improve this design? The answer requires judgment about what constitutes improvement. Generally desirable design objectives for a gear drive, stated at the beginning of Section 9–9, were that the drive should:

Be compact and small	Operate smoothly and quietly
Have long life	Be low in cost
Be easy to manufacture	

Be compatible with other elements in the machine, such as bearings, shafts, the housing, the driver, and the driven machine.

In a given project, some of these objectives may be given higher priority than others. Furthermore, some objectives are counter to others; a highly compact and small drive will not likely be the lowest cost or the easiest to manufacture. However, it may be "worth the cost" to produce a smaller drive. More must be known about the application before making such judgments.

Let's proceed with the premise that the smallest practical gear drive for the wood chipper is desired while cost and ease of manufacture are secondary considerations. Referring to the *Guidelines,* we can conclude that the following changes will facilitate the design of a smaller, yet still safe design:

1. Obviously, each gear must be smaller than the initial design from Example Problem 9–7.

2. Using a higher value of diametral pitch with the same number of teeth will result in smaller gears and a correspondingly smaller center distance.
3. Smaller gears typically result in higher bending and contact stresses in the teeth, requiring higher strength materials.
4. The number of teeth (18) cannot be made much smaller without risking interference.
5. The face width can be used to *fine-tune* the design.
6. More accurate teeth (higher value of A_v) can also be used to *fine-tune* the design.

Design decisions for second trial design: The primary change to achieve a smaller design is to raise the value of diametral pitch. We can try $P_d = 16$ instead of the $P_d = 12$ used in Example Problem 9–7. Now we can show the advantage of using a spreadsheet as a calculation aid. Figure 9–24 shows the final result called Example Problem 9–9,—a much smaller design that is still safe. ***Reaching this redesign took only a few minutes.*** The changed data are highlighted within bold boxes. Those that are design decisions are in the gray-shaded boxes while those computed by the spreadsheet have white backgrounds. User-controlled changes are:

1. $P_d = 16$.
2. $F = 1.00$ in [At the maximum end of the recommended range; $F_{max} = 16/P_d$].
3. $A_v = 10$ [Modestly more precise than the original value of $A_v = 11$; note that $K_v = 1.24$ is moderately lower than the value of 1.35 in the initial design].
4. $K_m = 1.33$ [Modestly higher than the value of 1.31; caused by the change in $C_{pf} = 0.064$ from 0.042 for the initial trial because the ratio F/D_P changed. K_v was computed by the spreadsheet after the computed adjustment to C_{pf} was displayed and selected by the user].

Comparison of Results:

	Design 1	Design 2
	Example Problem 9–7	Example Problem 9–9
a. Pinion diameter: D_P	1.500 in	1.125 in
b. Gear diameter: D_G	5.667 in	4.250 in
c. Center distance: C	3.583 in	2.688 in
d. Pitch line speed: v_t	687 ft/min	515 ft/min
e. Transmitted load: W_t	144 lb	192 lb
f. Required s_{atP}:	17 102 psi	28 496 psi
g. Required s_{atG}:	13 280 psi	22 127 psi
h. Required s_{acP}:	133 471 psi	172 288 psi
i. Required s_{acPG}:	129 256 psi	166 847 psi
j. Material:	SAE 4140 OQT 1000 Through hardened	SAE 6150 OQT 1200, Case hardened by induction hardening

Notes about the changes and their effects:

(a) Smaller gears do produce higher stresses.

(b) The size of the gears and the overall drive were reduced significantly as can be seen in Figure 9–25 that shows a scale drawing of the two designs. One measure is the set of minimum inside dimensions of the housing to enclose the two gears, shown as x and y.

$x_1 = 7.333$ in $x_2 = 5.500$ in (25% smaller)
$y_1 = 5.833$ in $y_2 = 4.375$ in (25% smaller)

Where $x = C + D_{oP}/2 + D_{oG}/2$ and $y = D_{oG}$

(c) The contact stress in the pinion (the governing stress value) increased by 29% for design #2 as compared with design #1. This was the result of the finer teeth and the smaller pitch diameter for the gears.

(d) The value of the contact stress in the pinion of approximately 172 ksi made it impractical to use through-hardened steel. Note that the calculation built into the spreadsheet indicates that a hardness of 445 HB would be required. However, Figure 9–19 indicates that no design should rely on a hardness over 400 HB.

(e) Therefore, resisting $s_{acP} = 172$ ksi requires some kind of case hardening, either *induction hardening*, *flame hardening*, or *case hardening by carburizing* as indicated in Table 9–9.

(f) Use of induction hardening was a design decision but induction is a very frequent choice in the gear industry. See Internet site 19 for one provider of commercially available production-oriented induction hardening systems.

(g) Specification of SAE 6150 steel for the gears was a design decision based on the need for a highly hardenable steel that can be hardened to 54 HRC minimum (as noted in Table 9–9). Referring to Figure A4–6 indicates that this alloy in the as-quenched condition can be in the range of 627 HB, corresponding to approximately 58 HRC (Appendix 17).

(h) Induction hardening is considered a tertiary process for gears because:
   (a) Gears are first machined, typically by hobbing.
   (b) Then the gears are heat-treated to produce desirable properties in the core of the teeth. In this case, by referring to Figure A4–6, we chose to use the OQT 1200 heat treatment that will produce a core hardness of HB 293 and 20% elongation, a highly ductile condition.
   (c) Then the gears are induction hardened for a moderate depth to achieve the high pitting resistance on the faces of the teeth. Well-designed processes will also produce case hardening in the root area where the highest bending stresses occur.

(i) Because the heat treatment may cause distortion, final grinding or other finishing processes may be needed to produce the final gear quality number; in the case $A_v = 11$.

DESIGN OF SPUR GEARS

APPLICATION:	*Example Problem 9–9 –Redesign of* Example Problem 9–7		
Wood Chipper driven by an electric motor			
Initial Input Data:			
Input Power:	$P =$	*3.00*	hp
Input Speed:	$n_P =$	*1750*	rpm
Diametral Pitch:	$P_d =$	*16*	
Number of Pinion Teeth:	$N_P =$	*18*	
Desired Output Speed:	$n_G =$	*462.5*	rpm
Computed number of gear teeth:		68.1	
Enter: Chosen No. of Gear Teeth:	$N_G =$	*68*	
Computed data:			
Actual Output Speed:	$n_G =$	463.2	rpm
Gear Ratio:	$m_G =$	3.78	
Pitch Diameter - Pinion:	$D_P =$	1.125	in
Pitch Diameter - Gear:	$D_G =$	4.250	in
Center Distance:	$C =$	2.688	in
Pitch Line Speed:	$v_t =$	515	ft/min
Transmitted Load:	$W_t =$	192	lb
Secondary Input Data:			
	Min	**Nom**	**Max**
Face Width Guidelines (in):	0.500	0.750	1.000
Enter: Face Width:	$F =$	*1.000* in	
Ratio: Face width/pinion diameter:	$F/D_P =$	0.89	
Recommended ratio	$F/D_P <$	2.00	
Enter: Elastic Coefficient:	$Cp =$	2300	Table 9–7
Enter: Quality Number:	$A_v =$	*10*	Table 9–5
Enter: Bending Geometry Factors:			
Pinion:	$J_P =$	*0.325*	Fig. 9–10
Gear:	$J_G =$	*0.410*	Fig. 9–17
Enter: Pitting Geometry Factor:	$I =$	*0.104*	Fig. 9–17
REF:	$m_G =$	3.78	

Factors in Design Analysis:				
Alignment Factor, $K_m = 1.0 + C_{pf} + C_{ma}$	If $F < 1.0$	If $F > 1.0$	$F/D_P =$	0.89
Pinion Proportion Factor, $C_{pf} =$	0.064	0.064	$[0.50 < F/D_P < 2.00]$	
Enter: $C_{pf} =$	*0.064*	Figure 9–12		
Type of gearing:	Open	Commer.	Precision	Ex. Prec.
Mesh Alignment Factor, $C_{ma} =$	0.264	0.143	0.080	0.048
Enter: $C_{ma} =$	*0.264*	Figure 9–13		
Alignment Factor: $K_m =$	1.33	[Computed]		
Overload Factor: $K_o =$	*1.75*	Table 9–1		
Size Factor: $K_s =$	*1.00*	Table 9–2: Use 1.00 if $P_d >= 5$		
Pinion Rim Thickness Factor: $K_{BP} =$	*1.00*	Fig. 9–14: Use 1.00 if solid blank		
Gear Rim Thickness Factor: $K_{BG} =$	*1.00*	Fig. 9–14: Use 1.00 if solid blank		
Dynamic Factor: $K_v =$	1.24	[Computed: See Table 9–16]		
Service Factor: $SF =$	*1.00*	Use 1.00 if no unusual conditions		
Reliability Factor: $K_R =$	*1.00*	Table 9–11 Use 1.00 for $R = .99$		
Enter: Design Life:	*3000*	hours	See Table 9–12	
Pinion - Number of load cycles: $N_P =$	3.15E+08	Guidelines: Y_N, Z_N		
Gear - Number of load cycles: $N_G =$	8.34E+07	10^7 cycles	$>10^7$	$<10^7$
Bending Stress Cycle Factor: $Y_{NP} =$	*0.96*	1.00	0.96	Fig. 9–21
Bending Stress Cycle Factor: $Y_{NG} =$	*0.98*	1.00	0.98	Fig. 9–21
Pitting Stress Cycle Factor: $Z_{NP} =$	*0.92*	1.00	0.92	Fig. 9–22
Pitting Stress Cycle Factor: $Z_{NG} =$	*0.95*	1.00	0.95	Fig. 9–22
Stress Analysis: Bending				
Pinion: Required $s_{at} =$	28,496	psi	See Fig. 9–18 or	
Gear: Required $s_{at} =$	22,127	psi	Table 9–9 or 9–10	
Stress Analysis: Pitting				
Pinion: Required $s_{ac} =$	172,288	psi	See Fig. 9–19 or	
Gear: Required $s_{ac} =$	166,847	psi	Table 9–9 or 9–10	
Specify materials, alloy and heat treatment, for most severe requirement.				
Required hardnesses for pinion and gear are too high for through hardening.				
Specifying case hardening by induction to HRC 54 minimum-Both pinion and gear				
Pinion and Gear: SAE 6150; Core heat treatment OQT 1200; HB 293.				

FIGURE 9–24 Spreadsheet solution for Example Problems 9–9; the alternate spur gear design using data for Example Problem 9–7.

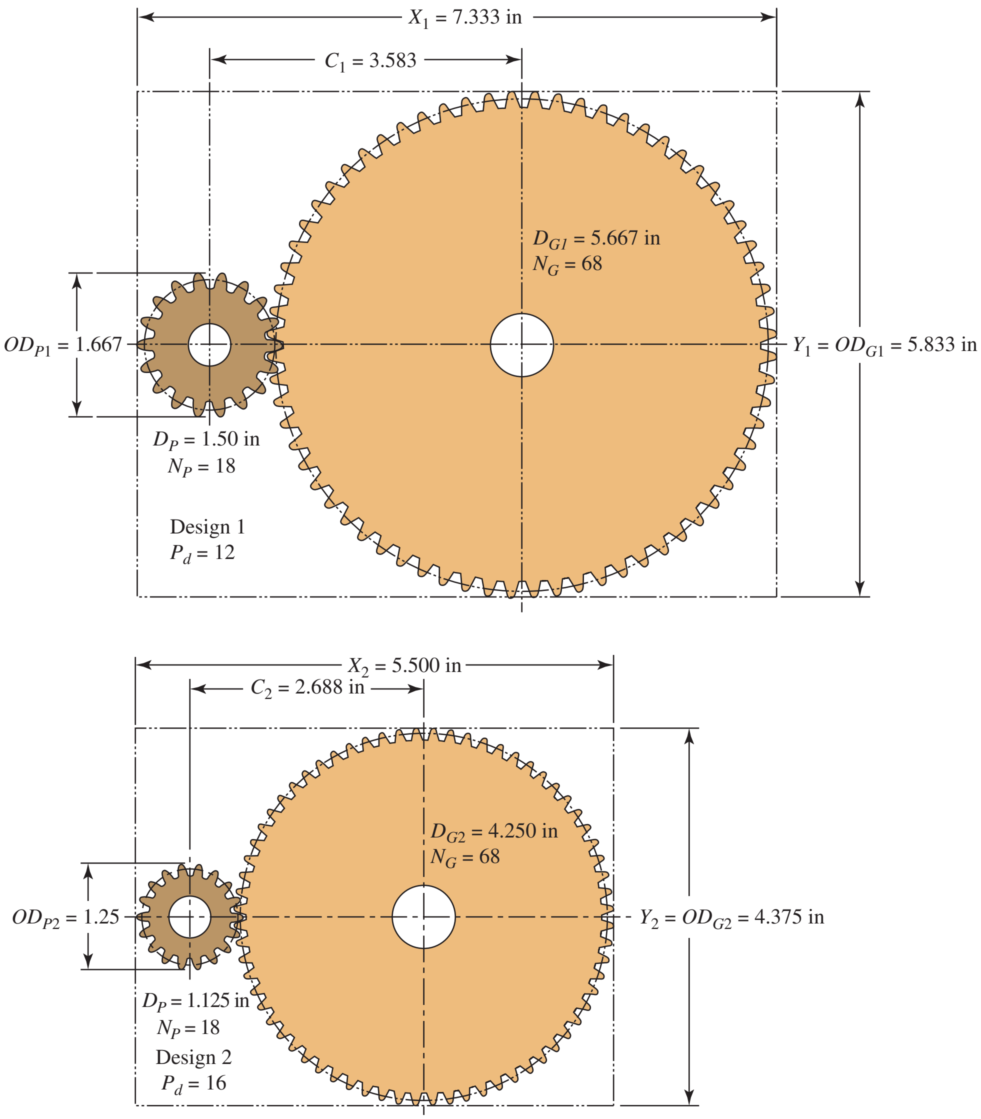


FIGURE 9–25 Comparison of sizes for gear designs for Example Problems 9–7 and 9–9 with minimum inside dimensions for housings

(j) This long list of processing steps makes the choice of case hardening by induction hardening much more costly than basic through hardening. The designer must make the judgment that the value of the benefit of the smaller system exceeds the added costs.

It is expected that readers of this book use similar decision-making processes when designing gear drives and specifying materials and heat treatments in their designs.

9–13 POWER-TRANSMITTING CAPACITY

It is sometimes desirable to compute the amount of power that a gear pair can safely transmit after it has been completely defined. The *power-transmitting capacity* P_{cap} is the capacity when the tangential load causes the expected stress to equal the allowable stress number with all of the modifying factors considered. The capacity should be computed for both bending and pitting resistance and for both the pinion and the gear.

When similar materials are used for both the pinion and the gear, it is likely that the pinion will be critical for bending stress. But the most critical condition is usually pitting resistance. The following relationships can be used to compute the power-transmitting capacity. In this analysis, it is assumed that the operating temperature of the gears and their lubricants is 250°F and that gears are produced with the appropriate surface finish.

Bending

We start with Equations (9–16) and (9–30) in which the computed bending stress number is compared with the modified allowable bending stress number for the gear:

$$s_{at} = s_t \frac{(SF)(KR)}{Y_N} = \frac{W_t P_d}{FJ} K_o K_s K_m K_B K_v \times \frac{(SF)(KR)}{Y_N}$$

But solving for W_t gives

$$W_t = \frac{s_{at} Y_N F J}{(SF) K_R K_o K_s K_m K_B K_v P_d} \quad (9\text{–}34)$$

It was shown in Equation (9–8) that

$$W_t = (126\ 000)(P)/(n_P D_P)$$

Then substituting into Equation (9–34) and calling the power P_{cap} gives

$$\frac{(126\ 000)(P_{cap})}{n_P D_P} = \frac{s_{at} Y_N F J}{(SF) K_R K_o K_s K_m K_B K_v P_d}$$

Solving for P_{cap}, we have

$$P_{cap} = \frac{s_{at} Y_N F J n_P D_P}{(126\ 000)(P_d)(SF) K_R K_o K_s K_m K_B K_v} \quad (9\text{–}35)$$

This equation should be solved for both the pinion and the gear. Most variables will be the same except for s_{at}, Y_N, J, and possibly K_B.

Pitting Resistance

Here we start with Equations (9–23) and (9–31) in which the computed contact stress number is compared with the modified allowable contact stress number for the gear. Equation (9–26) can be expressed in the form

$$s_{ac} = s_c \frac{(SF)(KR)}{Z_N} = C_P \sqrt{\frac{W_t K_o K_s K_m K_v}{D_P F I}} \times \frac{(SF)(KR)}{Z_N}$$

Squaring both sides of this equation and solving for W_t gives

$$\frac{W_t K_o K_s K_m K_v}{D_P F I} = \left[\frac{s_{ac} Z_N}{(SF) K_R C_P}\right]^2$$

$$W_t = \frac{D_P F I}{K_o K_s K_m K_v}\left[\frac{s_{ac} Z_N}{(SF) K_R C_P}\right]^2 \quad (9\text{–}36)$$

Now substituting this into Equation (9–8) and solving for the power P_{cap} gives

$$P_{cap} = \frac{W_t D_P n_P}{126\ 000} = \frac{D_P n_P D_P F I}{126\ 000\ K_o K_s K_m K_v}\left[\frac{s_{ac} Z_N}{(SF) K_R C_P}\right]^2$$

$$P_{cap} = \frac{n_P F I}{126\ 000\ K_o K_s K_m K_v}\left[\frac{s_{ac} D_P Z_N}{(SF) K_R C_P}\right]^2 \quad (9\text{–}37)$$

Equations (9–35) and (9–37) should be used to compute the power-transmitting capacity for a pair of gears of known design with particular materials. From the material specification along with its condition (typically a heat treatment or case-hardening process), the limiting values for s_{at} and s_{ac} can be found from Figures 9–18 and 9–19 and from Tables 9–9 and 9–10. Entering those values along with the known data for the proposed gear design permits the calculation of the power-transmitting capacity, P_{cap}. This value can be shared with colleagues and customers to verify the suitability of a design for a particular application.

9–14 PLASTICS GEARING

Plastics are satisfying an important and growing part of the applications for gearing. Some of the numerous advantages of plastics in gearing systems compared with steels and other metals are:

- Lighter weight.
- Lower inertia.
- Possibility of running with little or no external lubrication.
- Quieter operation.
- Low sliding friction, which results in efficient gear meshing.
- Chemical resistance and ability to operate in corrosive environments.
- Ability to operate well under conditions of moderate vibration, shock, and impact.
- Relatively low cost when made in large quantities.
- Ability to combine several features into one part.
- Accommodation of larger tolerances because of resiliency.
- Material properties that can be tailored to meet the needs of the application.
- Less wear among some plastics compared to metals in certain applications.

The advantages must be weighed against disadvantages such as:

- Relatively lower strength of plastics as compared with metals.
- Lower modulus of elasticity.
- Higher coefficients of thermal expansion.
- Difficulty operating at high temperatures.
- Initial high cost for design, development, and mold manufacture.
- Dimensional change with moisture absorption that varies with conditions.
- Wide range of possible material formulations, which makes design more difficult.

Some plastic gears are cut using hobbing or shaping processes similar to those used to cut metallic gears. However, most plastic gears are produced with the injection molding process because of its ability to make large quantities rapidly with low unit cost. Mold design is critical because it must accommodate the shrinking that occurs as the molten plastic solidifies. The typical successful approach accounts for predicted shrinkage by making the die larger than the required finished gear size. However, the allowance is not uniform throughout the gear, and significant amounts of data are required about the material molding properties and the molding process itself to produce plastic gears with high dimensional accuracy. Computer-assisted mold design software that simulates the flow of molten plastic through the mold cavities and the curing process is often used. The gear mold or the gear cutting tools are designed to produce dimensionally accurate gear teeth with tooth thickness controlled to produce a proper amount of backlash during operation. The electrical discharge machining process (EDM) is typically used to produce accurate gear tooth forms in molds made from high-hardness, wear-resistant steels to ensure that large production runs can be made without replacing tooling.

Plastic Materials for Gears

The great variety of plastics available makes material selection difficult, and it is recommended that gear system designers consult with material suppliers, mold designers, and manufacturing staff during the design process. While simulation can aid in reaching a suitable design, it is recommended that testing be done in realistic conditions before committing the design to production. Some of the more popular types of materials used for gears are:

Nylon	Acetal	ABS (acrylonitrile-butadiene-styrene)
Polycarbonate	Polyurethane	Polyester thermoplastic
Polyimide	Phenolic	Polyphenylene sulfide
Polysulfones	Phenylene oxides	Styrene-acrylonitrile (SAN)

Designers must seek a balance of material characteristics appropriate to the application, considering, for example:

- Strength in flexure under fatigue conditions.
- High modulus of elasticity for stiffness.
- Impact strength and toughness.
- Wear and abrasion resistance.
- Dimensional stability under expected temperatures.
- Dimensional stability due to moisture absorption from liquids and humidity.
- Frictional performance and need for lubrication, if any.
- Operation in vibration environments.
- Chemical resistance and compatibility with the operating environment.
- Sensitivity to ultraviolet radiation.
- Creep resistance if operated under load for long periods of time.
- Flame retarding ability.
- Cost.
- Ease of processing and molding.
- Assembly and disassembly considerations.
- Compatibility with mating parts.
- Environmental impact during processing, use, recycling, and disposal.

The basic plastic materials listed previously are typically modified with fillers and additives to produce optimum as-molded properties. Some of these are:

> ***Reinforcements for strength, toughness, moldability, long-term stability, thermal conductivity, and dimensional stability:*** Long glass fibers, chopped glass fibers, milled glass, woven glass fibers, carbon fibers, glass beads, aluminum flake, mineral, cellulose, rubber modifiers, wood flour, cotton, fabric, mica, talc, and calcium carbonate.
>
> ***Fillers to improve lubricity and overall frictional performance:*** PTFE (polytetrafluoroethylene), silicone, carbon fibers, graphite powders, and molybdenum disulfide (MoS_2).

Refer to Section 2-17 for additional discussion about plastic materials, their properties, and special considerations for selecting plastics. See Internet sites 1, 8, 18, and 20.

Design Strength for Plastic Gear Materials

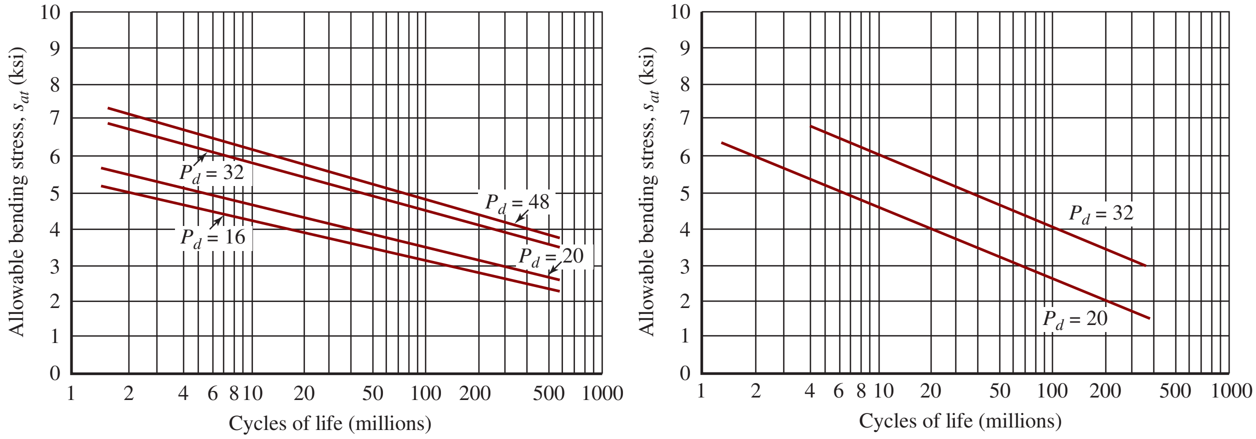
Data are provided here for typical plastic materials used for gears. They can be applied to problem solving in this book. However, verification of properties for materials to be actually used in a commercial application, with due regard for the operating conditions, should be acquired from the material supplier. The effects of temperature on strength, modulus, toughness, chemical stability, and dimensional precision are particularly important. Manufacturing processes must be controlled to ensure that final properties are consistent with prescribed values.

Table 9–14 lists some selected data for allowable tooth bending stress, s_{at}, in plastic gears. Much additional data for other materials can be found in References 19 and 21. Note the significant increase in allowable strength provided by the glass reinforcement. The combination of glass fibers and the basic plastic matrix performs like a composite material with the amount of reinforcement typically ranging from 20% to 50%.

TABLE 9–14 Approximate Allowable Tooth Bending Stress, s_{at}, in Plastic Gears

	Approximate allowable bending stress, s_{at}, psi (MPa)	
Material	**Unfilled**	**Glass-filled**
ABS	3000 (21)	6000 (41)
Acetal	5000 (34)	7000 (48)
Nylon	6000 (41)	12 000 (83)
Polycarbonate	6000 (41)	9000 (62)
Polyester	3500 (24)	8000 (55)
Polyurethane	2500 (17)	

Source: Plastics Gearing. Manchester, CT: ABA/PGT Publishing, 1994.



(*a*) Fatigue life data for DuPont© Zytel nylon resin

(*b*) Fatigue life data for DuPont© Delrin acetal resin

FIGURE 9–26 Fatigue life data for two types of plastic materials used for gears

Material suppliers may be able to provide fatigue data for plastics in charts such as those shown in Figure 9–26, showing allowable bending stress versus number of cycles to failure for DuPont Zytel© nylon resin and Delrin© acetal resin. These data are for molded gears operating at room temperature with diametral pitches shown, pitch line velocity below 4000 ft/min, and continuous lubrication. Reductions should be applied for cut gears, higher temperatures, different pitches, and different lubrication conditions. See Reference 21.

Tooth Geometry

In general, standard tooth geometry for plastic gears conforms to the configurations described in Section 8–4. Standard diametral pitches from Table 8–3 and standard metric modules from Table 8–4 should be used unless there are major advantages to using other values. Suppliers' ability to provide nonstandard pitches should be investigated. Pressure angles of $14\frac{1}{2}°$, 20°, and 25° are used, with 20° usually preferred. Standard formulas for addendum, dedendum, and clearance for full-depth involute teeth are listed in Table 8–1. Gear quality values are set similarly to those for metallic gears as discussed in Section 9–9. The typical AGMA quality number produced by injection molding is in range of A11 to A7.

Designers sometimes use special tooth forms to tailor the strength of plastic gear teeth to the demands of particular applications. The 20° stub tooth system provides a shorter, broader tooth than the standard 20° full-depth tooth system, decreasing tooth-bending stress. The Plastics Gearing Technology unit of the ABA-PGT company has developed another system that is finding favor with some designers. See References 1, 2, 12, and 13.

Many designers of plastic gears prefer to use a longer addendum on the pinion and a shorter addendum on the mating gear to produce more favorable operation because of the greater flexibility of plastics as compared with metals. Tooth thickness is typically thinned on either or both of the pinion and gear to provide acceptable backlash and to ensure that mating gears do not bind. Binding may result from deflection of the teeth under load or from expansions due to increased temperature or moisture absorption from exposure to water or high humidity. Enlarging the center distance is another method employed to adjust for backlash. Designers must specify these feature sizes on drawings and in specifications. Consult AGMA Standard 1006 (Reference 12) *Tooth Proportions for Plastic Gears* for details. Reference 2 provides useful tables of formulas and data for adjustments to tooth form and center distance. Reference 21 recommends the range of backlash values shown in Figure 9–27.

Shrinkage

During manufacture of plastic gears using injection molding, enlarging the effective diametral pitch and the pitch diameter of the gear teeth cut into the mold accommodates shrinkage. The pressure angle is also

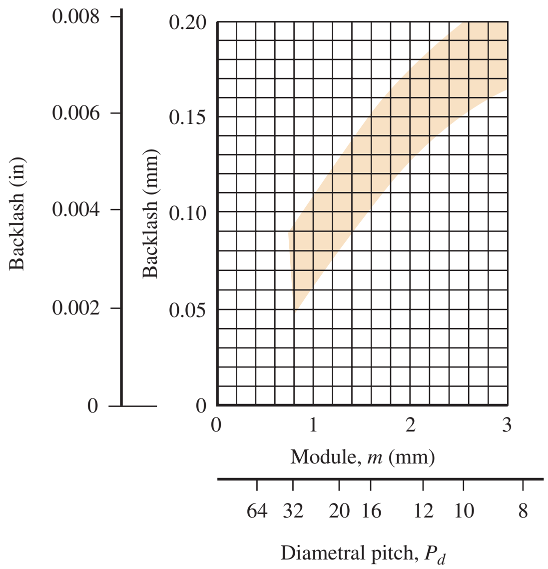


FIGURE 9–27 Recommended backlash for plastic gears

adjusted. The nominal corrections are computed as follows:

$$P_{dc} = \frac{P_d}{(1 + S)} \tag{9–38}$$

$$\cos \phi_1 = \frac{\cos \phi}{(1 + S)} \tag{9–39}$$

$$D_c = N/P_{dc} \tag{9–40}$$

where S = shrinkage of material
P_d = standard diametral pitch for the gear
P_{dc} = modified diametral pitch of the teeth in the mold
ϕ = standard pressure angle for the gear
ϕ_1 = modified pressure angle of the teeth in the mold
N = number of teeth
D_c = modified pitch diameter of teeth in the mold

After molding, the teeth should very nearly conform to standard geometry. Additional adjustments are sometimes made, relieving the tips of the teeth for smoother engagement and increasing the tooth width at the base near the point of highest bending stress.

Stress Analysis

Bending stress analysis for plastic gears relies on the basic Lewis formula introduced in Section 9–5, Equation (9–13). The modifying factors called for by the AGMA standards for steel gears are not specified for plastic gears at this time. We can account for uncertainty or shock loading by inserting a safety factor. The overload factor from Table 9–1 can be used as a guide. Testing of the proposed design in realistic conditions should be completed. The bending stress equation then becomes

$$\sigma_t = \frac{W_t P_d (SF)}{FY} \tag{9–41}$$

Values for the Lewis form factor, Y, shown in Table 9–15, describe the geometry of the involute gear teeth acting as a cantilever beam with the load applied near the pitch point. Thus Equation (9–41) gives the bending stress at the root of the tooth. Most plastic gear designs call for a generous fillet radius between the start of the active involute profile on the flank of the tooth and the root, resulting in little, if any, stress concentration.

TABLE 9–15 Lewis Tooth Form Factor, Y, for Load Near the Pitch Point

	Tooth form		
Number teeth	$14\frac{1}{2}°$ Full depth	20° Full depth	20° Stub
14	–	–	0.540
15	–	–	0.566
16	–	–	0.578
17	–	0.512	0.587
18	–	0.521	0.603
19	–	0.534	0.616
20	–	0.544	0.628
22	–	0.559	0.648
24	0.509	0.572	0.664
26	0.522	0.588	0.678
28	0.535	0.597	0.688
30	0.540	0.606	0.698
34	0.553	0.628	0.714
38	0.566	0.651	0.729
43	0.575	0.672	0.739
50	0.588	0.694	0.758
60	0.604	0.713	0.774
75	0.613	0.735	0.792
100	0.622	0.757	0.808
150	0.635	0.779	0.830
300	0.650	0.801	0.855
Rack	0.660	0.823	0.881

Wear Considerations

Wear of tooth surfaces in plastic gear teeth is a function of the contact stress between mating teeth as it is with metal teeth. Equation (9–18) can be used to compute the contact stress. However, published data are lacking for allowable contact stress values.

In reality, lubrication and the *combination of materials in mating gears* play major roles in the wear life of the pair. Communication with material suppliers and testing of proposed designs are recommended.

Presented here are some general guidelines from References 2 and 21.

- Continuously lubricated gearing promotes the longest life.
- With continuous lubrication and light loads, fatigue resistance, not wear, typically determines life.
- Unlubricated gears tend to fail by wear, not fatigue, provided proper design bending stresses are used.
- When continuous lubrication is not practical, initially lubricating the gearing can aid in the run-in process and add life compared with gears that are never lubricated.
- When continuous lubrication is not practical, the combination of a nylon pinion and an acetal gear exhibits low friction and wear.
- Excellent wear performance for relatively high loads and pitch line speeds can be obtained by using a lubricated pair of a hardened steel pinion (HRC $>$ 50)

mating with a plastic gear made from nylon, acetal, or polyurethane.

- Wear accelerates when operating temperatures rise. Cooling to promote heat dissipation can increase life.

Gear Shapes and Assembly

References 2 and 21 include many recommendations for the geometric design of gears considering strength, inertia, and molding conditions. Many smaller gears are simply made with uniform thickness equal to the face width of the gear teeth. Larger gears often have a rim to support the teeth, a thinned web for lightening and material savings, and a hub to facilitate mounting on a shaft. Figure 9–28 shows recommended proportions. Symmetrical cross sections are preferred, along with balanced section thicknesses to promote good flow of material and to minimize distortion during molding.

Fastening gears to shafts requires careful design. Keys placed in shaft key seats and keyways in the hub of the gear provide reliable transmission of torque. For light torques, setscrews can be used, but slippage and damage of the shaft surface are possible. The bore of the gear hub can be lightly press-fit onto the shaft with care to ensure that a sufficient torque can be transmitted while not overstressing the plastic hub. Knurling the shaft before pressing the gear on increases the torque capability. Some designers prefer to use metal hubs to facilitate the use of keys. Plastic is then molded onto the hub to form the rim and gear teeth.

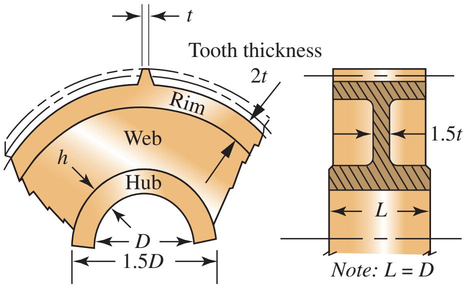


FIGURE 9–28 Suggested plastic gear proportions

Design Procedure

Design of plastic gearing should consider a variety of possibilities, and it is likely to be an iterative process. The following procedure outlines the steps for a given trial using U.S. Customary units for use in this book.

PROCEDURE FOR DESIGNING PLASTIC GEARS

1. Determine the required horsepower, P, to be transmitted and the speed of rotation, n_P, of the pinion in rpm.
2. Specify the number of teeth, N, and select a trial diametral pitch for the pinion.
3. Compute the pinion diameter from $D_P = N_P/P_d$.
4. Compute the transmitted load, W_t (in lb), from Equation (9–8), repeated here.

$$W_t = (126\,000)(P)/(n_P D_P)$$

5. Specify the tooth form and determine the Lewis form factor, Y, from Table 9–15.
6. Specify a safety factor, SF. Refer to Table 9–1 for guidance.
7. Specify the material to be used and determine the allowable stress, s_{at}, from Table 9–15 or Figure 9–31.
8. Solve Equation (9–41) for the face width, F, and compute its value from,

$$F = \frac{W_t P_d (SF)}{s_{at} Y} \quad \textbf{(9–42)}$$

9. Judge the suitability of the computed face width as it relates to the application. Consider its mounting on a shaft, space available in the diametral and axial directions, and whether the general proportions are acceptable for injection molding. See References 2 and 21. No general recommendations are published for the face width of plastic gears and often they are narrower than similar metallic gears.
10. Repeat steps 2 to 9 until a satisfactory design for the pinion is achieved. Specify convenient dimensions for the final value of the face width and other features of the pinion.
11. Considering the desired velocity ratio between the pinion and the gear, compute the required number of teeth in the gear and repeat steps 3 to 9 using the same diametral pitch as the pinion. Using the same face width as for the pinion, the stress in the gear teeth will always be lower than in the pinion because the form factor Y will increase and all other factors will be the same. When the same material is to be used for the gear, it will always be safe. Alternatively, you could compute the bending stress directly from Equation (9–41) and specify a different material for the gear that has a suitable allowable bending stress.

Example Problem 9–10 Design a pair of plastic gears for a paper shredder to transmit 0.25 horsepower at a pinion speed of 1160 rpm. The pinion will be mounted on the shaft of an electric motor that has a diameter of 0.625 in with a keyway for a 3/16 × 3/16 in key. The gear is to rotate approximately 300 rpm.

Given Data $P = 0.25$ hp, $n_P = 1160$ rpm,

Shaft diameter $= D_s = 0.625$ in, Keyway for a 3/16 × 3/16 in key.

Approximate gear speed $= n_G = 300$ rpm

Solution Use the design procedure outlined in this section.

Step 1. Consider the given data.

Step 2. Specify $N_P = 18$ and $P_d = 16$ (Design decisions)

Step 3. $D_P = N_P/P_d = 18/16 = 1.125$ *in*. This seems reasonable for mounting on the 0.625 in motor shaft.

Step 4. Compute the transmitted load,

$$W_t = (126\,000)(P)/(n_P D_P) = (126\,000)(0.25)/[(1160)(1.125)] = 24.1 \text{ lb}$$

Step 5. Specify 20° full-depth teeth. Then $Y = 0.521$ for 18 teeth from Table 9–15.

Step 6. Specify a safety factor, SF. The shredder will likely experience light shock; the preference is to operate the gears without lubrication. Specify $SF = 1.50$ from Table 9–1.

Step 7. Specify unfilled nylon. From Table 9–14, $s_{at} = 6000$ psi.

Step 8. Compute the required face width using Equation (9–42).

$$F = \frac{W_t P_d (SF)}{s_{at} Y} = \frac{(24.1)(16)(1.50)}{(6000)(0.521)} = 0.185 \text{ in}$$

Step 9. The dimensions seem reasonable.

Step 10. Appendix 2 lists a preferred size for the face width of 0.200 in.

Comment: In summary, the proposed pinion has the following features:

$$P_d = 16,\ N_P = 18 \text{ teeth},\ D_P = 1.125 \text{ in},\ F = 0.200 \text{ in, Bore} = 0.625 \text{ in},$$

Keyway for a 3/16 × 3/16 in key. Unfilled nylon material.

Step 11. Gear design: Specify $F = 0.200$ in, $P_d = 16$. Compute the number of teeth in the gear.

$N_G = N_P(n_P/n_G) = 18(1160/300) = 69.6$ teeth

Specify $N_G = 70$ teeth

Pitch diameter of gear $= D_G = N_G/P_d = 70/16 = 4.375$ in

From Table 9–15, $Y_G = 0.728$ by interpolation.

Stress in gear teeth using Equation (9–41):

$$\sigma_t = \frac{W_t P_d (SF)}{FY} = \frac{(24.1)(16)(1.50)}{(0.200)(0.728)} = 3973 \text{ psi}$$

Comment: This stress level is safe for nylon. The gear could also be made from acetal to achieve better wear performance.

9–15 PRACTICAL CONSIDERATIONS FOR GEARS AND INTERFACES WITH OTHER ELEMENTS

It is important to consider the design of the entire gear system when designing the gears because they must work in harmony with the other elements in the system. This section will briefly discuss some of these practical considerations and will show commercially available speed reducers.

Our discussion so far has been concerned primarily with the gear teeth, including the tooth form, pitch, face width, material selection, and heat treatment. Also to be considered is the type of gear blank. Figures 8–2 and 8–4 show several styles of blanks. Smaller gears and lightly loaded gears are typically made in the plain style. Gears with pitch diameters of approximately 5.0 in through 8.0 in are frequently made with thinned webs between the rim and the hub for lightening, with some having holes bored in the webs for additional lightening. Larger gears, typically with pitch diameters greater than 8.0 in, are made from cast blanks with spokes between the rim and the hub.

In many precision special machines and gear systems produced in large quantities, the gears are machined integral with the shaft carrying the gears. This, of course, eliminates some of the problems associated with mounting and location of the gears, but it may complicate the machining operations.

In general machine design, gears are usually mounted on separate shafts, with the torque transmitted from the shaft to the gear through a key. This setup provides a positive means of transmitting the torque while permitting easy assembly and disassembly. The axial location of the gear must be provided by another means, such as a shoulder on the shaft, a retaining ring, or a spacer (see Chapters 11 and 12).

Other considerations include the forces exerted on the shaft and the bearings that are due to the action of the gears. These subjects are discussed in Section 9–3. The housing design must provide adequate support for the bearings and protection of the interior components. Normally, it must also provide a means of lubricating the gears.

See References 9, 19, 20, 22, and 25 for additional practical considerations.

Lubrication

The action of spur gear teeth is a combination of rolling and sliding. Because of the relative motion, and because of the high local forces exerted at the gear faces, adequate lubrication is critical to smoothness of operation and gear life. A continuous supply of oil at the pitch line is desirable for most gears unless they are lightly loaded or operate only intermittently.

In splash-type lubrication, one of the gears in a pair dips into an oil supply sump and carries the oil to the pitch line. At higher speeds, the oil may be thrown onto the inside surfaces of the case; then it flows down, in a controlled fashion, onto the pitch line. Simultaneously, the oil can be directed to the bearings that support the shafts. One difficulty with the splash type of lubrication is that the oil is churned; at high gear speeds, excessive heat can be generated, and foaming can occur.

A positive oil circulation system is used for high-speed and high-capacity systems. A separate pump draws the oil from the sump and delivers it at a controlled rate to the meshing teeth.

The primary functions of gear lubricants are to reduce friction at the mesh and to keep operating temperatures at acceptable levels. It is essential that a continuous film of lubricant be maintained between the mating tooth surfaces of highly loaded gears and that there be a sufficient flow rate and total quantity of oil to maintain cool temperatures. Heat is generated by the meshing gear teeth, by the bearings, and by the churning of the oil. This heat must be dissipated from the oil to the case or to some other external heat-exchange device in order to keep the oil itself below 200°F (approximately 93°C). Above this temperature, the lubricating ability of the oil, as indicated by its viscosity, is severely decreased. Also, chemical changes can be produced in the oil, decreasing its lubricity. Because of the wide variety of lubricants available and the many different conditions under which they must operate, it is recommended that suppliers of lubricants be consulted for proper selection. (See also References 10, 16, 27–29.)

The AGMA, in Reference 10, defines several types of lubricants for use in gear drives.

- ***Rust and oxidation inhibited gear oils*** (called R&O) are petroleum based with chemical additives.
- ***Compounded gear lubricants*** (CP) blend 3% to 10% of fatty oils with petroleum oils.
- ***Extreme pressure lubricants*** (EP) include chemical additives that inhibit scuffing of gear tooth faces.
- ***Synthetic gear lubricants*** (S) are special chemical formulations applied mostly in severe operating conditions.

R&O lubricants are supplied in 10 ISO viscosity grades where the lower numbers refer to the lower viscosities. Similar numbers are used for the other types with modified grade designations carrying suffixes *CP*, *EP*, or *S*. The recommended lubricant grade depends on the ambient temperature around the drive and the pitch line velocity of the lowest speed pair of gears in a reducer. See Tables 9–16 and 9–17. Wormgear drives call for higher viscosity grades.

TABLE 9–16 Recommended Lubricant Viscosity Grades for Enclosed Gear Drives

ISO viscosity grade	Midpoint viscosity at 40°C (mm^2/s)	Former AGMA grade equivalent
ISO VG 32	32	0
ISO VG 46	46	1
ISO VG 68	68	2
ISO VG 100	100	3
ISO VG 150	150	4
ISO VG 220	220	5
ISO VG 320	320	6
ISO VG 460	460	7
ISO VG 680	680	8
ISO VG 1000	1000	8A

Notes:
1. Viscostiy unit of mm^2/s is commonly referred to as centistokes (cSt).
2. ISO standard prescribes minimum and maximum kinematic viscosity limits for each grade.

Adapted from AGMA 6013 (Reference 9) *Standard for Industrial Enclosed Drives*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA

TABLE 9–17 Viscosity Grade Guidelines for Enclosed Gear Drives

Approximate temperature range				Approximate pitch line velocity of final stage of the drive, ft/min (m/s)						
Ambient		Oil sump		<1000	1000–2000	2000–3000	3000–4000	4000–5000	5000–6000	6000–7000
°F	°C	°F	°C	(<5)	(5–10)	(10–15)	(15–20)	(20–25)	(25–30)	(30–35)
−40 to −10	−40 to −23.3	<60	<15.6	68S	68S	46S	46S	46S	32S	32S
−10 to +20	−23.3 to −6.7	60 to 90	15.6 to 32.2	100S	100S	68S	68S	46S	32S	32S
20 to 40	−6.7 to 4.5	90 to 150	32.2 to 65.6	150	150	150	68	68	68	46
40 to 80	4.5 to 26.7	150 to 190	65.6 to 87.8	320	220	220	150	100	100	100
80 to 120	26.7 to 48.9	190 to 210	87.8 to 98.9	460	460	320	320	220	150	100
>120	>48.9	>210	>98.9	Not recommended						

Notes: Viscosity grades are for R&O type, unless Synthetic (S) is specified.

See Table 9–16 for listing of ISO viscosity grades.

Adapted from AGMA 6013 (Reference 9) *Standard for Industrial Enclosed Drives*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA

Commercially Available Gear-Type Speed Reducers

By studying the design of commercially available gear-type speed reducers, you should get a better feel for design details and the relationships among the component parts: the gears, the shafts, the bearings, the housing, the means of providing lubrication, and the coupling to the driving and driven machines.

Figure 9–29 shows a double-reduction spur gear speed reducer with an electric motor rigidly attached. Such a unit is often called a *gear motor*. Figure 9–30 shows a triple-reduction unit with spur gears in the final two stages and helical gears in the first stage (as discussed in Chapter 10). The cross-sectional drawing shown with Figure 9–31 gives a clear picture of the several components of a reducer.

The planetary reducer in Figure 9–32 has quite a different design to accommodate the placement of the sun, planet, and ring gears. Figure 9–33 shows the eight-speed transmission from a large farm tractor and illustrates the high degree of complexity that may be involved in the design of transmissions.

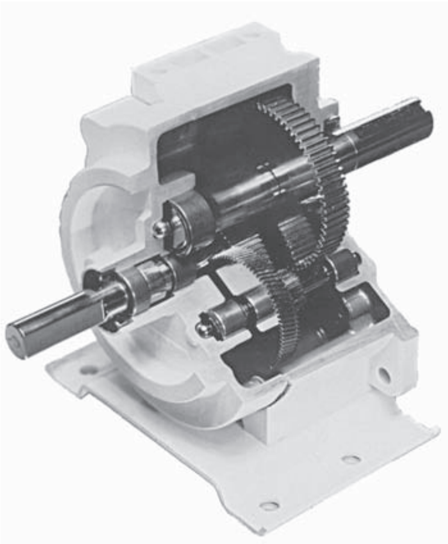


FIGURE 9–30 Triple-reduction gear reducer. Helical gears for stage one; spur gears for stages two and three. The pinion for stage three is on the lower left and not visible. (Bison Gear & Engineering Corporation, St. Charles, IL)

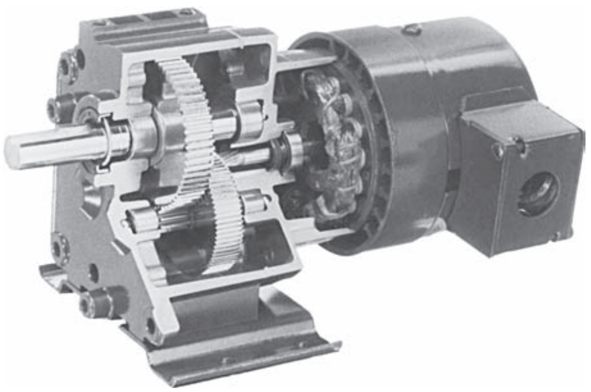
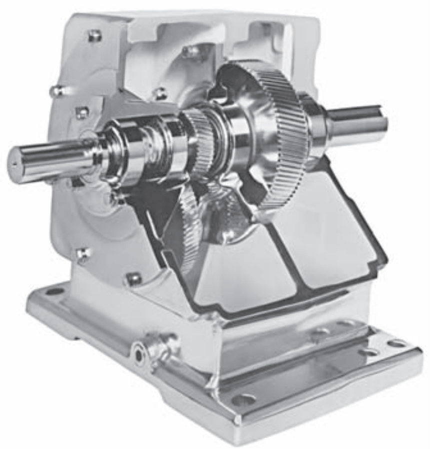
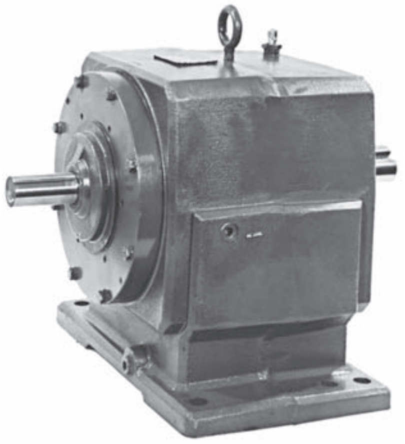


FIGURE 9–29 Double-reduction spur gear reducer (Bison Gear & Engineering Corporation, St. Charles, IL)



(*a*) Cutaway of a concentric helical gear reducer



(*b*) Complete reducer

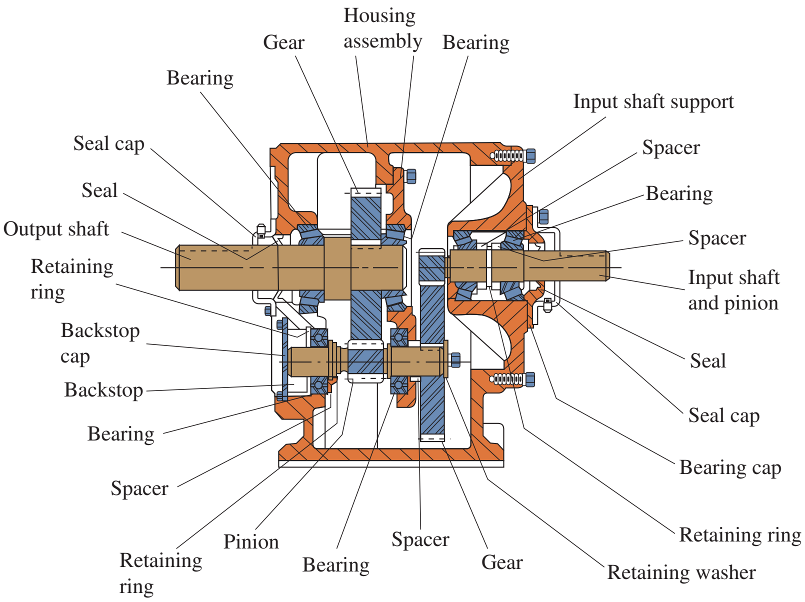
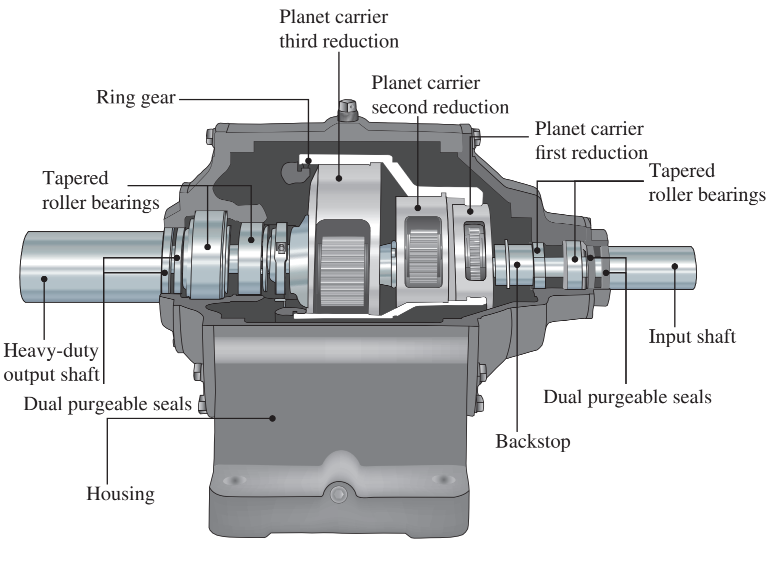
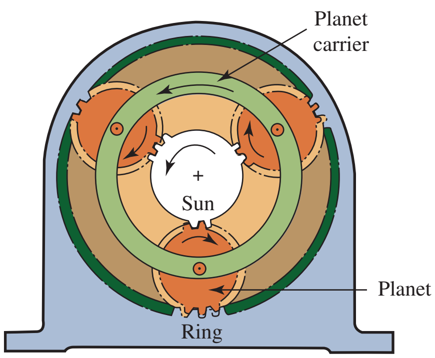


FIGURE 9–31 Concentric helical gear reducer



(*a*) Cutaway model with key features labeled



(*b*) Schematic arrangement of planetary gearing

FIGURE 9–32 Cutaway model of a triple reduction planetary gear reducer

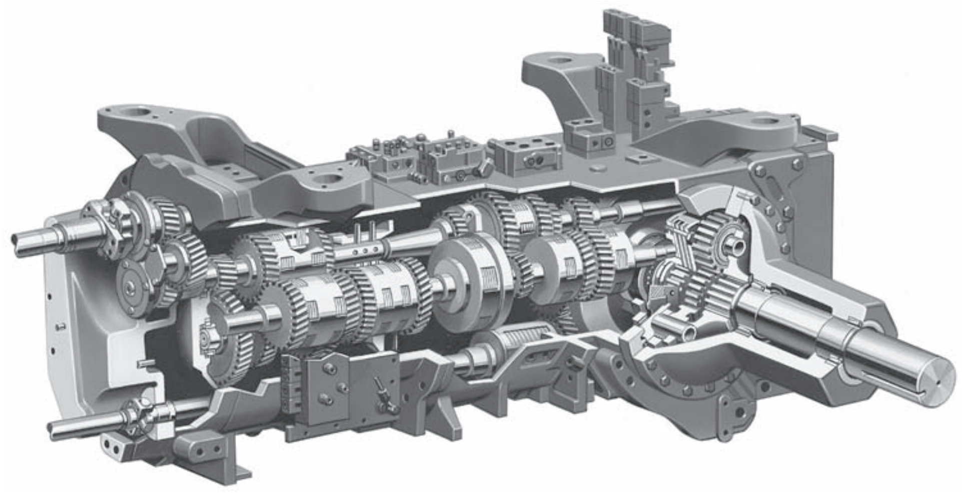


FIGURE 9–33 Eight-speed tractor transmission (Case IH, Racine, WI)

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INTERNET SITES RELATED TO SPUR GEAR DESIGN

1. **ABA-PGT, Inc.** The ABA division produces molds for making plastic gears using injection molding; the PGT division is dedicated to plastic gearing technology.
2. **American Gear Manufacturers Association (AGMA).** Develops and publishes voluntary, consensus standards for gears and gear drives.
3. **Baldor/Dodge.** Manufacturer of many power transmission components, including complete gear-type speed reducers, bearings, and components such as belt drives, chain drives, clutches, brakes, and couplings.
4. **Bison Gear, Inc.** Manufacturer of fractional horsepower gear reducers and gear motors. Site includes several videos showing Bison Gear products.
5. **Boston Gear, Company.** Manufacturer of gears and complete gear drives. Part of Altra Industrial Motion, Inc. Data provided for spur, helical, miter, bevel, and worm gearing.
6. **Bourn & Koch, Inc.** Manufacturer of hobbing, grinding, shaping, and other types of machines to produce gears, including the Barber-Colman line and Fellows shapers. Also provides remanufacturing services for a wide variety of existing machine tools.
7. **Drivetrain Technology Center.** Research center for gear-type drivetrain technology. Part of the Applied Research Laboratory of Penn State University.
8. **DuPont Polymers.** Information and data on plastics and their properties. A database by type of plastic or application.
9. **Regal Beloit Corporation.** The Browning and Morse Divisions produce spur, helical, bevel, and wormgearing and complete gear drives. Several other gear drive companies are part of Regal Beloit.
10. **Gear Technology Magazine.** Information source for many companies that manufacture or use gears or gearing systems. Includes gear machinery, gear cutting tools, gear materials, gear drives, open gearing, tooling and supplies, software, training and education.
11. **Gleason Corporation.** Manufacturer of many types of gear cutting machines for hobbing, milling, shaping, and grinding. The Gleason Cutting Tools Corporation manufactures a wide variety of milling cutters, hobs, shaper cutters, shaving cutters, and grinding wheels for gear production equipment. Producer of analytical gear inspection systems.
12. **International Organization for Standardization.** Organization that establishes standards for numerous types of products and devices including gearing. Recognized in most parts of the world. Most standards are presented in SI metric system units.
13. **Peerless-Winsmith, Inc.** Manufacturer of a wide variety of gear reducers and power transmission products, including wormgearing, planetary gearing, and combined helical/wormgearing. Subsidiary of HBD Industries, Inc.
14. **Power Transmission Engineering.** Clearinghouse on the Internet for buyers, users, and sellers of power transmission-related products and services. Included are gears, gear drives, and gear motors. Site includes several videos of power transmission products.
15. **Penta Gear Metrology.** Producer of innovative machines, products, systems, and services, including gear measurement systems. Provides gear inspection services also.
16. **QTC Metric Gears.** Supplier of stock metric gears.
17. **Star-SU, Inc.** Manufacturer of a wide range of gear-production systems including hobbing, grinding, shaping, and shaving, along with cutting tools for the gear industry.
18. **Stock Drive Products—Sterling Instruments.** Manufacturer and distributor of commercial and precision mechanical components, including gear reducers. Site includes an extensive handbook of design and information on metallic and plastic gears.
19. **Eldec Induction USA, Inc.** Producer of complete induction-hardening systems for production operations with special capabilities in gear manufacturing.
20. **Celanese Engineering Polymers.** Producer of plastic gears made from numerous high-performance polymers for many applications such as automotive, HVAC, industrial power tools, electronics components, medical, and home and garden equipment. Search under *Industrial–Gears*.

PROBLEMS

Forces on Spur Gear Teeth

1. A pair of spur gears with 20°, full-depth, involute teeth transmits 7.5 hp. The pinion is mounted on the shaft of an electric motor operating at 1750 rpm. The pinion has 20 teeth and a diametral pitch of 12. The gear has 72 teeth. Compute the following:
   a. The rotational speed of the gear
   b. The velocity ratio and the gear ratio for the gear pair
   c. The pitch diameter of the pinion and the gear
   d. The center distance between the shafts carrying the pinion and the gear
   e. The pitch line speed for both the pinion and the gear
   f. The torque on the pinion shaft and on the gear shaft
   g. The tangential force acting on the teeth of each gear
   h. The radial force acting on the teeth of each gear
   i. The normal force acting on the teeth of each gear

2. A pair of spur gears with 20°, full-depth, involute teeth transmits 50 hp. The pinion is mounted on the shaft of an electric motor operating at 1150 rpm. The pinion has 18 teeth and a diametral pitch of 5. The gear has 68 teeth. Compute the following:
   a. The rotational speed of the gear
   b. The velocity ratio and the gear ratio for the gear pair
   c. The pitch diameter of the pinion and the gear
   d. The center distance between the shafts carrying the pinion and the gear
   e. The pitch line speed for both the pinion and the gear
   f. The torque on the pinion shaft and on the gear shaft
   g. The tangential force acting on the teeth of each gear
   h. The radial force acting on the teeth of each gear
   i. The normal force acting on the teeth of each gear
3. A pair of spur gears with 20°, full-depth, involute teeth transmits 0.75 hp. The pinion is mounted on the shaft of an electric motor operating at 3450 rpm. The pinion has 24 teeth and a diametral pitch of 24. The gear has 110 teeth. Compute the following:
   a. The rotational speed of the gear
   b. The velocity ratio and the gear ratio for the gear pair
   c. The pitch diameter of the pinion and the gear
   d. The center distance between the shafts carrying the pinion and the gear
   e. The pitch line speed for both the pinion and the gear
   f. The torque on the pinion shaft and on the gear shaft
   g. The tangential force acting on the teeth of each gear
   h. The radial force acting on the teeth of each gear
   i. The normal force acting on the teeth of each gear
4. For the data of Problem 1, repeat Parts (g), (h), and (i) if the teeth have 25° full depth instead of 20°.
5. For the data of Problem 2, repeat Parts (g), (h), and (i) if the teeth have 25° full depth instead of 20°.
6. For the data of Problem 3, repeat Parts (g), (h), and (i) if the teeth have 25° full depth instead of 20°.
7. Figure P9–7 shows a drive system in which a 20-hp electric motor drives three separate output shafts. Gear A is mounted on the motor shaft that has a rotational speed of 1750 rpm clockwise. Gear A drives a gear train consisting of gears B, C, and D that deliver power through the shafts on which they are mounted. All gears have a diametral pitch of $P_d = 8$. The following data are given for the gear system:

   Power delivered by gears B, C, and D: P_B = 8 hp, $P_C = 7$ hp, $P_D = 5$ hp

   Numbers of teeth for all gears: N_A = 24, N_B = 48, $N_C = 96$, $N_D = 24$

   Determine the following of the drive system:
   a. The pitch diameter of each gear.
   b. The center distance of each gear mesh.
   c. The rotational speed of each output shaft.
   d. The torque through each output shaft.
   e. The tangential force on the teeth of each gear.

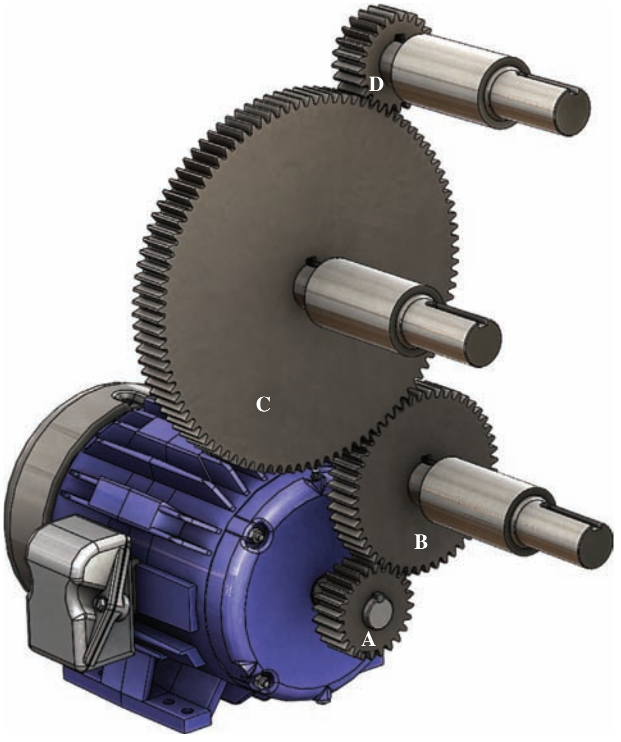


FIGURE P9–7 Gear drive with multiple output shafts

Gear Quality

8. Specify a suitable quality number for the gears in the drive for a grain harvester.
9. Specify a suitable quality number for the gears in the drive for a high-speed printing press.
10. Specify a suitable quality number for the gears in the drive for an automotive transmission.
11. Specify a suitable quality number for the gears in the drive for a gyroscope used in the guidance system for a spacecraft.
12. List five geometric factors measured by analytical gear quality measurement devices.
13. Identify the AGMA standard that is the basis for gear quality measurements and describe the range of quality numbers it includes. Compare that list with the two most recent predecessor standards that had been in use.
14. Specify a suitable quality number for the gears of Problem 1 if the drive is part of a precision machine tool.
15. Specify a suitable quality number for the gears of Problem 2 if the drive is part of a precision machine tool.
16. Specify a suitable quality number for the gears of Problem 3 if the drive is part of a precision machine tool.

Gear Materials

17. Identify the two major types of stresses that are created in gear teeth as they transmit power. Describe how the stresses are produced and where the maximum values of such stresses are expected to occur.
18. Describe the nature of the data contained in AGMA standards that relate to the ability of a given gear tooth to withstand the major types of stresses that it sees in operation.

19. Describe the general nature of steels that are typically used for gears, and list at least five examples of suitable alloys.
20. Describe the range of hardness that can typically be produced by through-hardening techniques and used successfully in steel gears.
21. Describe the general nature of the differences among steels produced as Grade 1, Grade 2, and Grade 3.
22. Suggest at least three applications in which Grade 2 or Grade 3 steel might be appropriate.
23. Describe three methods of producing gear teeth with strengths greater than can be achieved with through-hardening.
24. What AGMA standard should be consulted for data on the allowable stresses for steels used for gears?
25. In the AGMA standard identified in Problem 24, for what other materials besides steels are strength data provided?
26. Determine the allowable bending stress number and the allowable contact stress number for the following materials:
    - **a.** Through-hardened, Grade 1 steel with a hardness of 200 HB
    - **b.** Through-hardened, Grade 1 steel with a hardness of 300 HB
    - **c.** Through-hardened, Grade 1 steel with a hardness of 400 HB
    - **d.** Through-hardened, Grade 1 steel with a hardness of 450 HB
    - **e.** Through-hardened, Grade 2 steel with a hardness of 200 HB
    - **f.** Through-hardened, Grade 2 steel with a hardness of 300 HB
    - **g.** Through-hardened, Grade 2 steel with a hardness of 400 HB
27. If the design of a steel gear indicates that an allowable bending stress number of 36 000 psi is needed, specify a suitable hardness level for Grade 1 steel. What hardness level would be required for Grade 2 steel?
28. What level of hardness can be expected for gear teeth that are case-hardened by carburizing?
29. Name three typical steels that are used in carburizing.
30. What is the level of hardness that can be expected for gear teeth that are case-hardened by flame or induction hardening?
31. Name three typical steels that are used for flame or induction hardening. What is an important property of such steels?
32. State the minimum hardness level at the surface of gear teeth made from ASTM A536 ductile iron, Grade 80-55-06.
33. Determine the allowable bending stress number and the allowable contact stress number for the following materials:
    - **a.** Flame-hardened SAE 4140 steel, Grade 1, with a surface hardness of 50 HRC
    - **b.** Flame-hardened SAE 4140 steel, Grade 1, with a surface hardness of 54 HRC
    - **c.** Carburized and case-hardened SAE 4620 Grade 1 steel, DOQT 300
    - **d.** Carburized and case-hardened SAE 4620 Grade 2 steel, DOQT 300
    - **e.** Carburized and case-hardened SAE 1118 Grade 1 steel, SWQT 350
    - **f.** Gray cast iron, class 20
    - **g.** Gray cast iron, class 40
    - **h.** Ductile iron, 100-70-03
    - **i.** Sand-cast bronze with a minimum tensile strength of 40 ksi (275 MPa)
    - **j.** Heat-treated bronze with a minimum tensile strength of 90 ksi (620 MPa)
    - **k.** Glass-filled nylon
    - **l.** Glass-filled polycarbonate
34. What depth should be specified for the case for a carburized gear tooth having a diametral pitch of 6?
35. What depth should be specified for the case for a carburized gear tooth having a metric module of 6?

Bending Stresses in Gear Teeth

For Problems 36–41, compute the bending stress number, s_t, using Equation (9–16). Assume that the gear blank is solid unless otherwise stated. (*Note that the data in these problems are used in later problems through Problem 59. You are advised to keep solutions to earlier problems accessible so that you can use data and results in later problems. The four problems that are keyed to the same set of data require the analysis of bending stress and contact stress and the corresponding specification of suitable materials based on those stresses. Later design problems, 60–70, use the complete analysis within each problem.*)

36. A pair of gears with 20°, full-depth, involute teeth transmits 10.0 hp while the pinion rotates at 1750 rpm. The diametral pitch is 12, and the quality number is A11. The pinion has 18 teeth, and the gear has 85 teeth. The face width is 1.25 in. The input power is from an electric motor, and the drive is for an industrial conveyor. The drive is a commercial enclosed gear unit.
37. A pair of gears with 20°, full-depth, involute teeth transmits 40 hp while the pinion rotates at 1150 rpm. The diametral pitch is 6, and the quality number is A11. The pinion has 20 teeth, and the gear has 48 teeth. The face width is 2.25 in. The input power is from an electric motor, and the drive is for a cement kiln. The drive is a commercial enclosed gear unit.
38. A pair of gears with 20°, full-depth, involute teeth transmits 0.50 hp while the pinion rotates at 3450 rpm. The diametral pitch is 32, and the quality number is A7. The pinion has 24 teeth, and the gear has 120 teeth. The face width is 0.50 in. The input power is from an electric motor, and the drive is for a small machine tool. The drive is a precision enclosed gear unit.
39. A pair of gears with 25° full-depth, involute teeth transmits 15.0 hp while the pinion rotates at 6500 rpm. The diametral pitch is 10, and the quality number is A5. The pinion has 30 teeth, and the gear has 88 teeth. The face width is 1.50 in. The input power is from a universal electric motor, and the drive is for an actuator on an aircraft. The drive is an extra-precision, enclosed gear unit.
40. A pair of gears with 25°, full-depth, involute teeth transmits 125 hp while the pinion rotates at 2500 rpm. The diametral pitch is 4, and the quality number is A9. The pinion has 32 teeth, and the gear has 76 teeth. The face width is 1.50 in. The input power is from a gasoline engine, and the drive is for a portable industrial water pump. The drive is a commercial enclosed gear unit.

41. A pair of gears with 25°, full-depth, involute teeth transmits 2.50 hp while the pinion rotates at 680 rpm. The diametral pitch is 10, and the quality number is A11. The pinion has 24 teeth, and the gear has 62 teeth. The face width is 1.25 in. The input power is from a vane-type fluid motor, and the drive is for a small lawn and garden tractor. The drive is a commercial enclosed gear unit.

Required Allowable Bending Stress Number

For Problems 42–47, compute the required allowable bending stress number, s_{at}, using Equation (9–30). Assume that no unusual conditions exist unless stated otherwise. That is, use a service factor, *SF*, of 1.00. Then specify a suitable steel and its heat treatment for both the pinion and the gear based on bending stress.

42. Use the data and results from Problem 36. Design for a reliability of 0.99 and a design life of 20 000 h.
43. Use the data and results from Problem 37. Design for a reliability of 0.99 and a design life of 8000 h.
44. Use the data and results from Problem 38. Design for a reliability of 0.9999 and a design life of 12 000 h. Consider that the machine tool is a critical part of a production system calling for a service factor of 1.25 to avoid unexpected down time.
45. Use the data and results from Problem 39. Design for a reliability of 0.9999 and a design life of 4000 h.
46. Use the data and results from Problem 40. Design for a reliability of 0.99 and a design life of 8000 h.
47. Use the data and results from Problem 41. Design for a reliability of 0.90 and a design life of 2000 h. The uncertainty of the actual use of the tractor calls for a service factor of 1.25. Consider using cast iron or bronze if the conditions permit.

Pitting Resistance

For Problems 48–53, compute the expected contact stress number, s_c, using Equation (9–23). Assume that both gears are to be steel unless stated otherwise.

48. Use the data and results from Problems 36 and 42.
49. Use the data and results from Problems 37 and 43.
50. Use the data and results from Problems 38 and 44.
51. Use the data and results from Problems 39 and 45.
52. Use the data and results from Problems 40 and 46.
53. Use the data and results from Problems 41 and 47.

Required Allowable Contact Stress Number

For Problems 54–59, compute the required allowable contact stress number, s_{ac}, using Equation (9–31). Use a service factor, *SF*, of 1.00 unless stated otherwise. Then specify suitable material for the pinion and the gear based on pitting resistance. Use steel unless an earlier decision has been made to use another material. Then evaluate whether the earlier decision is still valid. If not, specify a different material according to the most severe requirement. If no suitable material can be found, consider redesigning the original gears to enable reasonable materials to be used.

54. Use the data and results from Problems 36, 42, and 48.
55. Use the data and results from Problems 37, 43, and 49.
56. Use the data and results from Problems 38, 44, and 50.
57. Use the data and results from Problems 39, 45, and 51.
58. Use the data and results from Problems 40, 46, and 52.
59. Use the data and results from Problems 41, 47, and 53.

Design Problems

Problems 60–70 describe design situations. For each, design a pair of spur gears, specifying (at least) the diametral pitch, the number of teeth in each gear, the pitch diameters of each gear, the center distance, the face width, and the material from which the gears are to be made. Design for recommended life with regard to both strength and pitting resistance. Work toward designs that are compact. Use standard values of diametral pitch, and avoid designs for which interference could occur. See Example Problem 9–7. Assume that the input to the gear pair is from an electric motor unless otherwise stated.

If the data are given in SI units, complete the design in the metric module system with dimensions in millimeters, forces in newtons, and stresses in megapascals. See Example Problem 9–8.

60. A pair of spur gears is to be designed to transmit 5.0 hp while the pinion rotates at 1200 rpm. The gear must rotate between 385 and 390 rpm. The gear drives a reciprocating compressor.
61. A gear pair is to be a part of the drive for a milling machine requiring 20.0 hp with the pinion speed at 550 rpm and the gear speed to be between 180 and 190 rpm.
62. A drive for a punch press requires 50.0 hp with the pinion speed of 900 rpm and the gear speed of 225 to 230 rpm.
63. A single-cylinder gasoline engine has the pinion of a gear pair on its output shaft. The gear is attached to the shaft of a small cement mixer. The mixer requires 2.5 hp while rotating at approximately 75 rpm. The engine is governed to run at approximately 900 rpm.
64. A four-cylinder industrial engine runs at 2200 rpm and delivers 75 hp to the input gear of a drive for a large wood chipper used to prepare pulpwood chips for paper making. The output gear must run between 4500 and 4600 rpm.
65. A small commercial tractor is being designed for chores such as lawn mowing and snow removal. The wheel drive system is to be through a gear pair in which the pinion runs at 600 rpm while the gear, mounted on the hub of the wheel, runs at 170 to 180 rpm. The wheel is 300 mm in diameter. The gasoline engine delivers 3.0 kW of power to the gear pair.
66. A water turbine transmits 75 kW of power to a pair of gears at 4500 rpm. The output of the gear pair must drive an electric power generator at 3600 rpm. The center distance for the gear pair must not exceed 150 mm.

67. A drive system for a large commercial band saw is to be designed to transmit 12.0 hp. The saw will be used to cut steel tubing for automotive exhaust pipes. The pinion rotates at 3450 rpm, while the gear must rotate between 725 and 735 rpm. It has been specified that the gears are to be made from SAE 4340 steel, oil quenched and tempered. Case hardening is *not* to be used.
68. Repeat Problem 67, but consider a case-hardened carburized steel from Appendix 5. Try to achieve the smallest practical design. Compare the result with the design from Problem 67.
69. A gear drive for a special-purpose, dedicated machine tool is being designed to mill a surface on a rough steel casting. The drive must transmit 20 hp with a pinion speed of 650 rpm and an output speed between 110 and 115 rpm. The mill is to be used continuously, two shifts per day, six days per week, for at least five years. Design the drive to be as small as practical to permit its being mounted close to the milling head.
70. A cable drum for a crane is to rotate between 160 and 166 rpm. Design a gear drive for 25 hp in which the input pinion rotates at 925 rpm and the output rotates with the drum. The crane is expected to operate with a 50% duty cycle for 120 hours per week for at least 10 years. The pinion and the gear of the drive must fit within the 24-in inside diameter of the drum, with the gear mounted on the drum shaft.

Power-Transmitting Capacity

71. Determine the power-transmitting capacity for a pair of spur gears having 20°, full-depth teeth, a diametral pitch of 10, a face width of 1.25 in, 25 teeth in the pinion, 60 teeth in the gear, and an AGMA quality class of A9. The pinion is made from SAE 4140 OQT 1000, and the gear is made from SAE 4140 OQT 1100. The pinion will rotate at 1725 rpm on the shaft of an electric motor. The gear will drive a centrifugal pump.
72. Determine the power-transmitting capacity for a pair of spur gears having 20°, full-depth teeth, a diametral pitch of 6, 35 teeth in the pinion, 100 teeth in the gear, a face width of 2.00 in, and an AGMA quality class of A11. A gasoline engine drives the pinion at 1500 rpm. The gear drives a conveyor for crushed rock in a quarry. The pinion is made from SAE 1040 WQT 800. The gear is made from gray cast iron, ASTM A48-83, class 30. Design for 15 000 hr life.
73. It was found that the gear pair described in Problem 72 wore out when driven by a 25-hp engine. Propose a redesign that would be expected to give 15 000 hr life under the conditions described.

Design of Double-Reduction Drives

74. Design a double-reduction gear train that will transmit 10.0 hp from an electric motor running at 1750 rpm to an assembly conveyor whose drive-shaft must rotate between 146 and 150 rpm. Note that this will require the design of two pairs of gears. Sketch the arrangement of the train, and compute the actual output speed.
75. A commercial food waste grinder in which the final shaft rotates at between 40 and 44 rpm is to be designed. The input is from an electric motor running at 850 rpm and delivering 0.50 hp. Design a double-reduction spur gear train for the grinder.
76. A small, powered hand drill is driven by an electric motor running at 3000 rpm. The drill speed is to be approximately 550 rpm. Design the speed reduction for the drill. The power transmitted is 0.25 hp.
77. The output from the drill described in Problem 76 provides the drive for a small bench-scale band saw used in a home shop. The saw blade is to move with a linear velocity of 375 ft/min. The saw blade rides on 9.0-in-diameter wheels. Design a spur gear reduction to drive the band saw. Consider using plastic gears.
78. Design a rack-and-pinion drive to lift a heavy access panel on a furnace. A fluid power motor rotating 1500 rpm will provide 5.0 hp at the input to the drive. The linear speed of the rack is to be at least 2.0 ft/s. The rack moves 6.0 ft each way during the opening and closing of the furnace doors. More than one stage of reduction may be used, but attempt to design with the fewest number of gears. The drive is expected to operate at least six times per hour for three shifts per day, seven days per week, for at least 15 years.
79. Design the gear drive for the wheels of an industrial lift truck. Its top speed is to be 20 mph. It has been decided that the wheels will have a diameter of 12.0 in. A DC motor supplies 20 hp at a speed of 3000 rpm. The design life is 16 hours per day, six days per week, for 20 years.

Plastics Gearing

80. Design a pair of plastic gears to drive a small band saw. The input is from a 0.50 hp electric motor rotating at 860 rpm, and the pinion will be mounted on its 0.75-inch diameter shaft with a keyway for a 0.1875×0.1875 in key. The gear is to rotate between 265 and 267 rpm.
81. Design a pair of plastic gears to drive a paper feed roll for an office printer. The pinion rotates at 88 rpm and the gear must rotate between 20 and 22 rpm. The power required is 0.06 hp. Work toward the smallest practical size.
82. Design a pair of plastic gears to drive the wheels of a small remote control car. The gear is mounted on the axle of the wheel and must rotate between 120 and 122 rpm. The pinion rotates at 430 rpm. The power required is 0.025 hp. Work toward the smallest practical size using unfilled nylon.
83. Design a pair of plastic gears to drive a commercial food-chopping machine. The input is from a 0.65 hp electric motor rotating at 1560 rpm, and the pinion will be mounted on its 0.875-inch diameter shaft with a keyway for a 0.1875×0.1875 in key. The gear is to rotate between 468 and 470 rpm.

CHAPTER TEN

HELICAL GEARS, BEVEL GEARS, AND WORMGEARING

THE BIG PICTURE — Helical Gears, Bevel Gears, and Wormgearing

Discussion Map

- The geometry of helical gears, bevel gears, and wormgearing was described in Chapter 8.
- The principles of stress analysis of gears were discussed in Chapter 9 for spur gears. Much of that information is applicable to the types of gears discussed in this chapter.

Discover

Review Chapters 8 and 9 now.

Recall some of the discussion at the beginning of Chapter 8 about uses for gears that you see in your world. Review that information now, and focus your discussion on helical gears, bevel gears, and wormgearing.

In this chapter, you acquire the skills to perform the necessary analyses to design safe gear drives that use helical gears, bevel gears, and wormgearing and that demonstrate long life.

Much was said in Chapter 8 about the geometry of spur gears, helical gears, bevel gears, and wormgearing and the kinematics of a single pair of gears and trains made from two or more pairs of gears. Also, parts of Chapters 8 and 9 discussed the types of metallic materials commonly used for power transmission gears, methods of manufacturing gears, principles of gear quality, and measurements required to ensure that quality. Then you learned how to analyze the bending stress in the fillet at the base of spur gear teeth and the contact stress along the face of the teeth near the pitch line. This was combined with determining the required bending strength of the gear material to avoid fatigue failure and the required hardness of the face of the gear to provide adequate resistance to surface pitting. The result led to a method of designing spur gear drives to achieve satisfactory performance and life. Section 9–14 applied gear design principles to plastics gearing.

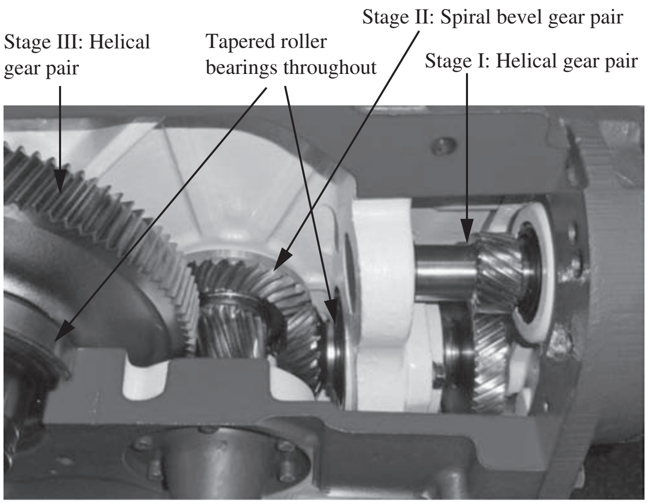
Similar analyses and design approaches are developed in this chapter for helical gears, bevel gears, and wormgearing. You will need to refer back to Chapters 5, 8, and 9 as we proceed. More detailed information can be found in the AGMA standards listed in References 1–14.

Figure 10–1 shows a combination of a gear-type speed reducer driven by a close-coupled electric motor; the combination is often called a *gearmotor.* Part (b) of the figure shows the internal arrangement of the three-stage speed reducer. At the right end, a helical pinion shaft is driven directly by the motor and it then drives its mating gear for the first stage of reduction. The same shaft with the output helical gear carries the bevel pinion for the second stage of reduction. The helical pinion for the third stage is placed close to the output bevel gear on the same shaft. The large output helical gear then delivers the power at the greatly reduced speed and at a correspondingly higher torque to the driven machine. The design of the output shaft employs a hollow shaft with taper-lock bushing that facilitates connection to the input shaft of the driven machine.

Figure 10–2 shows a double-reduction, helical gear reducer that receives power from the electric motor mounted above and through the belt drive that produces an initial speed reduction. Take note of the arrangement of the gears, shafts, and bearings in the housing shown to the left. Tapered roller bearings are used on all shafts to carry the combination of radial and axial forces that are inherently produced by helical gears.

Refer back to Figure 8–25 for an example of a wormgear reducer.

(*a*) Assembly of a gear reducer and a drive motor, called a gearmotor



(*b*) Cutaway showing the internal arrangement of the helical-bevel-helical three-stage reduction

FIGURE 10–1 Gearmotor assembly employing a three-stage gear reducer (Baldor/Dodge, Greenville, SC)

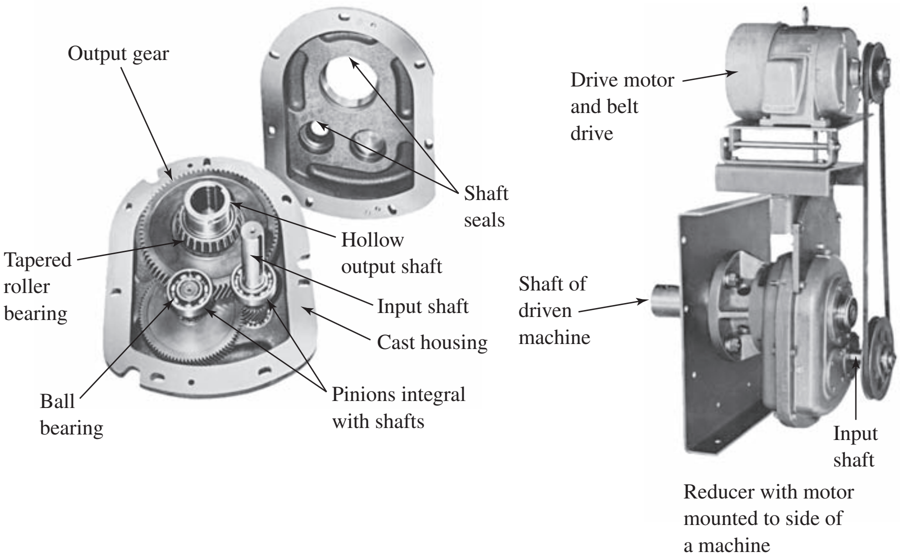


FIGURE 10–2 Helical shaft-mount reducer (Power Transmission Solutions, a business unit of Emerson Industrial Automation)

YOU ARE THE DESIGNER

The gear drives that were designed in Chapter 9 all assumed that spur gears would be used to accomplish the speed reduction or speed increase between the input and the output of the drive. But many other types of gears could have been used. Assume that you are the designer of the drive for the wood chipper described in Example Problem 9–7. How would the design be different if helical gears were used instead of spur gears? What forces would be created and transferred to the shafts carrying the gears and to the bearings carrying the shafts? Would you be able to use smaller gears? How is the geometry of helical gears different from that of spur gears?

Rather than having the input and output shafts parallel as they were in designs up to this time, how can we design drives that deliver power to an output shaft at right angles to the input shaft? What special analysis techniques are applied to bevel gears and wormgearing?

The information in this chapter will help you answer these and other questions. ■

10–1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Describe the geometry of helical gears and compute the dimensions of key features.
2. Compute the forces exerted by one helical gear on its mating gear.
3. Compute the stress due to bending in helical gear teeth and specify suitable materials to withstand such stresses.
4. Design helical gears for surface durability.
5. Describe the geometry of bevel gears and compute the dimensions of key features.
6. Analyze the forces exerted by one bevel gear on another and show how those forces are transferred to the shafts carrying the gears.
7. Design and analyze bevel gear teeth for strength and surface durability.
8. Describe the geometry of worms and wormgears.
9. Compute the forces created by a wormgear drive system and analyze their effect on the shafts carrying the worm and the wormgear.
10. Compute the efficiency of wormgear drives.
11. Design and analyze wormgear drives to be safe for bending strength and wear.

References at the end of the chapter are recommended for additional information for design and application of helical gears, bevel gears, and wormgearing.

10–2 FORCES ON HELICAL GEAR TEETH

Figure 10–3 shows a drawing of two helical gears in mesh and designed to be mounted on parallel shafts. This is the basic configuration that we analyze in this chapter. Refer to Figure 10–4 for a representation of the force system that acts between the teeth of two helical

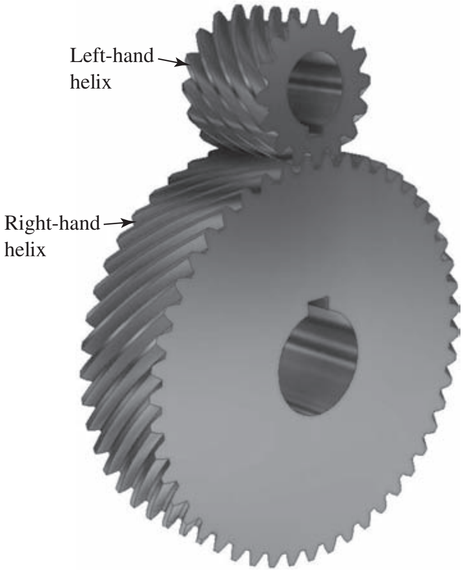
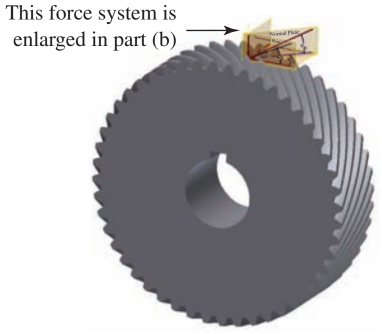


FIGURE 10–3 Helical gears in mesh. These gears have a 45° helix angle

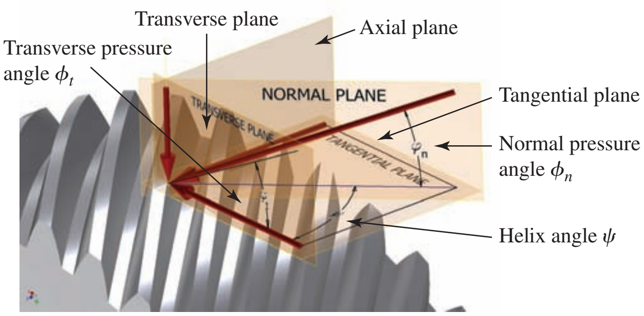
gears in mesh. Also, it is useful to review Section 8–7 in Chapter 8 on *Helical Gear Geometry*.

We now define the array of forces acting on helical gear teeth as shown in Figure 10–4.

- W_N is the *true normal force* that acts perpendicular to the face of the tooth in the plane normal to the surface of the tooth. We seldom need to use the value of W_N because its three orthogonal components, defined next, are used in the analyses performed for helical



(*a*) System of forces acting on a gear tooth



(*b*) Normal force and its three components

FIGURE 10–4 Helical gear geometry showing forces acting on gear teeth

gears. The values for the orthogonal components depend on the following three angles that help define the geometry of the helical gear teeth:

Normal pressure angle: ϕ_n
Transverse pressure angle: ϕ_t
Helix angle: ψ

For helical gears, the helix angle and one of the other two are specified. The third angle can be computed from

$$\tan \phi_n = \tan \phi_t \cos \psi \tag{10–1}$$

- W_t is the *tangential force* that acts in the transverse plane and tangent to the pitch circle of the helical gear and that causes the torque to be transmitted from the driver to the driven gear. Therefore, this force is often called the *transmitted force*. It is functionally similar to W_t used in the analysis of spur gears in Chapters 8 and 9. We can compute its value from the same equations, as follows:

If the torque being transmitted (T) and the size of the gear (D) are known,

➪ Tangential Force

$$W_t = T/(D/2) \tag{10–2}$$

If the power being transmitted (P) and the rotational speed (n) are known,

$$T = (P/n) \tag{10–3}$$

Power, torque, and forces for the U.S. unit system: Here we bring concepts and equations from Section 9–3 developed for spur gears and apply them for helical gears. *These are unit-specific equations.*

When power, P, is expressed in hp, rotational speed, n, is in rpm, and diameters, D, are in inches:

Torque: $$T = 63\,000(P)/n \text{ lb}\cdot\text{in} \tag{10–4}$$

Pitch line speed: $$v_t = \pi Dn/12 \text{ ft/min} \tag{10–5}$$

Tangential force: $$W_t = (126\,000)(P)/[(n)(D)] \text{ lb} \tag{10–6}$$

Or: $$W_t = (33\,000)(P)/v_t \text{ lb} \tag{10–7}$$

Power, torque, and forces for the SI metric unit system: Again, bringing concepts and equations from Sections 9–3 and 9–10 developed for spur gears, we apply them for helical gears. *These are unit-specific equations.*

When power, P, is expressed in kW, rotational speed, n, is in rpm, and diameters, D, are in mm:

Torque: $$T = 9549(P)/n \text{ N}\cdot\text{m} \tag{10–4M}$$

Pitch line speed: $$v_t = \pi Dn/60\,000 \text{ m/s} \tag{10–5M}$$

Tangential force: $$W_t = (19\,099)(P)/[(n)(D)] \text{ N} \tag{10–6M}$$

Or: $$W_t = (1000)(P)/v_t \text{ N} \tag{10–7M}$$

The value of the tangential load is the most fundamental of the three orthogonal components of the true normal force. The calculation of the bending stress number and the contact stress number of the gear teeth depends on W_t.

- W_r is the *radial force* that acts toward the center of the gear perpendicular to the pitch circle and to the tangential force. It tends to push the two gears apart. As can be seen in Figure 10–4(b)

➪ Radial Force

$$W_r = W_t \tan \phi_t \qquad (10\text{–}8)$$

- where ϕ_t = transverse pressure angle for the helical teeth
- W_x is the *axial force* that acts parallel to the axis of the gear and causes a thrust load that must be resisted by the bearings carrying the shaft. With the tangential force known, the axial force is computed from

➪ Axial Force

$$W_x = W_t \tan \psi \qquad (10\text{–}9)$$

Example Problem 10–1

A helical gear has a normal diametral pitch, P_{nd}, of 8, a normal pressure angle of 20°, 32 teeth, a face width of 3.00 in, and a helix angle of 15°. Compute the diametral pitch, the transverse pressure angle, and the pitch diameter. If the gear is rotating at 650 rpm while transmitting 7.50 hp, compute the pitch line speed, the tangential force, the axial force, and the radial force.

Solution

Diametral Pitch:

$$P_d = P_{nd} \cos \psi = 8 \cos (15°) = 7.727$$

Transverse Pressure Angle: [Equation (10–1)]

$$\phi_t = \tan^{-1}(\tan \phi_n / \cos \psi)$$

$$\phi_t = \tan^{-1}[\tan(20°)/\cos(15°)] = 20.65°$$

Pitch Diameter:

$$D = N/P_d = 32/7.727 = 4.141 \text{ in}$$

Pitch Line Speed, v_t: [Equation (10–5)]

$$v_t = \pi Dn/12 = \pi(4.141)(650)/12 = 704.7 \text{ ft/min}$$

Tangential Force, W_t: [Equation (10–7)]

$$W_t = 33\,000(P)/v_t = 33\,000(7.5)/704.7 = 351 \text{ lb}$$

Axial Force, W_x: [Equation (10–9)]

$$W_x = W_t \tan \psi = 351 \tan(15°) = 94 \text{ lb}$$

Radial Force, W_r: [Equation (10–8)]

$$W_r = W_t \tan \phi_t = 351 \tan(20.65°) = 132 \text{ lb}$$

The following example problem illustrates similar calculations for the SI metric system.

Example Problem 10–2

A helical gear has a normal module of 3 mm, a normal pressure angle of 25°, a helix angle of 22°, 32 teeth, and a face width of 75 mm. Compute the transverse module, the transverse pressure angle, and the pitch diameter. Then, if the gear is rotating at 650 rpm while transmitting 5.0 kW of power, compute the pitch line speed, the tangential force, the axial force, and the radial force.

Solution

Transverse Module:

$$m_n = m \cos \psi$$

$$m = m_n/\cos \psi = 3.00 \text{ mm}/\cos 22° = 3.236 \text{ mm}$$

Transverse Pressure Angle: [Equation (10–1)]

$$\phi_t = \tan^{-1}(\tan \phi_n/\cos \psi) = \tan^{-1}[\tan(25°)/\cos(22°)] = 23.38°$$

Pitch Diameter: $D = mN = (3.236\ \text{mm})(32) = 103.54\ \text{mm}$

Pitch Line Speed: [Equation (10–5M)]

$$v_t = \pi Dn/(60\,000) = \pi(103.54\ \text{mm})(650\ \text{rpm})/(60\,000) = 3.524\ \text{m/s}$$

Tangential Force: [Equation (10–6M)]

$$W_t = (1000)(P)/v_t = (1000)(5.0\ \text{kW})/(3.524\ \text{m/s}) = 1419\ \text{N}$$

Axial Force: [Equation (10–9)]

$$W_x = W_t \tan\psi = (1419\ \text{N})\tan(22°) = 573.3\ \text{N}$$

Radial Force: [Equation (10–8)]

$$W_r = W_t \tan\phi_t = (1419\ \text{N})\tan(23.38°) = 613.5\ \text{N}$$

10–3 STRESSES IN HELICAL GEAR TEETH

We will use the same basic equation for computing the bending stress number for helical gear teeth as we did for spur gear teeth in Chapter 9, given in Equation (9–16) and repeated here:

$$s_t = \frac{W_t P_d}{FJ} K_o K_s K_m K_B K_v$$

Figures 10–5 to 10–7 show the values for the geometry factor, J, for helical gear teeth with 15°, 20°, and 22° normal pressure angles, respectively.[1] The K factors are the same as those used for spur gears. See References 9 and 18 and the following locations for values:

K_o = overload factor (Table 9–1)

K_s = size factor (Table 9–2)

K_m = load-distribution factor [Figures 9–12 and 9–13 and Equation (9–17)]

K_B = rim thickness factor (Figure 9–14)

K_v = dynamic factor (Figure 9–16)

For design, a material must be specified that has an allowable bending stress number, s_{at}, greater than the computed bending stress number, s_t. Design values of s_{at} can be found in:

Figure 9–18: Steel, through-hardened, Grades 1 and 2

Table 9–9: Case-hardened steels

Table 9–10: Cast iron and bronze

(See also Reference 11 and 17–21.) The data for steel, iron, and bronze apply to a design life of 10^7 cycles at a reliability of 99% (fewer than one failure in 100). If other values for design life or reliability are desired, the allowable stress can be modified using the procedure described in Section 9–8.

10–4 PITTING RESISTANCE FOR HELICAL GEAR TEETH

Pitting resistance for helical gear teeth is evaluated using the same procedure as that discussed in Chapter 9 for spur gears. Equation (9–23) is repeated here:

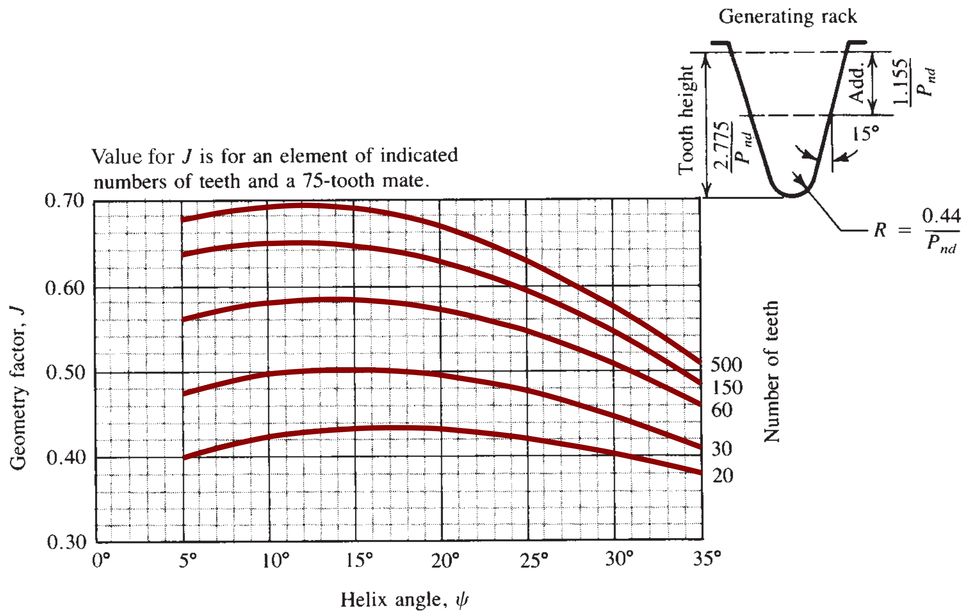
$$s_c = C_p \sqrt{\frac{W_t K_o K_s K_m K_v}{F D_p I}} \qquad (9\text{–}23)$$

All of the factors are the same for helical gears except the geometry factor for pitting resistance, I. The values for C_p are found in Table 9–7. Note that the other K factors have the same values as the K factors discussed and identified in Section 10–3.

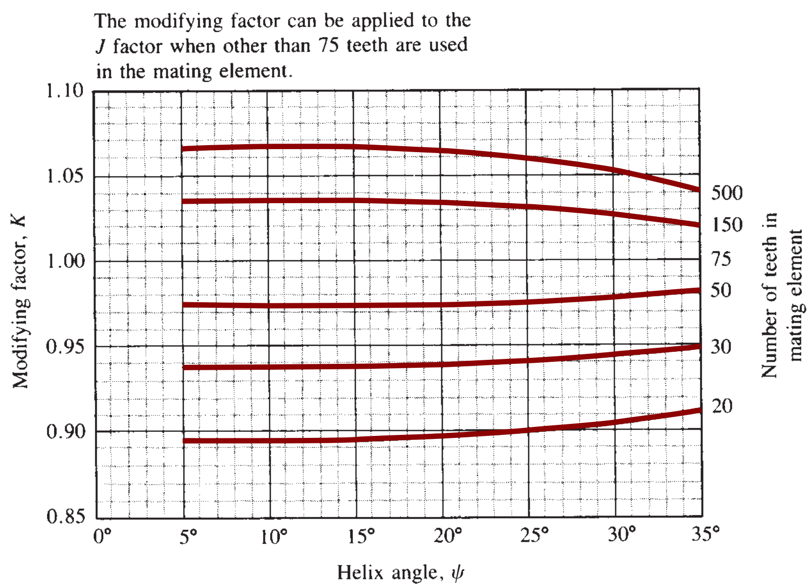
Because of the larger variety of geometric features needed to define the form of helical gears, it is not reasonable to reproduce all of the necessary tables of values or the complete formulas for computing I. Values change with the gear ratio, the number of teeth in the pinion, the tooth form, the helix angle, and the specific values for addendum, whole depth, and fillet radius. See References 6 and 13 for extensive discussions about the procedures. To facilitate problem solving in this book, Tables 10–1 and 10–2 give a few values for I.

For design, when the computed contact stress number is known, a material must be specified that has an

[1]Figures 10–5 to 10–7:
Graphs for the geometry factor, J, for helical gears have been taken from AGMA Standard 218.01-1982, *Standard for Rating the Pitting Resistance and Bending Strength of Spur and Helical Involute Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314. This standard has been superseded by two standards: (1) Standard 908-B89 (R 1999), *Geometry Factors for Determining the Pitting Resistance and Bending Strength of Spur, Helical, and Herringbone Gear Teeth*, 1999 and (2) Standard 2001-D04, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, 2004. The method of calculating the value for J has not been changed. However, the new standards do not contain the graphs. Users are cautioned to ensure that geometry factors for a given design conform to the specific cutter geometry used to manufacture the gears. Standards 908-B89 (R 1999) and 2001-D04 should be consulted for the details of computing the values for J and for rating the performance of the gear teeth.



(*a*) Geometry factor (J) for 15° normal pressure angle and indicated addendum



(*b*) J factor multipliers

FIGURE 10–5 Geometry factor (J) for 15° normal pressure angle

allowable contact stress number, s_{ac}, greater than s_c. Design values for s_{ac} can be found from the following:

Figure 9–19: Steel, through-hardened, Grades 1 and 2

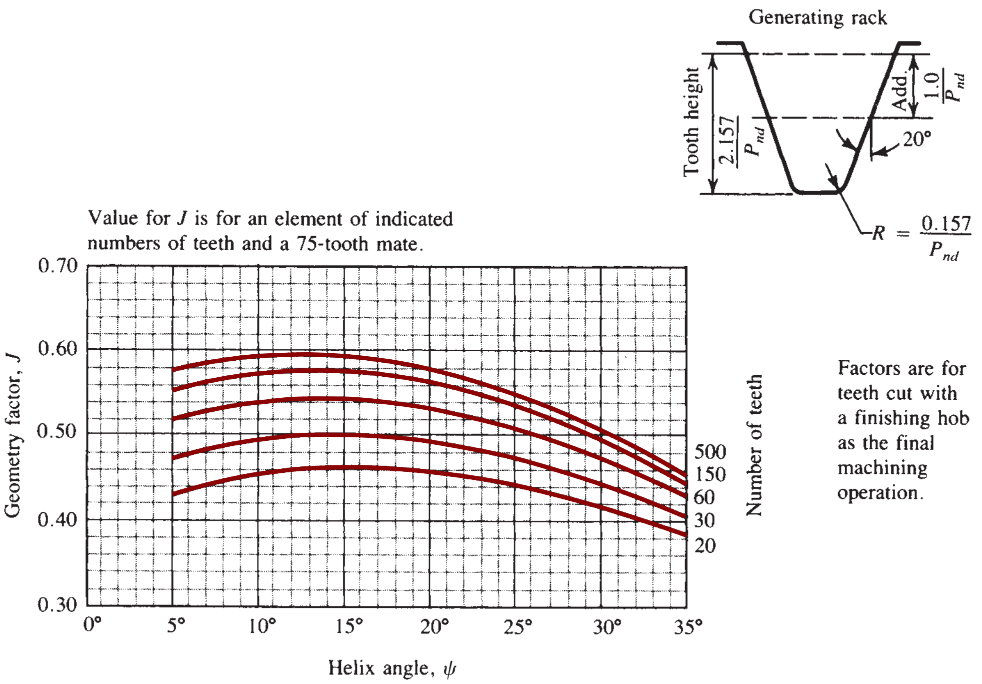
Table 9–9: Steel, case-hardened, Grade 1; flame- or induction-hardened, or carburized

Figure 9–10: Cast iron and bronze

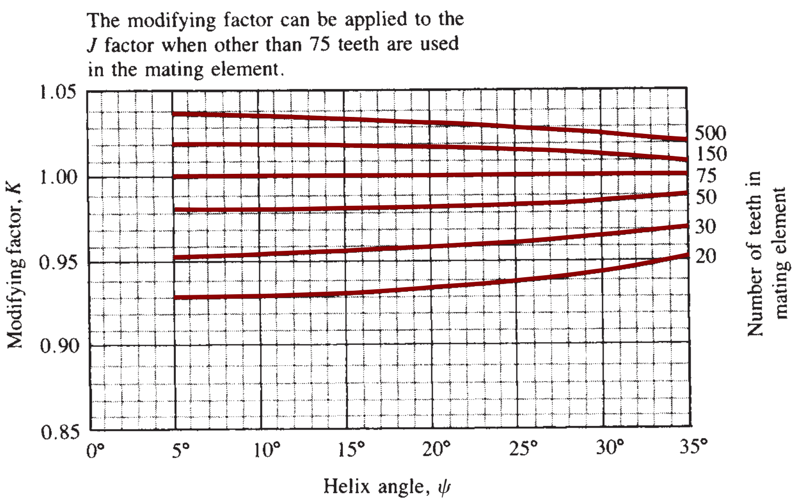
The data from these sources apply to a design life of 10^7 cycles at a reliability of 99% (fewer than one failure in 100). If other values for design life or reliability are desired, or if a service factor is to be applied, the allowable contact stress number can be modified using the procedure described in Section 9–8.

10–5 DESIGN OF HELICAL GEARS

The example problem that follows illustrates the procedure to design helical gears.



(a) Geometry factor (J) for 20° normal pressure angle, standard addendum, and finishing hob



(b) J factor multipliers

FIGURE 10–6 Geometry factor (J) for 20° normal pressure angle

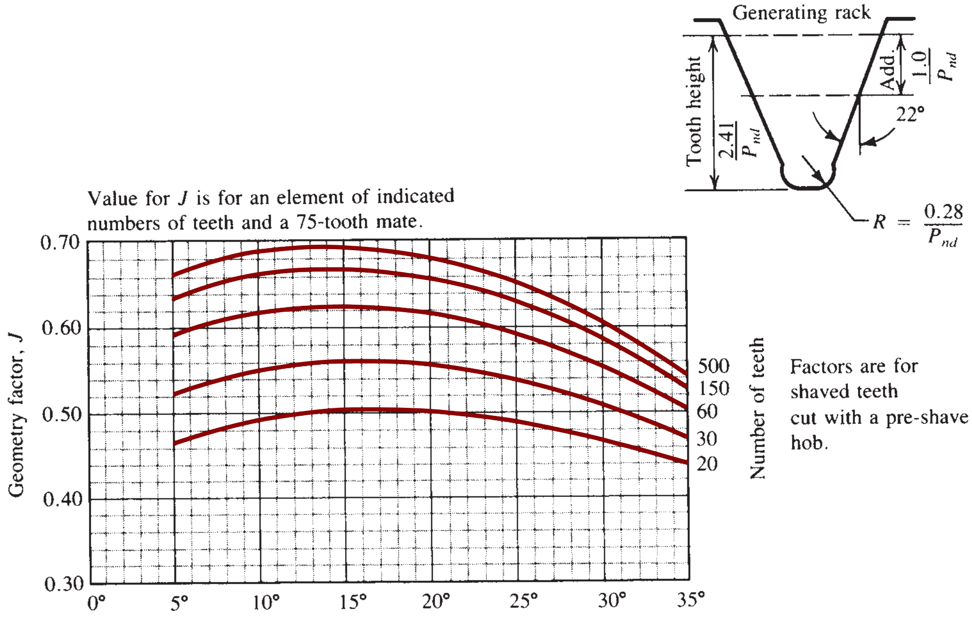
Example Problem 10–3 A pair of helical gears for a milling machine drive is to transmit 65 hp with a pinion speed of 3450 rpm and a gear speed of 1100 rpm. The power is from an electric motor. Design the gears.

Solution Of course, there are several possible solutions. Here is one. Let's try a normal diametral pitch of 12, 24 teeth in the pinion, a helix angle of 15°, a normal pressure angle of 20°, and a quality number of A9.

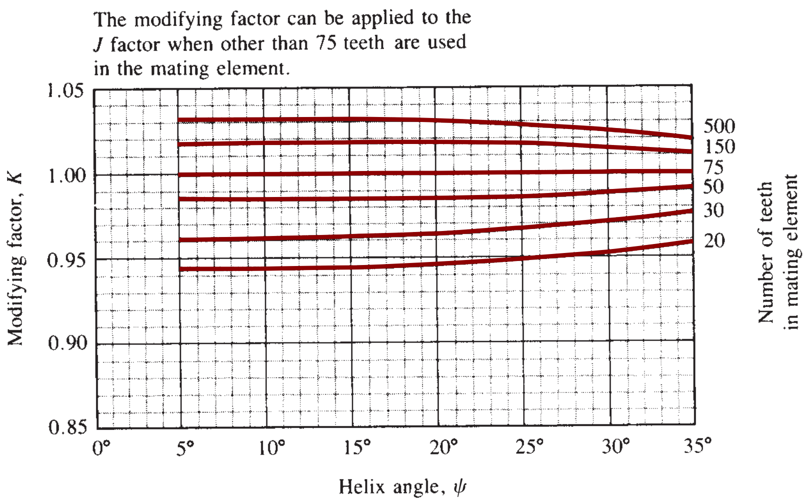
Now compute the transverse diametral pitch, the axial pitch, the transverse pressure angle, and the pitch diameter. Then we will choose a face width that will give at least two axial pitches to ensure true helical action.

$$P_d = P_{dn}\cos\psi = 12\cos(15°) = 11.59$$

$$P_x = \frac{\pi}{P_d \tan\psi} = \frac{\pi}{11.59\tan(15°)} = 1.012 \text{ in}$$



(a) Geometry factor (J), for 22° normal pressure angle, standard addendum, and pre-shave hob



(b) J factor multipliers

FIGURE 10–7 Geometry factor (J) for 22° normal pressure angle

$$\phi_t = \tan^{-1}(\tan \phi_n/\cos \psi) = \tan^{-1}[\tan(20°)/\cos(15°)] = 20.65°$$

$$D_p = N_p/P_d = 24/11.59 = 2.071 \text{ in}$$

$$F = 2P_x = 2(1.012) = 2.024 \text{ in} \quad \text{(nominal face width)}$$

Let's use 2.25 in, a more convenient value. The pitch line speed and the transmitted load are

$$v_t = \pi D_p\, n/12 = \pi(2.071)(3450)/12 = 1871 \text{ ft/min}$$

$$W_t = 33\,000(\text{hp})/v_t = (33\,000)(65)/1871 = 1146 \text{ lb}$$

Now we can calculate the number of teeth in the gear:

$$VR = N_G/N_P = n_P/n_G = 3450/1100 = 3.14$$

$$N_G = N_P\,(VR) = 24(3.14) = 75 \text{ teeth} \quad \text{(integer value)}$$

TABLE 10–1 Geometry Factors for Pitting Resistance, *I*, for Helical Gears with 20° Normal Pressure Angle and Standard Addendum

A. Helix angle $\psi = 15.0°$

Gear teeth	Pinion teeth 17	21	26	35	55
17	0.124				
21	0.139	0.128			
26	0.154	0.143	0.132		
35	0.175	0.165	0.154	0.137	
55	0.204	0.196	0.187	0.171	0.143
135	0.244	0.241	0.237	0.229	0.209

B. Helix angle $\psi = 25.0°$

Gear teeth	Pinion teeth 14	17	21	26	35	55
14	0.123					
17	0.137	0.126				
21	0.152	0.142	0.130			
26	0.167	0.157	0.146	0.134		
35	0.187	0.178	0.168	0.156	0.138	
55	0.213	0.207	0.199	0.189	0.173	0.144
135	0.248	0.247	0.244	0.239	0.230	0.210

Source: Extracted from AGMA Standard 908-B89 (R 1999), *Geometry Factors for Determining the Pitting Resistance and Bending Strength of Spur, Helical and Herringbone Gear Teeth,* with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314.

The values for the factors in Equation (9–6) must now be determined to enable the calculation of the bending stress. The geometry factor for the pinion is found in Figure 10–6 for 24 teeth in the pinion and 75 teeth in the gear: $J_P = 0.48$. The value of J_G will be greater than the value of J_P, resulting in a lower stress in the gear.

The K factors are as follows:

K_o = overload factor = 1.5 (moderate shock)
K_s = size factor = 1.0
K_m = load-distribution factor = 1.26 for $F/D_P = 1.09$ and commercial-quality, enclosed gearing
K_B = rim thickness factor = 1.0 (solid gears)
K_v = dynamic factor = 1.35 for $A_v = 9$ and $v_t = 1871$ ft/min

The bending stress in the pinion can now be computed:

$$s_{tP} = \frac{W_t P_d}{F J_P} K_o K_s K_m K_B K_v$$

$$s_{tP} = \frac{(1146)(11.59)}{(2.25)(0.48)}(1.50)(1.0)(1.26)(1.0)(1.35) = 31\,400 \text{ psi}$$

From Figure 9–18, a Grade 1 steel with a hardness of approximately 250 HB would be required. Let's proceed to the design for pitting resistance.

Use Equation (9–23):

$$s_c = C_p \sqrt{\frac{W_t K_o K_s K_m K_v}{F D_p I}}$$

TABLE 10–2 Geometry Factors for Pitting Resistance, *I*, for Helical Gears with 25° Normal Pressure Angle and Standard Addendum

A. Helix angle $\psi = 15.0°$

Gear teeth	Pinion teeth 14	17	21	26	35	55
14	0.130					
17	0.144	0.133				
21	0.160	0.149	0.137			
26	0.175	0.165	0.153	0.140		
35	0.195	0.186	0.175	0.163	0.143	
55	0.222	0.215	0.206	0.195	0.178	0.148
135	0.257	0.255	0.251	0.246	0.236	0.214

B. Helix angle $\psi = 25.0°$

Gear teeth	Pinion teeth 12	14	17	21	26	35	55
12	0.129						
14	0.141	0.132					
17	0.155	0.146	0.135				
21	0.170	0.162	0.151	0.138			
26	0.185	0.177	0.166	0.154	0.141		
35	0.203	0.197	0.188	0.176	0.163	0.144	
55	0.227	0.223	0.216	0.207	0.196	0.178	0.148
135	0.259	0.258	0.255	0.251	0.246	0.235	0.213

Source: Extracted from AGMA Standard 908-B89, *Geometry Factors for Determining the Pitting Resistance and Bending Strength of Spur, Helical and Herringbone Gear Teeth,* with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314.

For two steel gears, $C_p = 2300$. Rough interpolation from the data in Table 10–1 for $N_P = 24$ and $N_G = 75$ gives $I = 0.202$. It is recommended that the computational procedure described in the AGMA standards be used to compute a more precise value for critical work. The contact stress is then

$$s_c = 2300\sqrt{\frac{(1146)(1.50)(1.0)(1.26)(1.35)}{(2.25)(2.071)(0.202)}} = 128\,200 \text{ psi}$$

It is obvious that the contact stress governs this design. Let's adjust the solution for a higher reliability and to account for the expected number of cycles of operation. Certain design decisions must be made. For example, consider the following:

Design for a reliability of 0.999 (less than one failure in 1000): $K_R = 1.25$ (Table 9–11). Design life: Let's design for 10 000 h of life as suggested in Table 9–12 for multipurpose gearing. Then, using Equation (9–27), we can compute the number of cycles of loading. For the pinion rotating at 3450 rpm with one cycle of loading per revolution,

$$N_c = (60)(L)(n)(q) = (60)(10\,000)(3450)(1.0) = 2.1 \times 10^9 \text{ cycles}$$

From Figure 9–22, we find that $Z_N = 0.89$.

No unusual conditions seem to exist in this application beyond those already considered in the various *K* factors. Therefore, we use a service factor, *SF,* of 1.00.

We can use Equation (9–27) to apply these factors.

$$\frac{K_R(SF)}{Z_N} s_c = s_{ac} = \frac{(1.25)(1.00)}{(0.89)}(128\,200 \text{ psi}) = 180\,000 \text{ psi}$$

Table 9–9 indicates that Grade 1 steel, case hardened by carburizing, would be suitable. From Appendix 5, let's specify AISI 4320 SOQT 450, having a case hardness of HRC 59 and a core hardness of 415 HB. This should be satisfactory for both bending and pitting resistance. Both the pinion and the gear should be of this material.

10–6 FORCES ON STRAIGHT BEVEL GEARS

Review Section 8–8 and Figure 8–17 for the geometry of bevel gears. Also see References 1, 10, and 12.

Because of the conical shape of bevel gears and because of the involute-tooth form, a three-component set of forces acts on bevel gear teeth. Using notation similar to that for helical gears, we will compute the tangential force, W_t; the radial force, W_r; and the axial force, W_x. It is assumed that the three forces act concurrently at the midface of the teeth and on the pitch cone (see Figure 10–8). Although the actual point of application of the resultant force is a little displaced from the middle, no serious error results.

The tangential force acts tangential to the pitch cone and is the force that produces the torque on the pinion and the gear. The torque can be computed from the known power transmitted and the rotational speed:

$$T = 63\,000\ P/n$$

Then, using the pinion, for example, the transmitted load is

$$W_{tP} = T/r_m \tag{10–10}$$

where r_m = mean radius of the pinion.
The value of r_m can be computed from

$$r_m = d/2 - (F/2)\sin\gamma \tag{10–11}$$

Remember that the pitch diameter, d, is measured to the pitch line of the tooth at its large end. The angle, γ, is the pitch cone angle for the pinion as shown in Figure 10–8(a). The radial load acts toward the center of the pinion, perpendicular to its axis, causing bending of the pinion shaft. Thus,

$$W_{rP} = W_t \tan\phi \cos\gamma \tag{10–12}$$

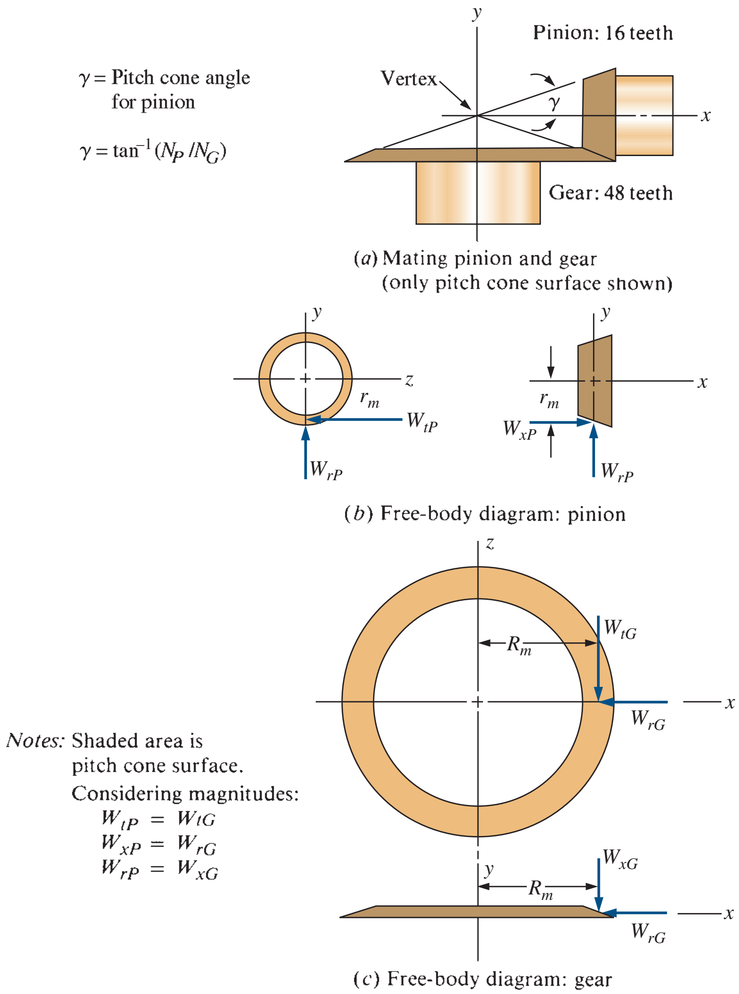


FIGURE 10–8 Forces on bevel gears

The angle, ϕ, is the pressure angle for the teeth.

The axial load acts parallel to the axis of the pinion, tending to push it away from the mating gear. It causes a thrust load on the shaft bearings. It also produces a bending moment on the shaft because it acts at the distance from the axis equal to the mean radius of the gear. Thus,

$$W_{xP} = W_t \tan\phi \sin\gamma \qquad (10\text{–}13)$$

The values for the forces on the gear can be calculated by the same equations shown here for the pinion, if the geometry for the gear is substituted for that of the pinion. Refer to Figure 10–8 for the relationships between the forces on the pinion and the gear in both magnitude and direction.

Example Problem 10–4

For the gear pair described in Example Problem 8–3, calculate the forces on the pinion and the gear if they are transmitting 2.50 hp with a pinion speed of 600 rpm. The geometry factors computed in Example Problem 8–3 apply. The data are summarized here.

Summary of Pertinent Results from Example Problem 8–3 and Given Data

Number of teeth in the pinion: $N_P = 16$

Number of teeth in the gear: $N_G = 48$

Diametral pitch: $P_d = 8$

Pitch diameter of pinion: $d = 2.000$ in

Pressure angle: $\phi = 20°$

Pinion pitch cone angle: $\gamma = 18.43°$

Gear pitch cone angle: $\Gamma = 71.57°$

Face width: $F = 1.00$ in

Rotational speed of pinion: $n_P = 600$ rpm

Power transmitted: $P = 2.50$ hp

Solution

Forces on the pinion are described by the following equations:

$$W_t = T/r_m$$

But

$$T_p = 63\,000(P)/n_P = [63\,000(2.50)]/600 = 263 \text{ lb}\cdot\text{in}$$

$$r_m = d/2 - (F/2)\sin\gamma$$

$$r_m = (2.000/2) - (1.00/2)\sin(18.43°) = 0.84 \text{ in}$$

Then

$$W_t = T_P/r_m = 263 \text{ lb}\cdot\text{in}/0.84 \text{ in} = 313 \text{ lb}$$

$$W_r = W_t \tan\phi\cos\gamma = 313 \text{ lb}\tan(20°)\cos(18.43°) = 108 \text{ lb}$$

$$W_x = W_t \tan\phi\sin\gamma = 313 \text{ lb}\tan(20°)\sin(18.43°) = 36 \text{ lb}$$

To determine the forces on the gear, first let's calculate the rotational speed of the gear:

$$n_G = n_P(N_P/N_G) = 600 \text{ rpm}(16/48) = 200 \text{ rpm}$$

Then

$$T_G = 63\,000(2.50)/200 = 788 \text{ lb}\cdot\text{in}$$

$$R_m = D/2 - (F/2)\sin\Gamma$$

$$R_m = 6.000/2 - (1.00/2)\sin(71.57°) = 2.53 \text{ in}$$

$$W_t = T_G/R_m = (788 \text{ lb}\cdot\text{in})/(2.53 \text{ in}) = 313 \text{ lb}$$

$$W_r = W_t \tan\phi\cos\Gamma = 313 \text{ lb}\tan(20°)\cos(71.57°) = 36 \text{ lb}$$

$$W_x = W_t \tan\phi\sin\Gamma = 313 \text{ lb}\tan(20°)\sin(71.57°) = 108 \text{ lb}$$

Note from Figure 10–8 that the forces on the pinion and the gear form an *action-reaction pair.* That is, the forces on the gear are equal to those on the pinion, but they act in the opposite direction. Also, because of the 90° orientation of the shafts, the radial force on the pinion becomes the axial thrust load on the gear, and the axial thrust load on the pinion becomes the radial load on the gear.

10–7 BEARING FORCES ON SHAFTS CARRYING BEVEL GEARS

Because of the three-dimensional force system that acts on bevel gears, the calculation of the forces on shaft bearings can be cumbersome. An example is worked out here to show the procedure. In order to obtain numerical data, the arrangement shown in Figure 10–9 is proposed for the bevel gear pair that was the subject of Example Problems 8–3 and 10–4. The locations for the bearings are given with respect to the vertex of the two pitch cones where the shaft axes intersect.

Note that both the pinion and the gear are *straddle mounted*; that is, each gear is positioned between the supporting bearings. This is the most preferred arrangement because it usually provides the greatest rigidity and maintains the alignment of the teeth during power transmission. Care should be exercised to provide rigid mountings and stiff shafts when using bevel gears.

The arrangement of Figure 10–9 is designed so that the bearing on the right resists the axial thrust load on the pinion, and the lower bearing resists the axial thrust load on the gear.

Example Problem 10–5 Compute the reaction forces on the bearings that support the shafts carrying the bevel gear pair shown in Figure 10–9. The values of Example Problems 8–3 and 10–4 apply.

Solution Referring to the results of Example Problem 10–4 and Figure 10–8, we have listed the forces acting on the gears:

Force	Pinion	Gear
Tangential	$W_{tP} = 313$ lb	$W_{tG} = 313$ lb
Radial	$W_{rP} = 108$ lb	$W_{rG} = 36$ lb
Axial	$W_{xP} = 36$ lb	$W_{xG} = 108$ lb

It is critical to be able to visualize the directions in which these forces are acting because of the three-dimensional force system. Notice in Figure 10–8 that a rectangular coordinate system has been set up. Figure 10–10 is an isometric sketch of the free-body diagrams of the pinion and the gear, simplified to represent the concurrent forces acting at the pinion/gear interface and at the bearing locations. Although the two free-body diagrams are separated for clarity, notice that you can bring them together by moving the point called *vertex* on each sketch together. This is the point in the actual gear system where the vertices of the two pitch cones lie at the same point. The two pitch points also coincide.

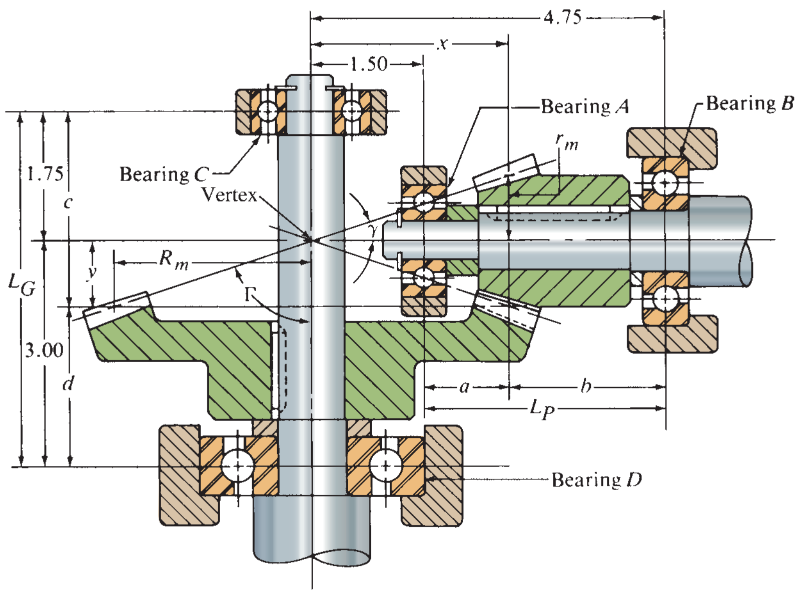
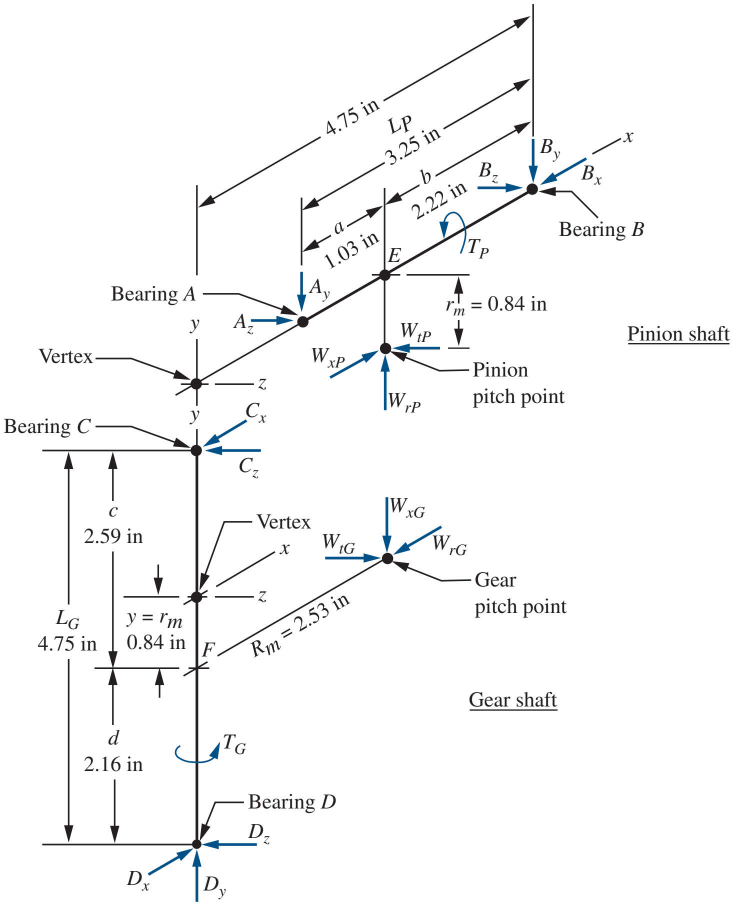


FIGURE 10–9 Layout of bevel gear pair for Example Problem 10–5

FIGURE 10–10 Free-body diagrams for pinion and gear shafts



For setting up the equations of static equilibrium needed to solve for the bearing reactions, the distances a, b, c, d, L_P, and L_G are needed, as shown in Figure 10–9. These require the two dimensions labeled x and y. Note from Example Problem 10–4 that

$$x = R_m = 2.53 \text{ in}$$

$$y = r_m = 0.84 \text{ in}$$

Then

$$a = x - 1.50 = 2.53 - 1.50 = 1.03 \text{ in}$$

$$b = 4.75 - x = 4.75 - 2.53 = 2.22 \text{ in}$$

$$c = 1.75 + y = 1.75 + 0.84 = 2.59 \text{ in}$$

$$d = 3.00 - y = 3.00 - 0.84 = 2.16 \text{ in}$$

$$L_P = 4.75 - 1.50 = 3.25 \text{ in}$$

$$L_G = 1.75 + 3.00 = 4.75 \text{ in}$$

These values are shown in Figure 10–10.

To solve for the reactions, we need to consider the horizontal (x-z) and the vertical (x-y) planes separately. It may help you to look also at Figure 10–11, which breaks out the forces on the pinion shaft in these two planes. Then we can analyze each plane using the fundamental equations of equilibrium.

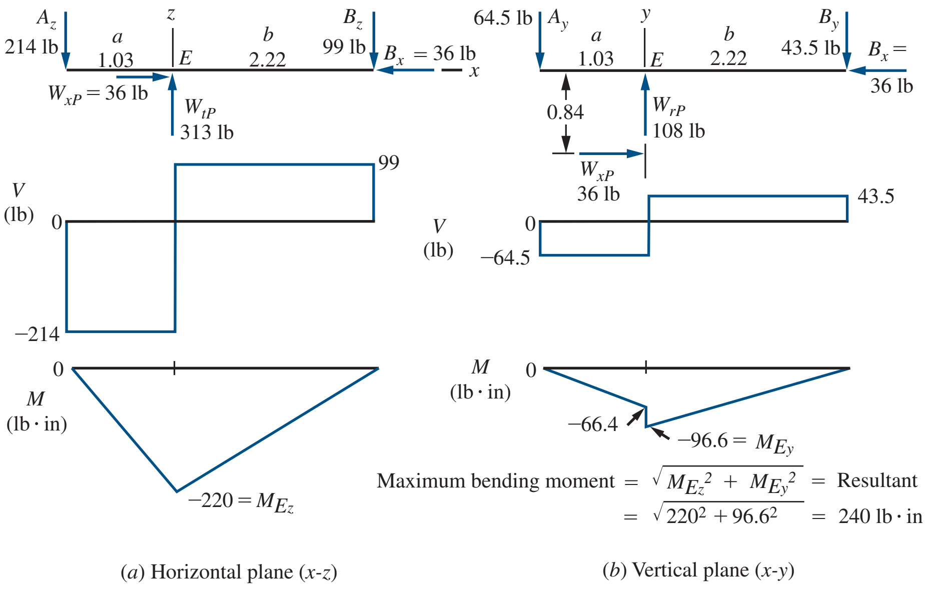
Bearing Reactions, Pinion Shaft: Bearings A and B

Step 1. To find B_z and A_z: In the x-z plane, only W_{tP} acts. Summing moments about A yields

$$0 = W_{tP}(a) - B_z(L_P) = 313(1.03) - B_z(3.25)$$

$$B_z = 99.2 \text{ lb}$$

FIGURE 10–11 Pinion shaft bending moments



Summing moments about B yields

$$0 = W_{tP}(b) - A_z(L_P) = 313(2.22) - A_z(3.25)$$

$$A_z = 214 \text{ lb}$$

Step 2. To find B_y and A_y: In the x-y plane, both W_{rP} and W_{xP} act. Summing moments about A yields

$$0 = w_{rP}(a) + W_{xP}(r_m) - B_y(L_P)$$

$$0 = 108(1.03) + 36(0.84) - B_y(3.25)$$

$$B_y = 43.5 \text{ lb}$$

Summing moments about B yields

$$0 = W_{rP}(b) + W_{xP}(r_m) - A_y(L_P)$$

$$0 = 108(2.22) - 36(0.84) - A_y(3.25)$$

$$A_y = 64.5 \text{ lb}$$

Step 3. To find B_x: Summing forces in the x-direction yields

$$B_x = W_{xP} = 36 \text{ lb}$$

This is the thrust force on bearing B.

Step 4. To find the total radial force on each bearing: Compute the resultant of the y- and z-components.

$$A = \sqrt{A_y^2 + A_z^2} = \sqrt{64.5^2 + 214^2} = 224 \text{ lb}$$

$$B = \sqrt{B_y^2 + B_z^2} = \sqrt{43.5^2 + 99.2^2} = 108 \text{ lb}$$

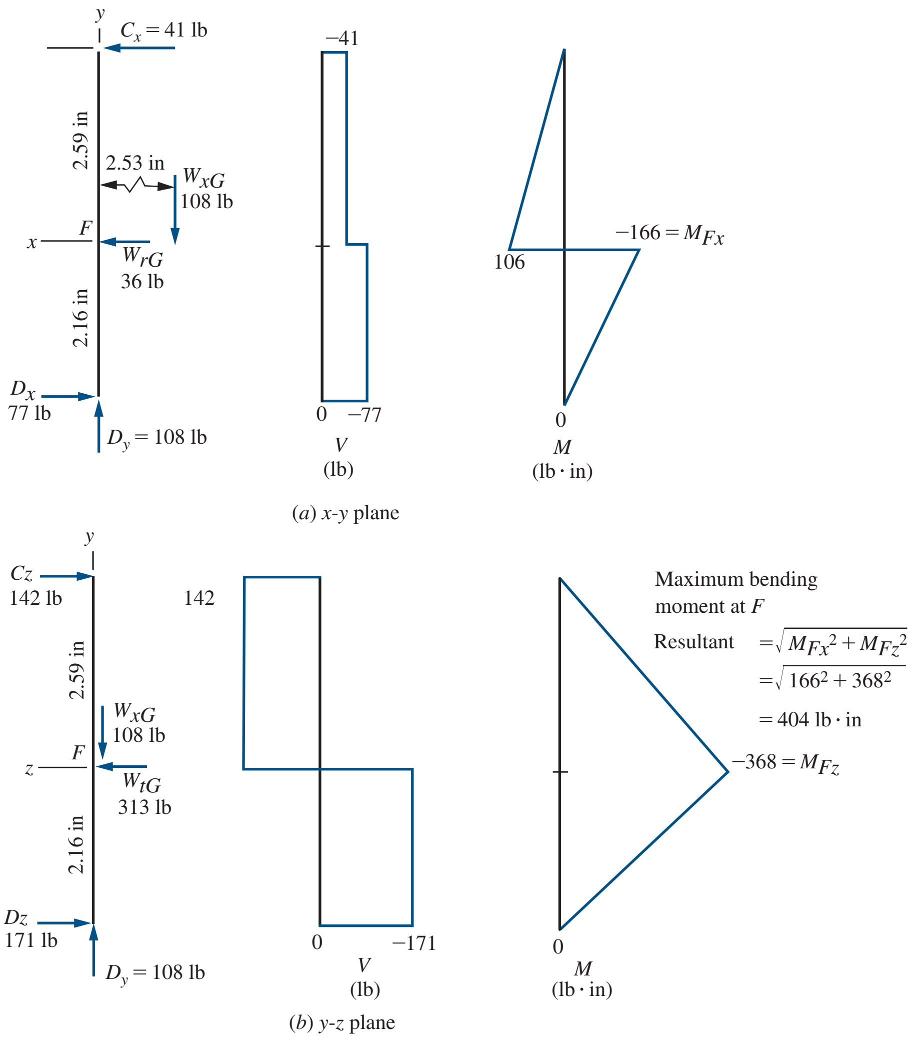
Bearing Reactions, Gear Shaft: Bearings C and D
Using similar methods, we can find the forces in Figure 10–12.

$$\left.\begin{array}{l} C_z = 142 \text{ lb} \\ C_x = 41.1 \text{ lb} \end{array}\right\} C = 148 \text{ lb (radial force on } C)$$

$$\left.\begin{array}{l} D_z = 171 \text{ lb} \\ D_x = 77.1 \text{ lb} \end{array}\right\} D = 188 \text{ lb (radial force on } D)$$

$$D_y = W_{xG} = 108 \text{ lb (thrust force on } D)$$

FIGURE 10–12 Gear shaft bending moments



Summary In selection of the bearings for these shafts, the following capacities are required:

Bearing *A:* 224-lb radial

Bearing *B:* 108-lb radial; 36-lb thrust

Bearing *C:* 148-lb radial

Bearing *D:* 188-lb radial; 108-lb thrust

10–8 BENDING MOMENTS ON SHAFTS CARRYING BEVEL GEARS

Because there are forces acting in two planes on bevel gears, as discussed in the preceding section, there is also bending in two planes. The analysis of the shearing force and bending moment diagrams for the shafts must take this into account.

Figure 10–11 and Figure 10–12 show the resulting diagrams for the pinion and the gear shafts, respectively, for the gear pair used for Example Problems 8–3, 10–4, and 10–5. Notice that the axial thrust load on each gear provides a concentrated moment to the shaft equal to the axial load times the distance that it is offset from the axis of the shaft. Also notice that the maximum bending moment for each shaft is the resultant of the moments in the two planes. On the pinion shaft, the maximum moment is 240 lb·in at *E,* where the lines of action for the radial and tangential forces intersect the shaft. Similarly, on the gear shaft, the maximum moment is 404 lb·in at *F.* These data are used in the shaft design (as discussed in Chapter 12).

10–9 STRESSES IN STRAIGHT BEVEL GEAR TEETH

This section generally follows AGMA Standard 2003 (Reference 10), *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth*, considered to be the primary standard in the United States. However, only straight bevel gears are treated here. The standard presents design analysis in both the U.S. unit system based on diametral pitch, P_d, and the SI Metric unit system based on metric module, m.

This book will maintain similar notations and symbols for gear tooth features, allowable stresses, and modifying factors that were initially presented in Chapter 9 for spur gear design to facilitate the comparison of design approaches. The reader should note that Standard 2003-C10 presents design analysis in SI units using terminology from ISO standards that employ radically different symbol sets. In this book, we maintain similar notations for factors in both systems except for the basic terms, diametral pitch, P_d, and metric module, m.

Furthermore, as introduced in Chapter 9, this book makes the following assumptions:

1. The gears are operating at temperatures between 32°F and 250°F (0°C and 120°C) for which the temperature factor $K_T = 1.0$ and, therefore, it is not written in the equations for stress analysis.
2. Gear teeth are made to normal standards without special modifications. This does, however, assume that crowning of the teeth is included in the manufacturing process, typical of the bevel gear industry.
3. Both the pinion and the gear of the bevel gear set are straddle mounted as shown in Figure 10–9, providing the most rigid arrangement. If either gear is not straddle mounted, the overhung arrangement, being generally less stiff, requires the application of additional modifying factors.
4. The hardness of both the pinion and the gear are nearly equal to enable each to withstand the contact stress without pitting, allowing the use of the hardness factor, $C_H = 1.0$, and not including that factor in the equations. The standard contains significant discussion and data to modify allowable strengths when the pinion is significantly harder than the gear, an approach used by some designers to promote *wearing in* of the gear teeth by the harder pinion. Also included in the standard with the hardness factor are adjustments based on the surface roughness of the teeth that are not included in this book.
5. When case hardening is used, as is frequently done, the case has sufficient depth and the core hardness and strength are sufficiently high to avoid case crushing or subsurface failure of the core due to either bending or contact stresses. The standard includes much discussion of these factors.
6. Residual stresses in gear teeth are not considered in this section. Note that beneficial compressive residual stresses in the root area from peening processes can significantly enhance the life of gears. Conversely, residual tensile stresses can be detrimental.

Following is the discussion of bending stress and contact stress calculations along with the corresponding parameters and modification factors:

Pitch Diameter, D: An important difference in the analysis of bevel gears is the definition of pitch diameter; it is measured at the large (outer) end of the gear rather than at the middle of the teeth as it was for spur and helical gears. This difference is accommodated in the determination of the geometry factors J and I that are shown later.

Pitches: The pitch for bevel gears is defined at the outer pitch diameter and is called the *outer transverse pitch*. Calculation is the same as for spur and helical gears with the exception of the definition of pitch diameter given above.

U.S. Units: Outer transverse diametral pitch = $P_d = N/D$ (units are in^{-1}; rarely reported)

SI Metric Units: Outer transverse metric module = $m = D/N$ mm

Tangential Force, W_t: As in Chapter 9, we will use the following unit-specific equations for torque on a gear, pitch line speed, and the resulting tangential force.

U.S. Units: Power, P, in hp; rotational speed, n, in rpm; diameters, D, in inches

Pitch line speed $= v_t = \pi Dn/12$ ft/min

Torque $= T = 63\,000\ P/n$ lb·in

Tangential force $= W_t = T/(D/2) = 126\,000\ P/(Dn)$ lb

Or, Tangential force $= W_t = 33\,000\ P/v_t$ lb

SI Metric Units: Power, P, in kW; rotational speed, n, in rpm; diameters, D, in mm

Pitch line speed $= v_t = \pi Dn/(60\,000)$ m/s

Torque $= T = 9550\ P/n$ N·m

Tangential force $= W_t = T/(D/2)$
$= 19.1 \times 10^6\ P/(Dn)$ N $= 19\,100\ P/(Dn)$ kN

Or, Tangential force $= W_t = 1000\ P/v_t$ N $= P/v_t$ kN

Bending Stress Number, s_t: The maximum bending stress number occurs in the root area of the teeth as it does in spur and helical gears; the equations are as follows:

U.S. Units:
$$s_t = \frac{W_t P_d K_o K_s K_m K_v}{FJ}\ \text{psi} \qquad \textbf{(10–14)}$$

SI Metric Units:
$$s_t = \frac{W_t K_o K_s K_m K_v}{FJm}\ \text{MPa} \qquad \textbf{(10–14M)}$$

Overload Factor, K_o: Use the same values given in Table 9–1.

Size Factor, K_s for Bending Strength: Use Figure 10–13, adapted from AGMA Standard 2003-C10.

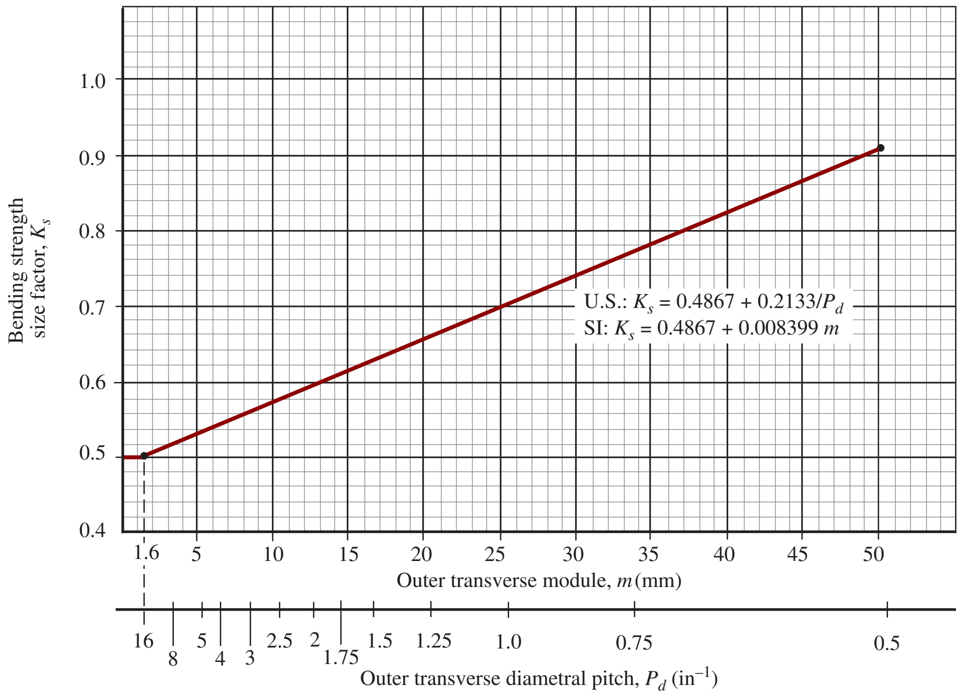


FIGURE 10–13 Size factor for bending stress, K_s, for bevel gears (Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth,* with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

For $P_d \geq 16\,(m \leq 1.6\text{ mm})$ use $K_s = 0.50$. The equation for the sloped portion of Figure 10–13 is

$$K_s = 0.4867 + 0.2133/P_d \qquad \textbf{(10–15)}$$

$$K_s = 0.4867 + 0.008399\, m \qquad \textbf{(10–15M)}$$

Load Distribution Factor, K_m: Typical bevel gear teeth are crowned along the profile and from end to end to ensure smooth engagement under foreseeable conditions of tooth accuracy and deflection under load. During the engagement/disengagement cycle, the contact pattern across the teeth should spread the load over the entire area of the faces. When additional misalignment occurs, this pattern is changed and the effects are more prominent with larger face widths and the manner of mounting. Use Figure 10–14 when discrete analysis of deformations is not practical. The three curves conform to the equations:

U.S. Units: $K_m = K_{mb} + 0.0036\ F^2$ **(10–16)**

SI Metric Units: $K_m = K_{mb} + 5.6 \times 10^{-6}\ F^2$ **(10–16M)**

where,

$K_{mb} = 1.00$ for both gears straddle mounted

$K_{mb} = 1.10$ for one gear straddle mounted

$K_{mb} = 1.25$ for neither gear straddle mounted

Problems in this book can assume that both gears are straddle mounted unless otherwise noted. Designers should take steps to provide a precise and rigid system comprised of the gears, shafts, bearings, and housing.

Dynamic Factor, K_v: Use the same chart shown for spur gears in Figure 9–16, based on the quality system defined in AGMA standards 2015 (Reference 5) and 2001 (Reference 9), in which quality numbers from A11 (least accurate) to A4 (most accurate) are used. Bevel gear standard AGMA 2003 (Reference 10) includes the former chart for K_v based on the Q-system of quality from Q5 (least accurate) to Q11 (most accurate). To a reasonable degree of precision for values of K_v in these two charts are compatible provided that the quality number in the A-system plus the quality number in the Q-system add to 17. For example, Q8 is similar to A9 and Q10 is similar to A7.

Geometry Factor for Bending Strength, J: Use Figure 10–15 for problems in this book. This figure is for straight-tooth bevel gears with a 20° pressure angle and a shaft angle between the shafts of the pinion and the gear of 90°. Standard AGMA 2003 (Reference 10) gives formulas from which the value for other designs can be computed when detailed geometry for the teeth is known.

Example Problem 10–6 illustrates the calculation for bending stress number for straight bevel gear teeth, using the same data from earlier problems. Then we will introduce allowable stresses and material selection to resist bending stresses.

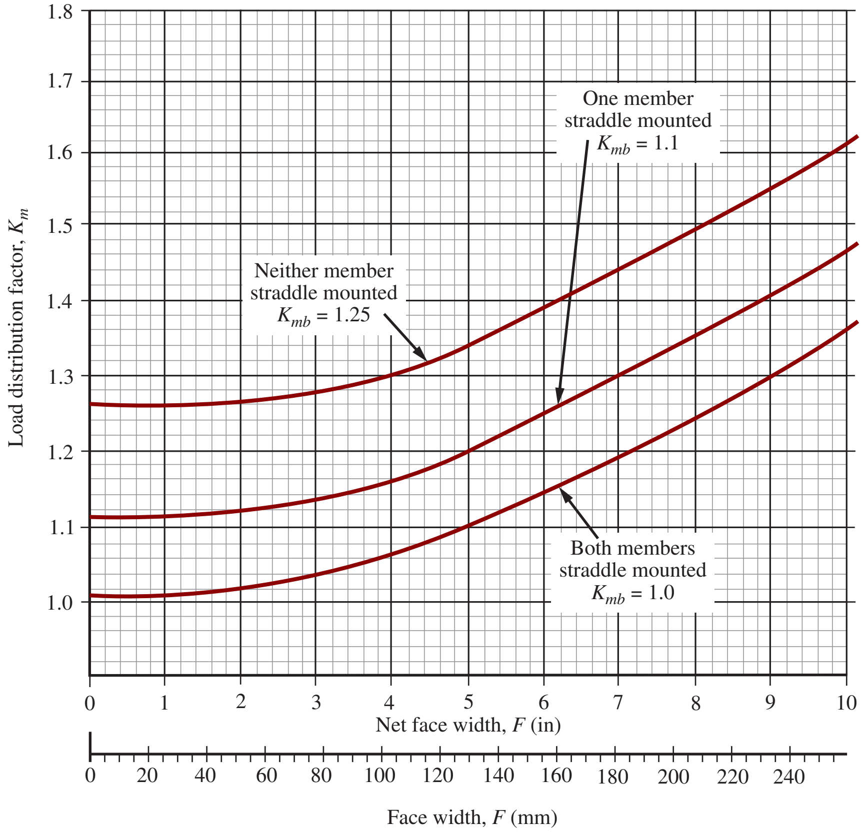


FIGURE 10–14 Load distribution factor, K_m, for crowned bevel gear teeth (Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth,* with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

Example Problem 10–6 Compute the bending stress in the teeth of the bevel pinion shown in Figure 10–9. The data from Example Problem 10–4 apply: $N_P = 16$; $N_G = 48$; $n_P = 600$ rpm; $P = 2.50$ hp; $P_d = 8$; $d = 2.000$ in; and $F = 1.00$ in. Assume that the pinion is driven by an electric motor and that the load provides moderate shock. The quality number, A_v, is to be 11.

Solution

$$W_t = \frac{T}{r} = \frac{63\,000(P)}{n_p}\,\frac{1}{d/2} = \frac{63\,000(2.50)}{600}\,\frac{1}{2.000/2} = 263 \text{ lb}$$

$$v_t = \pi d n_P/12 = \pi(2.000)(600)/12 = 314 \text{ ft/min}$$

$$K_o = 1.50 \text{ (from Table 9–1)}$$

$$K_s = 0.4867 + 0.2133/P_d = 0.4867 + 0.2133/8 = 0.513$$

$$K_m = 1.004 \text{ (both gears straddle mounted, general commercial quality)}$$

$$J_P = 0.230 \text{ (from Figure 10–5)}$$

$$K_v = 1.24 \text{ (Use } A_v = 11 \text{ and } v_t = 314 \text{ ft/min}$$

(Read from Figure 9–16 or computed from equations in Table 9–6)

Then, from Equation (10–14),

$$s_t = \frac{W_t P_d K_o K_s K_m K_v}{FJ} = \frac{(263)(8)(1.50)(0.5131)(1.004)(1.24)}{(1.00)(0.230)} = 8764 \text{ psi}$$

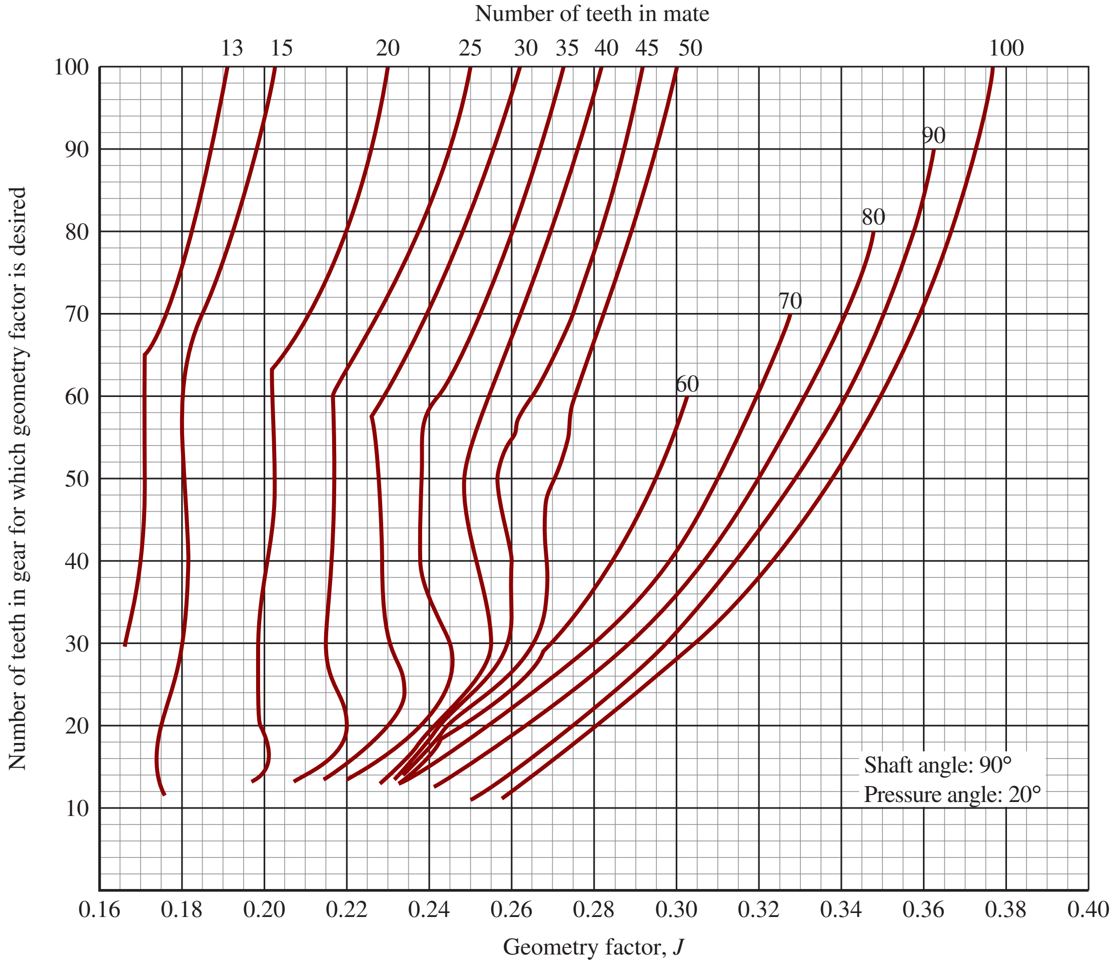


FIGURE 10–15 Geometry factor, J, for straight bevel gears with 20° pressure angle and 90° shaft angle (Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth,* with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

Allowable Bending Strength Number: The process of finding the required allowable bending strength, s_{at}, for bevel gears is similar to that used for spur and helical gears. The fundamental equation is

$$s_{wt} = \frac{s_{at}K_L}{(SF)(K_R)} \tag{10–17}$$

where

s_{wt} = permissible bending stress number considering design life and reliability

K_L = Stress cycle factor for bending
For carburized case-hardened steel bevel gears, use Figure 10–16. Most of the test data for these curves were developed for carburized case-hardened gears so use for through-hardened steel is only approximate. In Chapter 9, a similar factor was called Y_N and that is the term used in ISO standards as well. Note that the curve for life factors above about 3×10^6 is used for most commercial gear drives. The standard permits a lower value for some critical applications.

K_R = Reliability factor. Use Table 10–3. Note that for bending, the reliability factor for bevel gears is the same as for spur and helical gears in Table 9–11.

SF = Safety factor. Typically taken to be 1.00, but values up to about 1.50 can be used for greater uncertainty or for critical systems.

For design where the goal is to specify a suitable gear material, Equation (10–17) can be combined with the equation for s_t from Equation (10–14) and we can solve for the required value of the allowable bending strength, s_{at}. That is,

$$\text{Let } s_t = s_{wt} = \frac{s_{at}K_L}{(SF)(K_R)}$$

Then, the required value of s_{at} is

$$s_{at} = \frac{s_t(SF)(K_R)}{K_L} \tag{10–18}$$

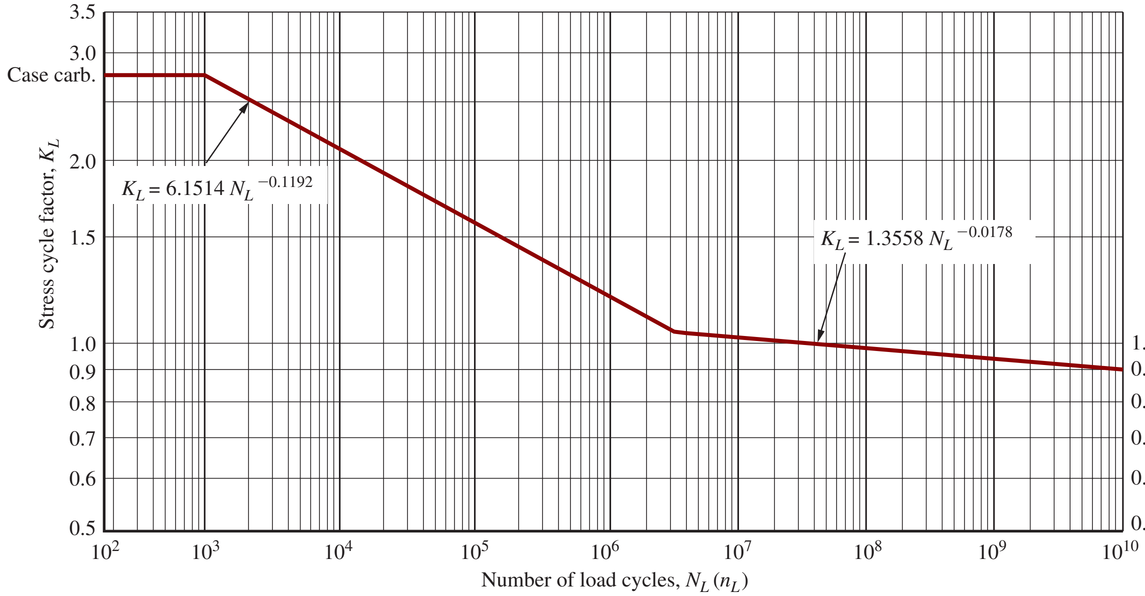


FIGURE 10–16 Stress cycle factor for bending strength, K_L (carburized case-hardened steel bevel gears) (Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth,* with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

TABLE 10–3 Reliability Factors for Allowable Bending and Contact Stresses

Reliability R	Interpretation	Reliability factors: Bending K_R	Reliability factors: Contact C_R
0.9	Fewer than one failure in 10	0.85	0.92
0.99	Fewer than one failure in 100	1.00	1.00
0.999	Fewer than one failure in 1000	1.25	1.12
0.9999	Fewer than one failure in 10 000	1.50	1.22

Source: Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth,* with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.

Use Figure 10–17 to determine the required hardness for through-hardened steel, recognizing that HB 400 is the maximum recommended value. Above that, flame, induction, or carburized case-hardened steel should be used up to the limits shown in Table 10–4.

Next, we develop the similar approach for determining the required allowable contact stress number, s_{ac}. Then we will show another example problem in which we identify the critical value on which material selection is based.

Contact Stress Number, s_c: The maximum contact stress number occurs on the face of the teeth as it does in spur and helical gears; the equation is the same for both U.S. and SI units with due attention to units for W_t, F, D_P, *and* C_p:

U.S. or SI Units:

$$s_c = C_p\sqrt{\frac{W_tK_oK_mK_vC_sC_{xc}}{FD_pI}}\ \text{psi or MPa} \qquad \textbf{(10–19)}$$

TABLE 10–4 Allowable Stress Numbers for Bevel Gears

Case hardened Grade 1 steel materials				
Hardness at surface	Allowable bending stress number, s_{at} (ksi)	(Mpa)	Allowable contact stress number, s_{ac} (ksi)	(Mpa)
Flame or induction hardened				
50 HRC— Unhardened roots	12.5	86	175	1207
50 HRC— hardened roots	22.5	155	175	1207
Carburized and case hardened				
55–64 HRC	30	207	200	1379

Source: Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, ZEROL Bevel and Spiral Bevel Gear Teeth,* with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.

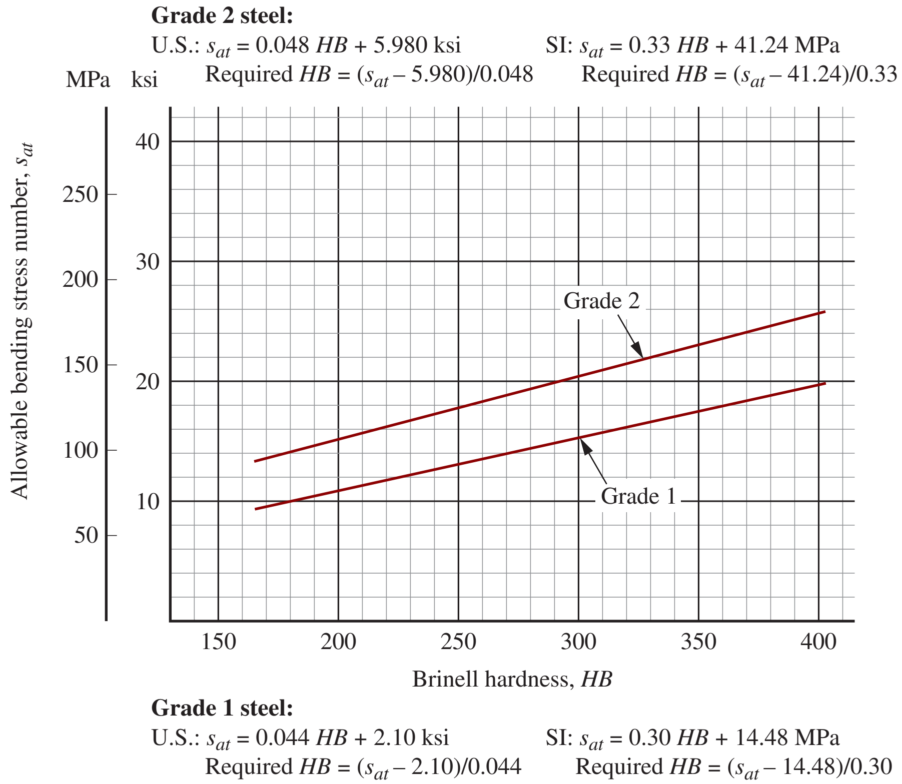


FIGURE 10–17 Allowable bending stress number, s_{at}, for through-hardened steel for bevel gears (Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth,* with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

Note that this equation is used for both the pinion and the gear because the basic contact stress is equal on each. *Do not use the pitch diameter of the gear in this equation.*

All variables in this equation have been discussed in regard to bending stress number except for I, C_p, C_s, *and* C_{xc}, discussed next.

C_p: The elastic coefficient depends on the modulus of elasticity and Poisson's ratio for the materials of the pinion and the gear. For two steel gears, $C_p = 2300\ \text{psi}^{0.5}(191\ \text{MPa}^{0.5})$. Use the values from Chapter 9 in Table 9–7 for other materials.

C_s: The *size factor for contact stress* is different from the value of K_s used for bending stress. See Figure 10–18. The factor is based on the face width, F. For $F \leq 0.50$ in (12.5 mm), use $C_s = 0.50$. For $F \geq 3.14$ in (80 mm), use $C_s = 0.83$. Between those limits, use the figure or compute the value from the equations given here:

$$\text{U.S. Units:} \quad C_s = 0.125F + 0.4375 \qquad \textbf{(10–20)}$$

$$\text{SI Metric Units:} \quad C_s = 0.00492F + 0.4375 \qquad \textbf{(10–20M)}$$

C_{xc}: The *crowning factor for pitting* accounts for the contact pattern between mating teeth. Typical bevel gear production processes create crowning both along the tooth flank and across the full width of the face. This is the preferred approach. However, some bevel gears are not crowned. AGMA Standard 2003 (Reference 10) recommends the following factors:

$$C_{xc} = 1.5 \text{ for properly crowned teeth}$$

$$C_{xc} = 2.0 \text{ or larger for non-crowned teeth}$$

I: The *geometry factor for pitting resistance* incorporates the radii of curvature of the pinion and gear teeth and the degree of load sharing between teeth. It is a function of the number of teeth in both the pinion and the gear and, therefore, to the gear ratio. Use Figure 10–19 for values for problem solving in this book. AGMA Standard 2003 (Reference 10) provides a formula for I and a detailed procedure for acquiring the necessary data and performing the calculations.

We now revisit the example design problem used throughout this chapter and compute the contact stress.

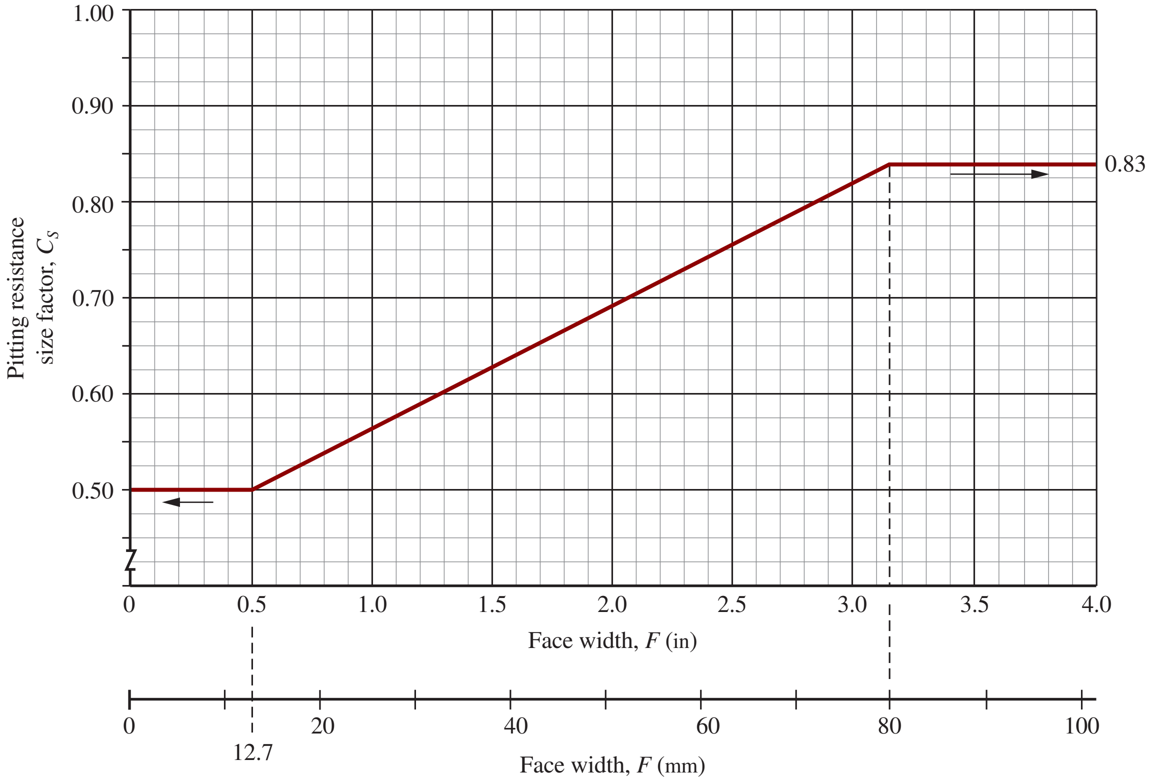


FIGURE 10–18 Pitting resistance size factor, C_s (Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth,* with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

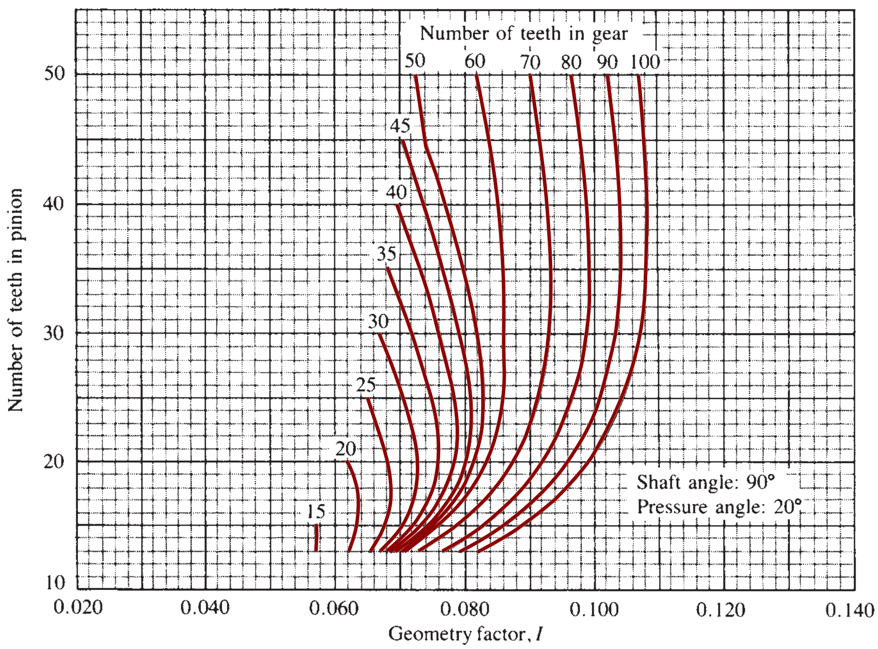


FIGURE 10–19 Geometry factors for straight and ZEROL® bevel gears (Extracted from AGMA 2003 (Reference 10), *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, ZEROL® Bevel and Spiral Bevel Gear Teeth,* with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314.)

Example Problem 10–7 Compute the contact stress for the gear pair in Figure 10–9 for the conditions used in Example Problem 10–5: $N_P = 16$; $N_G = 48$; $n_P = 600$ rpm; $P_d = 8$; $F = 1.00$ in; and $D_P = 2.000$ in. Both gears are to be steel.

Solution We use Equation (10–19) with U.S. units.

$$s_c = C_p\sqrt{\frac{W_t K_o K_m K_v C_s C_{xc}}{F D_p I}}\ \text{psi}$$

From Example Problem 10–5: $W_t = 263$ lb; $K_o = 1.50$; $K_m = 1.004$; and $K_v = 1.24$. Other factors are as follows:

Elastic coefficient $C_p = 2300\ \text{psi}^{0.5}$ for two steel gears

Size factor $C_s = 0.56$ (Figure 10–18)

Crowning factor $C_{xc} = 1.5$ (Specify properly crowned teeth)

Geometry factor for pitting resistance $I = 0.077$ (Figure 10–19)

Then

$$s_c = 2300\sqrt{\frac{(263)(1.50)(1.004)(1.24)(0.56)(1.5)}{(1.00)(2.000)(0.077)}}\ \text{psi} = 119\,044\ \text{psi}$$

Allowable Contact Stress Number: The process of finding the required allowable contact stress, s_{ac}, for bevel gears is similar to that used for spur and helical gears. The fundamental equation is

$$s_{wc} = \frac{s_{ac} C_L}{(SF)(C_R)} \tag{10–21}$$

where

s_{wc} = permissible contact stress number considering design life and reliability.

C_L = Stress cycle factor for contact stress. Use Figure 10–20, developed for carburized case-hardened steel bevel gears. Application to through-hardened gears is approximate.

C_R = Reliability factor. Use Table 10–3. Note that the reliability factor for pitting resistance is equal to the square root of that for bending. The AGMA standard adjusts the allowable bending strengths accordingly.

SF = Safety factor. Typically taken to be 1.00, but values up to about 1.50 can be used for greater uncertainty or for critical systems.

For design where the goal is to specify a suitable gear material, Equation (10–21) can be combined with the equation for s_c from Equation (10–19) and we can solve for the required value of the allowable bending strength, s_{ac}. That is,

$$\text{Let } s_c = s_{wc} = \frac{s_{ac} C_L}{(SF)(C_R)}$$

Then, the required value of s_{ac} is

$$s_{ac} = \frac{s_c (SF)(C_R)}{C_L} \tag{10–22}$$

Use Figure 10–21 to determine the required hardness for through-hardened steel, recognizing that HB 400 is the maximum recommended value. Above that, flame, induction, or carburized case-hardened steel should be used up to the limits shown in Table 10–4.

Now we look again at the design analysis example considered before and evaluate both the required value of s_{at} and s_{ac}. Finally, we identify the critical value and specify a suitable material for the two gears.

Example Problem 10–8 Specify suitable materials for the bevel pinion and gear for the data of Example Problems 10–5 to 10–7. Design for a life of 15 000 hours.

Solution From Example Problems 10–6 and 10–7, we find the following data:

Rotational speed of pinion $= n_P = 600$ rpm

Bending stress number, $s_t = 8764$ psi

Contact stress number, $s_c = 119\,044$ psi

We apply Equations (10–18) and (10–22).

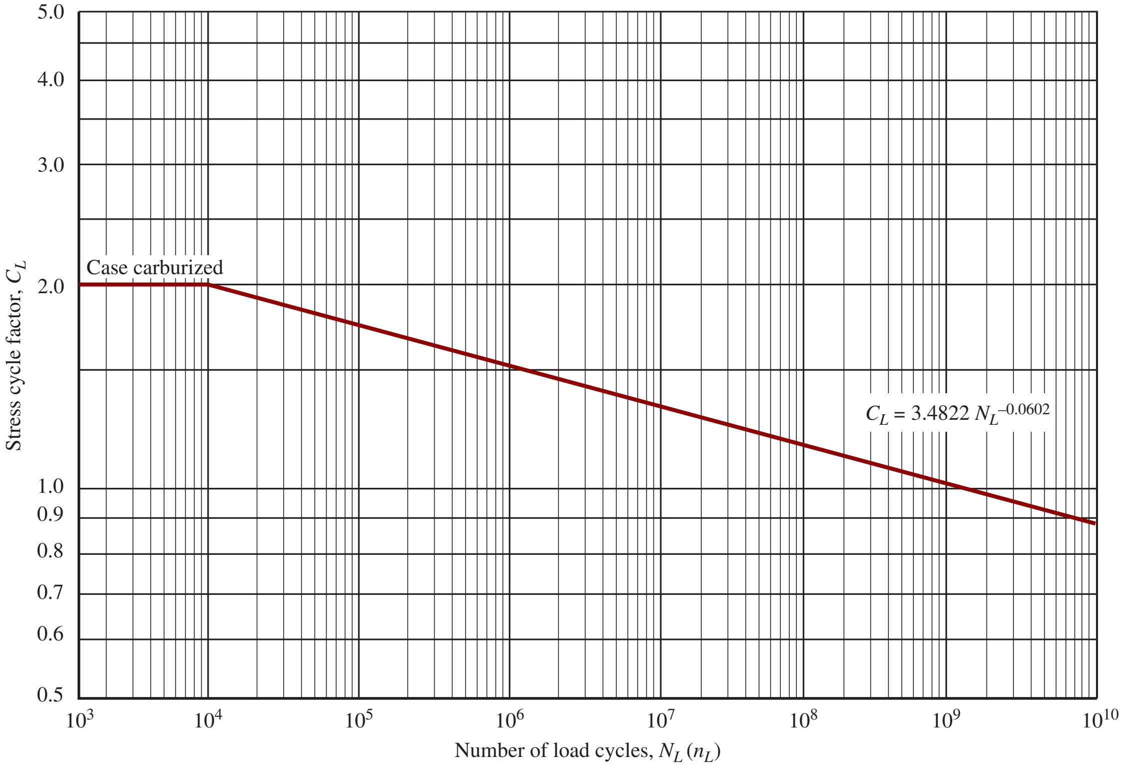


FIGURE 10–20 Stress cycle factor for pitting resistance, C_L (carburized case-hardened steel bevel gears) (Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth,* with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

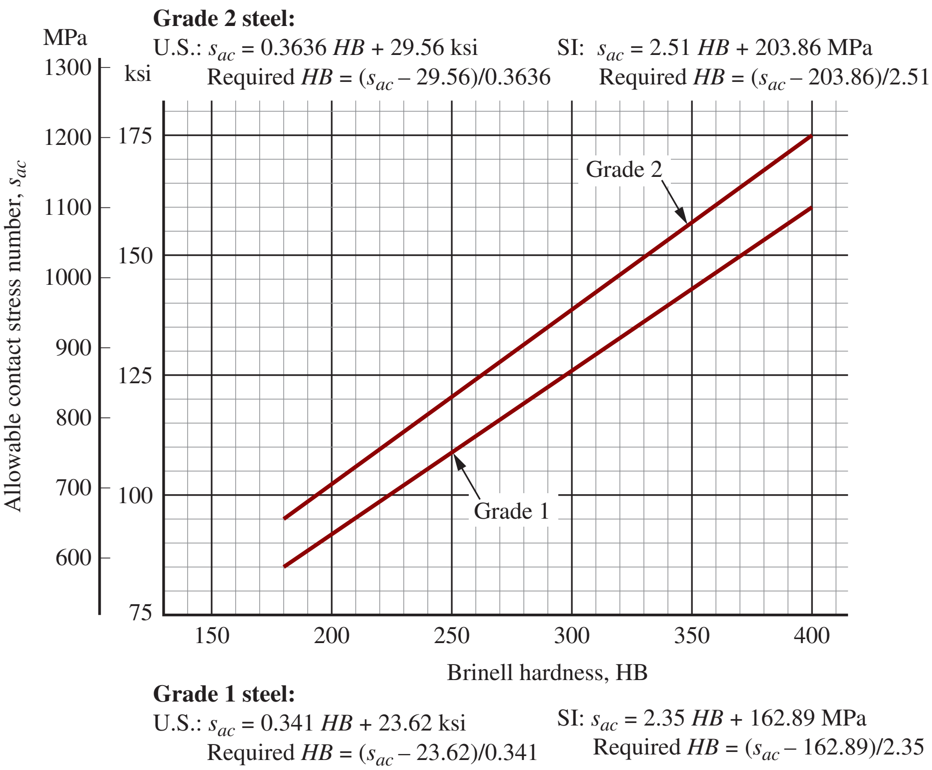


FIGURE 10–21 Allowable contact stress number, S_{ac}, for through-hardened steel for bevel gears (Adapted from AGMA 2003 (Reference 10), *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, Zerol Bevel and Spiral Bevel Gear Teeth,* with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

$$s_{at} = \frac{s_t(SF)(K_R)}{K_L} \tag{10–18}$$

$$s_{ac} = \frac{s_c(SF)(C_R)}{C_L} \tag{10–22}$$

Two of the factors in each equation are design decisions.

Safety factor *SF*:

No additional uncertainties or special requirements are known. Then $SF = 1.00$.

Reliability factors K_R and C_R: Assume a reliability of 0.99, one failure in 100.

Then, $K_R = 1.00$ and $C_R = 1.00$.

Life factors K_L and C_L: We use Figures 10–16 and 10–20 that require the expected number of load cycles, N_c. The equation shown below was used in Chapter 9 with q indicating the number of stress cycles per revolution of the gear, typically 1.0. For planetary or split power systems, q can be 2.0 or more.

$$N_c = 60(L)(n_P)(q) = (60)(15\,000)(600)(1) = 5.4 \times 10^8 \text{ load cycles}$$

For bending: $K_L = 0.948$ (computed from equation in Figure 10–16)
For pitting resistance: $C_L = 1.038$ (computed from equation in Figure 10–20)
We can then complete the calculations:

$$s_{at} = \frac{s_t(SF)(K_R)}{K_L} = \frac{(8764)(1.0)(1.0)}{0.948} = 9245 \text{ psi} = 9.245 \text{ ksi}$$

$$s_{ac} = \frac{s_c(SF)(C_R)}{C_L} = \frac{(119\,044)(1.0)(1.0)}{1.038} = 114\,686 \text{ psi} = 114.7 \text{ ksi}$$

Required hardness of steels and material specification:

Use Figure 10–17 for bending: Required $HB = 162$; Low; Use almost any steel.

Use Figure 10–21 for contact stress: Required $HB = 267$.

This value is quite suitable for through-hardened steel.

Use Figure A4–1: Specify SAE 1040 WQT 1000; $HB = 269$

22% elongation; $s_y = 88$ ksi; $s_u = 114$ ksi

Summary and comments: A bevel gear drive has been designed with straight teeth for transmission of power to shafts oriented 90° to each other. Key data were given and used in Chapter 8 and this chapter in Example Problems 8–3, and 10–4 to 10–8. Key data were given and used in this chapter and in Example Problems 10–4 to 10–8. Results are summarized as follows:

1. ***Example Problem 8–3***: Values for geometrical features were calculated for the following input data: $P_d = 8$; 20° pressure angle; $N_P = 16$; $N_G = 48$; 90° shaft angle.
   a. Output data used in subsequent problems include: Gear Ratio = 3.00; $D_P = 2.000$ in; $D_G = 6.000$ in; Pitch cone angle for the pinion = $\gamma = 18.43°$; Pitch cone angle for the gear = $\Gamma = 71.57°$; Face width = $F = 1.00$ in (design decision); Pinion outside diameter = $D_{oP} = 2.368$ in; Pinion outside diameter = $D_{oP} = 2.368$ in; Gear outside diameter = $D_{oG} = 6.041$ in. Several other detailed gear tooth features were also calculated.
2. ***Example Problem 10–4:*** Forces and torques on the pinion and gear were calculated for a given power transmitted = $P = 2.50$ hp at a pinion speed of 600 rpm.
   a. Results included: Torque on the pinion = $T_P = 263$ lb·in; Torque on the gear = $T_G = 788$ lb·in; Tangential transmitted force at the mean radii = $W_t = 313$ lb; Radial force = $W_r = 108$ lb; Axial force = $W_x = 36$ lb; Mean radius of the pinion teeth = $R_{mP} = 0.84$ in; Mean radius of the gear teeth = $R_{mG} = 2.53$ in;
   b. The forces at the mean radii, torques, and mean radii for the pinion and the gear were used in the analysis of forces on the shafts and bearings in Example Problem 10–5. Note that tangential forces are recalculated in Example Problem 10–6 to conform to conventions for stress analysis of bevel gear teeth.
3. ***Example Problem 10–5:*** See Figure 10–10 for a graphical display of the free-body diagrams of both shafts showing all forces and torques acting on the two shafts carrying the pinion and the gear along with bearing reactions. This required analysis of forces, shearing forces, and

bending moments on both shafts in two planes. All four bearings experience radial reaction forces in two planes. Resultants were calculated for each bearing. By a design decision, the thrust forces created by the bevel gears were taken at Bearings *B* and *D*. Results are as follows:

a. Bearing *A*: Radial load = 224 lb
b. Bearing *B*: Radial load = 108 lb; Thrust load = 36 lb
c. Bearing *C*: Radial load = 148 lb
d. Bearing *D*: Radial load = 188 lb; Thrust load = 108 lb

4. ***Section 10–8:*** The equilibrium analysis to determine forces in Example Problem 10–5 was extended to complete the shearing forces and bending moments in two planes for both shafts. See Figure 10–11 for the pinion shaft and Figure 10–12 for the gear shaft. Resultant bending moments were also calculated that would be needed for completing the design for the shafts, as will be discussed in Chapter 12.

5. ***Example Problem 10–6:*** The bending stress number for the teeth of the pinion was calculated using data from previous problems and additional given data:
   a. The pinion is driven by an electric motor and the load provides moderate shock. The quality number is $A_v = 11$, achievable by general commercial processing.
   b. Note that the transmitted forces on the pinion and the gear are recalculated based on the torque transmitted and the outer pitch radii for the pinion and the gear. The differences between these forces and those used in the shaft force analysis are accounted for in the geometry factors *J* and *I*. Here we use $W_t = 263$ lb.
   c. Bending stress number on pinion teeth = $s_t = 8764$ psi. The stress on the gear teeth will be lower because of the relative value of the geometry factor.

6. ***Example Problem 10–7:*** The contact stress number for the teeth of the pinion was calculated using data from previous problems. A design decision chose to use steel for both the pinion and the gear.
   a. Contact stress number, $s_c = 119\,044$ psi

7. ***Example Problem 10–8:*** The minimum required allowable bending stress number and the minimum allowable contact stress numbers were calculated based on results from Example Problems 10–6 and 10–7 along with the following design decisions:
   a. Reliability = $R = 0.99\,(< 1$ failure in 100); $K_R = C_R = 1.0$
   b. Safety factor = $SF = 1.0$, assuming no unusual uncertainties beyond other *K*-factors
   c. Design for a life of 15 000 hours. This is reasonable for general industrial applications, fully utilized.
   d. Results: Bending: $s_{at} = 9245$ psi = 9.245 ksi; $s_{ac} = 114\,686$ psi = 114.7 ksi

8. ***Required hardness of steels and material specification:***

   Bending: Required $HB = 162$; Low; Use almost any steel.

   Contact stress–Pitting resistance: Required $HB = 267$.

   This value controls the material specification and is quite suitable for through-hardened steel.

   Using Figure A4–1 Specify SAE 1040 WQT 1000; $HB = 269$; 22% elongation;
   $s_y = 88$ ksi; and $s_u = 114$ksi

9. ***Comments on the analysis and design:*** Taken together, the procedures and results summarized here demonstrate a reasonable approach to designing bevel gear pairs with straight teeth. The design is satisfactory as shown. However, alternate designs are practical and other iterations are recommended to explore what improvements can be made. Possible choices for different design decisions are as follows:
   a. A smaller overall drive may be practical by permitting the use of case hardening by flame or induction hardening or carburizing. These would be capable of operating at higher bending and contact stresses produced by smaller gears, but with corresponding higher processing costs.
   b. A more accurate gear quality could be chosen instead of $A_v = 11$ to decrease stresses and improve smoothness of operation and noise, but this also has higher associated costs.
   c. Consider using shot peening or other means of improving the life of the gears.
   d. Spiral bevel gears may permit a more compact design and reference should be made to AGMA Standard 2003 (Reference 10) for analysis methodology and additional data required.

Practical Considerations for Bevel Gearing

Factors similar to those discussed for spur and helical gears should be considered in the design of systems using bevel gears. The accuracy of alignment and the accommodation of thrust loads discussed in the example problems are critical.

Figure 10–22 shows the exterior view of a heavy-duty gear reducer for an industrial right angle drive. The input shaft is to the left and the output shaft is the larger one at the right extending through the side of the housing. Figure 10–23 shows a reducer similar to that in Figure 10–22 but with the upper part of the housing removed so that the entire three-stage reduction can be seen. The first stage is a spiral bevel gear pair and stages two and three are helical gear pairs. The large speed reduction results in a correspondingly large increase in the torque on the output shaft, requiring its diameter to be larger as shown. Observe, also, the careful placement of the bearings that support all shafts in the housing and how the housing allows for lubrication of the gears and assembly of all components.

10–10 FORCES, FRICTION, AND EFFICIENCY IN WORMGEAR SETS

See Chapter 8 for the geometry of wormgear sets. Also see References 2, 14, 16–18, and 21.

The force system acting on the worm/wormgear set is usually considered to be made of three perpendicular components as was done for helical and bevel gears. There are a tangential force, a radial force, and an axial force acting on the worm and the wormgear. We will use the same notation here as in the bevel gear system.

Figure 10–24 shows two orthogonal views (front and side) of a worm/wormgear pair, showing only the pitch diameters of the gears. The figure shows the separate worm and wormgear with the forces acting on each. Note that because of the 90° orientation of the two shafts,

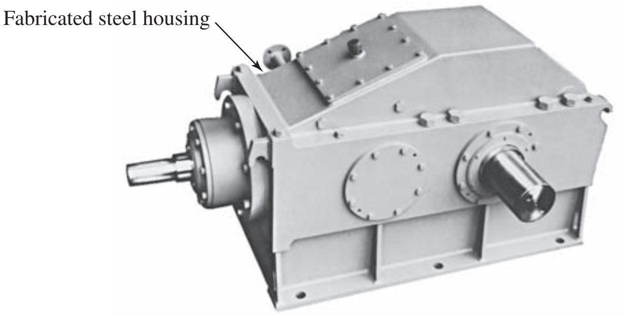


FIGURE 10–22 Heavy-duty right angle gear reducer (Sumitomo Machinery Corporation of America, Teterboro, NJ)

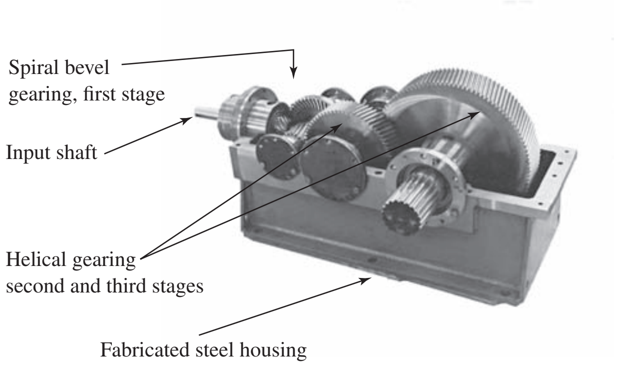


FIGURE 10–23 Three-stage industrial gear reducer employing spiral bevel and helical gears (Sumitomo Machinery Corporation of America, Teterboro, NJ)

➪ Forces on Worms and Wormgears

$$\left.\begin{aligned} W_{tG} &= W_{xW} \\ W_{xG} &= W_{tW} \\ W_{rG} &= W_{rW} \end{aligned}\right\} \qquad (10\text{–}23)$$

Of course, the directions of the paired forces are opposite because of the action/reaction principle.

The tangential force on the wormgear is computed first and is based on the required operating conditions of torque, power, and speed at the output shaft.

Pitch Line Speed, v_t

As stated in Chapter 8, the pitch line velocity is the linear velocity of a point on the pitch line for the worm or the wormgear. For the worm having a pitch diameter D_W in, rotating at n_W rpm,

➪ Pitch Line Velocity for Worm

$$v_{tW} = \frac{\pi D_W n_W}{12}\ \text{ft/min} \quad \text{or} \quad v_{tW} = \frac{\pi D_W n_W}{60\,000}\ \text{m/s}$$

For the wormgear having a pitch diameter D_G in, rotating at n_G rpm,

➪ Pitch Line Velocity for Gear

$$v_{tG} = \frac{\pi D_G n_G}{12}\ \text{ft/min} \quad \text{or} \quad v_{tG} = \frac{\pi D_G n_G}{60\,000}\ \text{m/s}$$

Note that these two values for pitch line velocity are *not* equal.

Velocity Ratio, *VR*

It is most convenient to calculate the velocity ratio of a worm and wormgear set from the ratio of the input rotational speed to the output rotational speed:

➪ Velocity Ratio for Worm/Sot

$$VR = \frac{\text{speed of worm}}{\text{speed of gear}} = \frac{n_W}{n_G} = \frac{N_G}{N_W}$$

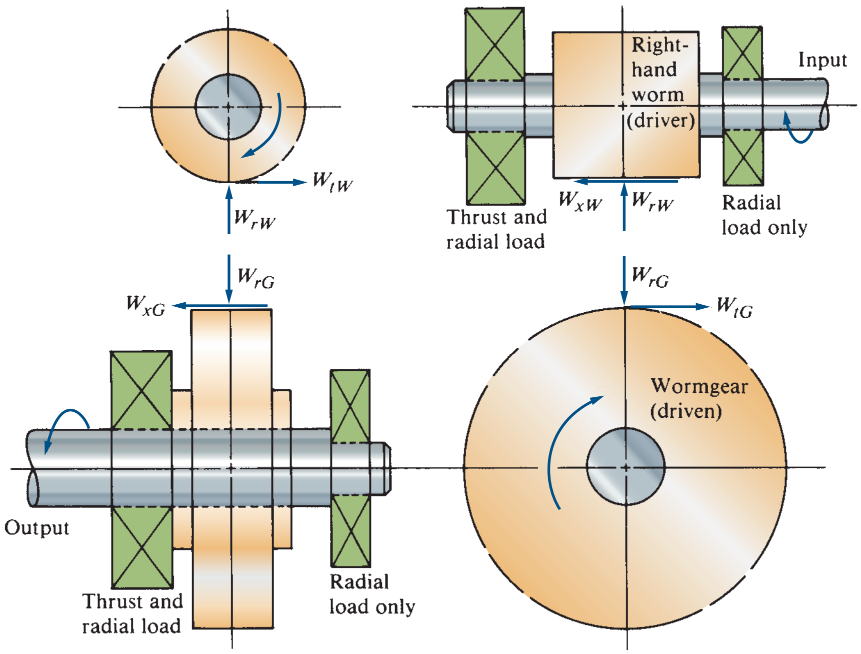


FIGURE 10–24 Forces on a worm and a wormgear

Coefficient of Friction, μ

Friction plays a major part in the operation of a wormgear set because there is inherently sliding contact between the worm threads and the wormgear teeth. The coefficient of friction is dependent on the materials used, the lubricant, and the sliding velocity. Based on the pitch line speed of the gear, the sliding velocity is

➪ Sliding Velocity for Gear

$$v_s = v_{tG}/\sin \lambda \qquad (10\text{–}24)$$

Based on the pitch line speed of the worm,

➪ Sliding Velocity for Worm

$$v_s = v_{tW}/\cos \lambda \qquad (10\text{–}25)$$

The term λ is the lead angle for the worm thread as defined in Equation (8–24).

The AGMA (see Reference 14) recommends the following formulas to estimate the coefficient of friction for a hardened steel worm (58 HRC minimum), smoothly ground, or polished, or rolled, or with an equivalent finish, operating on a bronze wormgear. The choice of formula depends on the sliding velocity. *Note:* v_s must be in ft/min in the formulas; 1.0 ft/min = 0.0051 m/s.

Static Condition: $v_s = 0$

$$\mu = 0.150$$

Low Speed: $v_s < 10$ **ft/min (0.051 m/s)**

$$\mu = 0.124e^{(-0.074 v_s^{0.645})} \qquad (10\text{–}26)$$

Higher Speed: $v_s > 10$ **ft/min**

$$\mu = 0.103e^{(-0.110 v_s^{0.450})} + 0.012 \qquad (10\text{–}27)$$

Figure 10–25 is a plot of the coefficient μ versus the sliding velocity v_s.

Output Torque from Wormgear Drive, T_o

In most design problems for wormgear drives, the output torque and the rotating speed of the output shaft will be known from the requirements of the driven machine. Torque and speed are related to the output power by

➪ Output Torque from Wormgear

$$T_o = \frac{63\,000(P_o)}{n_G} \qquad (10\text{–}28)$$

By referring to the end view of the wormgear in Figure 10–24, you can see that the output torque is

$$T_o = W_{tG} \cdot r_G = W_{tG}(D_G/2)$$

Then the following procedure can be used to compute the forces acting in a worm/wormgear drive system.

Procedure for Calculating the Forces on a Worm/Wormgear Set

Given:

Output torque, T_o, in lb·in
Output speed, n_G, in rpm
Pitch diameter of the wormgear, D_G, in inches
Lead angle, λ
Normal pressure angle, ϕ_n

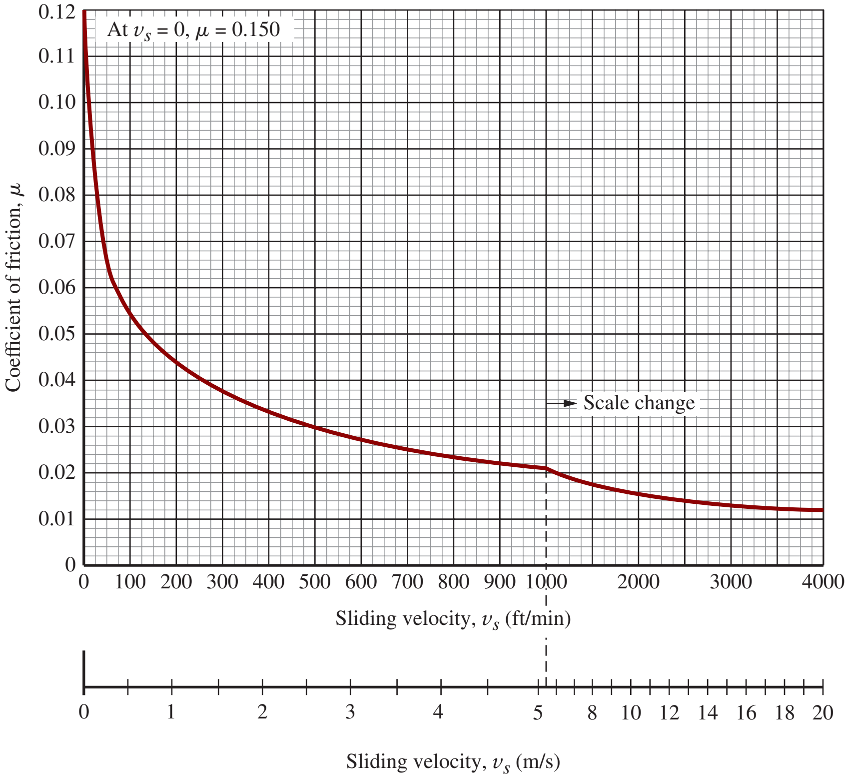


FIGURE 10–25 Coefficient of friction versus sliding velocity for steel worm and bronze wormgear

Compute:

$$W_{tG} = 2\ T_o/D_G \tag{10–29}$$

$$W_{xG} = W_{tG}\frac{\cos\phi_n \sin\lambda + \mu\cos\lambda}{[\cos\phi_n\cos\lambda - \mu\sin\lambda]} \tag{10–30}$$

$$W_{rG} = \frac{W_{tG}\sin\phi_n}{\cos\phi_n\cos\lambda - \mu\sin\lambda} \tag{10–31}$$

The forces on the worm can be obtained by observation, using Equation (10–23). Equations (10–30) and (10–31) were derived using the components of both the tangential driving force on the wormgear and the friction force at the location of the meshing worm threads and wormgear teeth. The complete development of the equations is shown in Reference 15.

Friction Force, W_f

The friction force, W_f, acts parallel to the face of the worm threads and the gear teeth and depends on the tangential force on the gear, the coefficient of friction, and the geometry of the teeth:

$$W_f = \frac{\mu W_{tG}}{(\cos\lambda)(\cos\phi_n) - \mu\sin\lambda} \tag{10–32}$$

Power Loss Due to Friction, P_L

Power loss is the product of the friction force and the sliding velocity at the mesh. That is,

$$P_L = \frac{v_s W_f}{33\ 000} \tag{10–33}$$

In this equation, the power loss is in hp, v_s is in ft/min, and W_f is in lb.

Input Power, P_i

The input power is the sum of the output power and the power loss due to friction:

$$P_i = P_o + P_L \tag{10–34}$$

Efficiency, η

Efficiency is defined as the ratio of the output power to the input power:

$$\eta = P_o/P_i \tag{10–35}$$

Efficiency for a wormgear drive with the usual case of the input coming through the worm can also be computed directly from the following equation.

$$\eta = \frac{\cos\phi_n - \mu\tan\lambda}{\cos\phi_n + \mu/\tan\lambda} \tag{10–36}$$

Factors Affecting Efficiency

As can be seen in Equation (10–32), the lead angle, the normal pressure angle, and the coefficient of friction all affect the efficiency. The one that has the largest effect, and the one over which the designer has the most control, is the lead angle, λ. The larger the lead angle, the higher the efficiency, up to approximately $\lambda = 45°$. (See Figure 10–26.)

Now, looking back to the definition of the lead angle, note that the number of threads in the worm has a major effect on the lead angle. Therefore, to obtain a high efficiency, use multiple-threaded worms. But there is a disadvantage to this conclusion. More worm threads require more gear teeth to achieve the same ratio, resulting in a larger system overall. The designer is often forced to compromise.

Example Problem: Forces and Efficiency in Wormgearing

Review now the results of Example Problem 8–4, in which the geometry factors for a particular worm and wormgear set were computed. The following example problem extends the analysis to include the forces acting on the system for a given output torque.

Example Problem 10–9 The wormgear drive described in Example Problem 8–4 is transmitting an output torque of 4168 lb · in. The transverse pressure angle is 20°. The worm is made from hardened and ground steel, and the wormgear is bronze. Compute the forces on the worm and the wormgear, the output power, the input power, and the efficiency.

Solution Recall from Example Problem 8–4 that

$$\lambda = 14.04° \qquad D_G = 8.667 \text{ in} \qquad n_G = 101 \text{ rpm}$$

$$n_W = 1750 \text{ rpm} \qquad D_W = 2.000 \text{ in}$$

The normal pressure angle is required. From Equation (8–26),

$$\phi_n = \tan^{-1}(\tan \phi_t \cos \lambda) = \tan^{-1}(\tan 20° \cos 14.04°) = 19.45°$$

Because they recur in several formulas, let's compute the following:

$$\sin \phi_n = \sin 19.45° = 0.333$$

$$\cos \phi_n = \cos 19.45° = 0.943$$

$$\cos \lambda = \cos 14.04° = 0.970$$

$$\sin \lambda = \sin 14.04° = 0.243$$

$$\tan \lambda = \tan 14.04° = 0.250$$

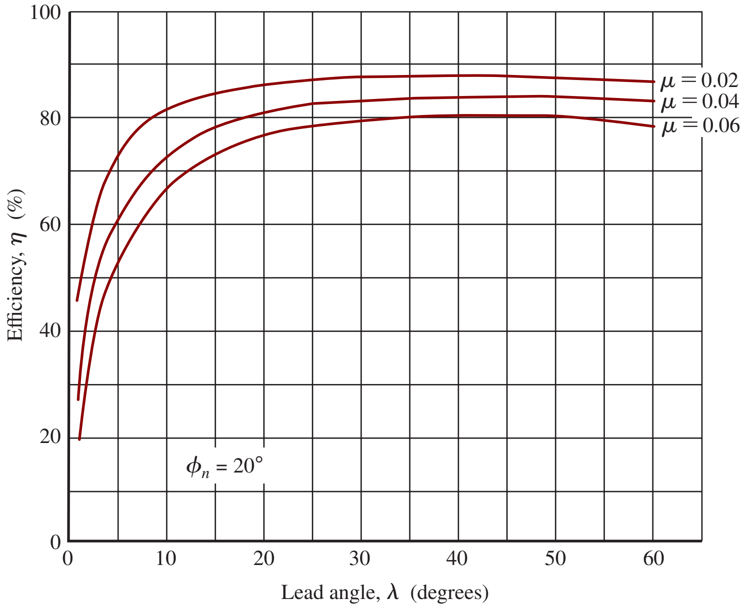


FIGURE 10–26 Efficiency of wormgear drive versus lead angle

We can now compute the tangential force on the wormgear using Equation (10–29)

$$W_{tG} = \frac{2T_o}{D_G} = \frac{(2)(4168 \text{ lb} \cdot \text{in})}{8.667 \text{ in}} = 962 \text{ lb}$$

The calculations of the axial and radial forces require a value for the coefficient of friction that, in turn, depends on the pitch line speed and the sliding velocity.

Pitch Line Speed of the Gear:

$$v_{tG} = \pi D_G n_G/12 = \pi(8.667)(101)/12 = 229 \text{ ft/min}$$

Sliding Velocity: [Equation (10–24)]

$$v_s = v_{tG}/\sin\lambda = 229/\sin 14.04° = 944 \text{ ft/min}$$

Coefficient of Friction: From Figure 10–25, at a sliding velocity of 944 ft/min, we can read $\mu = 0.022$.

Now the axial and radial forces on the wormgear can be computed.

Axial Force on the Wormgear: [Equation (10–30)]

$$W_{xG} = 962 \text{ lb}\left[\frac{(0.943)(0.243) + (0.022)(0.970)}{(0.943)(0.970) - (0.022)(0.243)}\right] = 265 \text{ lb}$$

Radial Force on the Wormgear: [Equation (10–31)]

$$W_{rG} = \left[\frac{(962)(0.333)}{(0.943)(0.970) - (0.022)(0.243)}\right] = 352 \text{ lb}$$

Now the output power, input power, and efficiency can be computed.

Output Power: [Equation (10–28)]

$$P_o = \frac{T_o n_G}{63\,000} = \frac{(4168 \text{ lb} \cdot \text{in})(101 \text{ rpm})}{63\,000} = 6.68 \text{ hp}$$

The input power depends on the friction force and the consequent power loss due to friction.

Friction Force: [Equation (10–32)]

$$W_f = \frac{\mu W_{tG}}{(\cos\lambda)(\cos\phi_n) - \mu\sin\lambda} = \frac{(0.022)(962 \text{ lb})}{(0.970)(0.943) - (0.022)(0.243)} = 23.3 \text{ lb}$$

Power Loss Due to Friction: [Equation (10–33)]

$$P_L = \frac{v_s W_f}{33\,000} = \frac{(944 \text{ ft/min})(23.3 \text{ lb})}{33\,000} = 0.666 \text{ hp}$$

Input Power: [Equation (10–34)]

$$P_i = P_o + P_L = 6.68 + 0.66 = 7.35 \text{ hp}$$

Efficiency: [Equation (10–35)]

$$\eta = \frac{P_o}{P_i}(100\%) = \frac{6.68 \text{ hp}}{7.35 \text{ hp}}(100\%) = 90.9\%$$

Equation (10–36) could also be used to compute efficiency directly without computing friction power loss.

Self-Locking Wormgear Sets

Self-locking is the condition in which the worm drives the wormgear, but, if torque is applied to the gear shaft, the worm does not turn. It is locked! The locking action is produced by the friction force between the worm threads and the wormgear teeth, and this is highly dependent on the lead angle. It is recommended that a lead angle no higher than about 5.0° be used in order to ensure that self-locking will occur. This lead angle usually requires the use of a single-threaded worm; the low lead angle results in a low efficiency, possibly as low as 60% or 70%.

10–11 STRESS IN WORMGEAR TEETH

We present here an approximate method of computing the bending stress in the teeth of the wormgear. Because the geometry of the teeth is not uniform across the face width, it is not possible to generate an exact solution. However, the method given here should predict the bending stress with sufficient accuracy to check a design because most worm/wormgear systems are limited by pitting, wear, or thermal considerations rather than strength.

The AGMA, in its Standard 6034-B92 (Reference 14), does not include a method of analyzing wormgears for strength. The method shown here was adapted from Reference 15. Only the wormgear teeth are analyzed because the worm threads are inherently stronger and are typically made from a stronger material.

The stress in the gear teeth can be computed from

$$\sigma = \frac{W_d}{yFp_n} \qquad \textbf{(10–37)}$$

where W_d = dynamic load on the gear teeth

y = Lewis form factor (see Table 10–5)

F = face width of the gear

p_n = normal circular pitch = $p\cos\lambda = \pi \cos\lambda/P_d$ **(10–38)**

The dynamic load can be estimated from

$$W_d = W_{tG}/K_v \qquad \textbf{(10–39)}$$

and

$$K_v = 1200/(1200 + v_{tG}) \qquad \textbf{(10–40)}$$

$v_{tG} = \pi D_G n_G/12$ = pitch line speed of the gear **(10–41)**

Only one value is given for the Lewis form factor for a given pressure angle because the actual value is very difficult to calculate precisely and does not vary much with the number of teeth. The actual face width should be used, up to the limit of two-thirds of the pitch diameter of the worm.

The computed value of tooth bending stress from Equation (10–37) can be compared with the fatigue strength of the material of the gear. For manganese gear bronze, use a fatigue strength of 17 000 psi; for phosphor gear bronze, use 24 000 psi. For cast iron, use approximately 0.35 times the ultimate strength, unless specific data are available for fatigue strength.

TABLE 10–5 Approximate Lewis Form Factor for Wormgear Teeth

ϕ_n	y
$14\frac{1}{2}°$	0.100
20°	0.125
25°	0.150
30°	0.175

10–12 SURFACE DURABILITY OF WORMGEAR DRIVES

AGMA Standard 6034-B92 (Reference 14) gives a method for rating the surface durability of hardened steel worms operating with bronze gears. The ratings are based on the ability of the gears to operate without significant damage from pitting or wear. All equations from the standard are in the U.S. system. For designs in the SI system, values of parameters should be converted to U.S. units for which the equations were written to determine the corresponding factors. The units are as follows:

W_{tR}—Rated tangential load: lb

Diameters and face width: in

v_s—Sliding velocity: ft/min

Equivalent values in SI units are shown only for reference.

The procedure calls for the calculation of a *rated tangential load*, W_{tR}, from

➩ Rated Tangential Load on Wormgears

$$W_{tR} = C_s D_G^{0.8} F_e C_m C_v \qquad \textbf{(10–42)}$$

where C_s = materials factor (from Figure 10–27)

D_G = pitch diameter of the wormgear, in inches

F_e = effective face width, in inches. Use the actual face width of the wormgear up to a maximum of 0.67 D_W

C_m = ratio correction factor (from Figure 10–28)

C_v = velocity factor (from Figure 10–29)

Conditions on the Use of Equation (10–42)

1. The analysis is valid only for a hardened steel worm (58 HRC minimum) operating with gear bronzes specified in AGMA Standard 6034-B92. The classes of bronzes typically used are tin bronze, phosphor bronze, manganese bronze, and aluminum bronze. The materials factor, C_s, is dependent on the method of casting the bronze, as indicated in Figure 10–27. The values for C_s can be computed from the following formulas.

Sand-Cast Bronzes:

For $D_G > 2.5$ in (64 mm),

$$C_s = 1189.636 - 476.545 \log_{10}(D_G) \qquad \textbf{(10–43)}$$

For $D_G < 2.5$ in (64 mm),

$$C_s = 1000$$

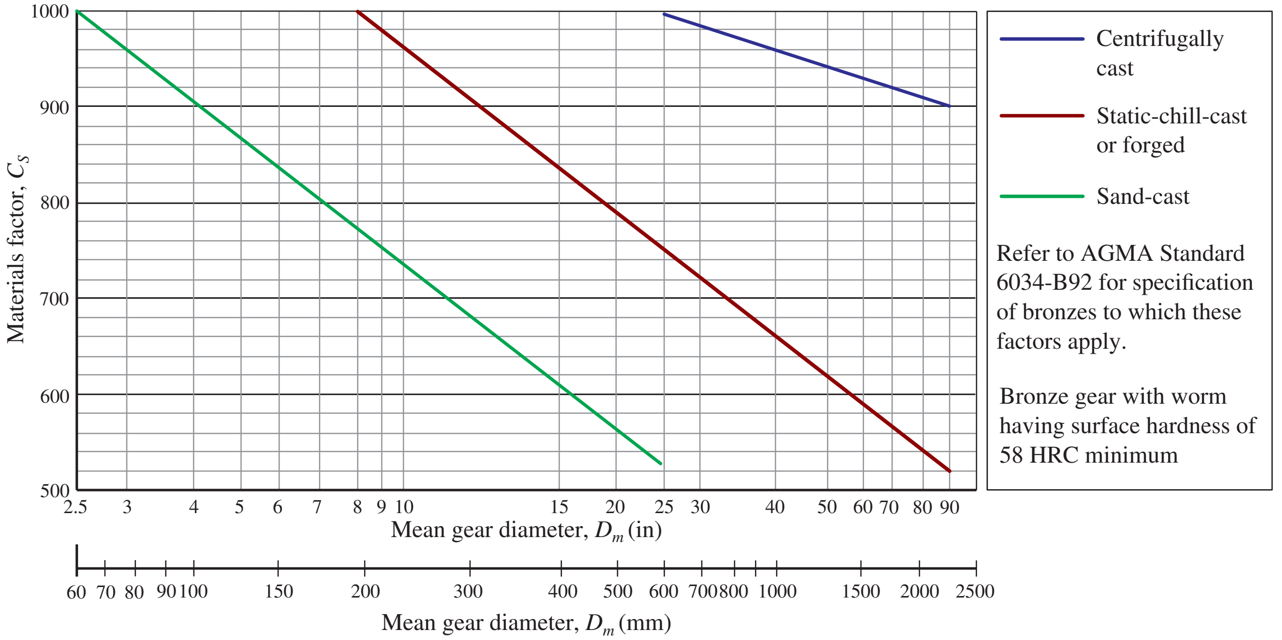


FIGURE 10–27 Materials factor, C_s, for hardened steel worms operating with bronze gears for center distance $C > 3.0$ in (76 mm) (Extracted from AGMA Standard 6034–B92, *Practice for Enclosed Cylindrical Wormgear Speed Reducers and Gearmotors,* with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA 22314.)

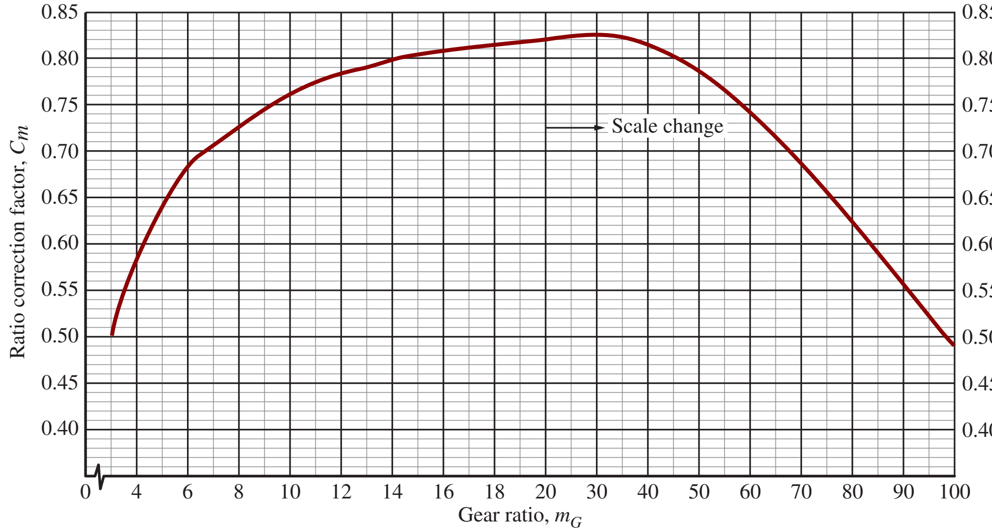


FIGURE 10–28 Ratio correction factor, C_m, versus gear ratio, m_G

Static-Chill-Cast or Forged Bronzes:

For $D_G > 8.0$ in (203 mm),

$$C_s = 1411.651 - 455.825 \log_{10}(D_G) \quad \textbf{(10–44)}$$

For $D_G < 8.0$ in (203 mm)

$$C_s = 1000$$

Centrifugally Cast Bronzes:

For $D_G > 25$ in (635 mm),

$$C_s = 1251.291 - 179.750 \log_{10}(D_G) \quad \textbf{(10–45)}$$

For $D_G < 25$ in (635 mm),

$$C_s = 1000$$

2. The wormgear diameter is the second factor in determining C_s. The *mean diameter* at the midpoint of the working depth of the gear teeth should be used. If standard addendum gears are used, the mean diameter is equal to the pitch diameter.
3. Use the actual face width, F, of the wormgear as F_e if $F < 0.667(D_w)$. For larger face widths, use $F_e = 0.67(D_w)$, because the excess width is not effective.

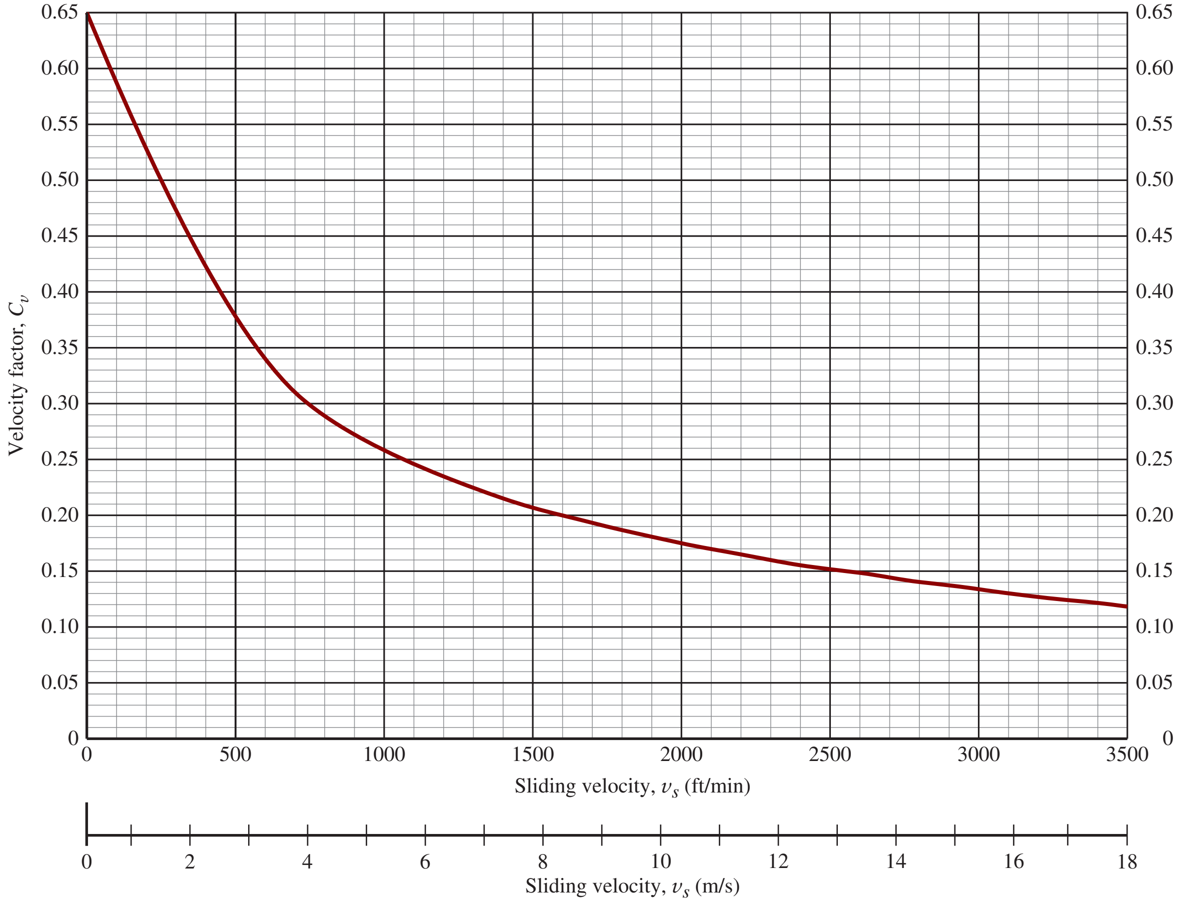


FIGURE 10–29 Velocity factor, C_v, versus sliding velocity

4. The ratio correction factor, C_m, can be computed from the following formulas.

For Gear Ratios, m_G, from 6 to 20

$$C_m = 0.0200(-m_G^2 + 40m_G - 76)^{0.5} + 0.46 \quad \textbf{(10–46)}$$

For Gear Ratios, m_G, from 20 to 76

$$C_m = 0.0107(-m_G^2 + 56m_G + 5145)^{0.5} \quad \textbf{(10–47)}$$

For $m_G > 76$

$$C_m = 1.1483 - 0.00658m_G \quad \textbf{(10–48)}$$

5. The velocity factor depends on the sliding velocity, v_s, computed from Equation (10–24) or (10–25). Values for C_v can be computed from the following formulas.

For v_s, from 0 to 700 ft/min (0–3.56 m/s)

$$C_v = 0.659e^{(-0.0011v_s)} \quad \textbf{(10–49)}$$

For v_s, from 700 to 3000 ft/min (3.56 to 15.24 m/s)

$$C_v = 13.31v_s^{(-0.571)} \quad \textbf{(10–50)}$$

For $v_s > 3000$ ft/min (>15.24 m/s)

$$C_v = 65.52v_s^{(-0.774)} \quad \textbf{(10–51)}$$

6. The proportions of the worm and the wormgear must conform to the following limits defining the maximum and minimum pitch diameters of the worm in relation to the center distance, C, for the gear set. All dimensions are in inches.

$$\text{Maximum } D_W = C^{0.875}/1.6 \quad \textbf{(10–52)}$$

$$\text{Minimum } D_W = C^{0.875}/3.0 \quad \textbf{(10–53)}$$

7. The shaft carrying the worm must be sufficiently rigid to limit the deflection of the worm at the pitch point to the maximum value of $0.005\sqrt{P_x}$, where P_x is the axial pitch of the worm, which is numerically equal to the circular pitch, p, of the wormgear.
8. When you are analyzing a given wormgear set, the value of the rated tangential load, W_{tR}, must be greater than the actual tangential load, W_t, for satisfactory life.
9. Ratings given in this section are valid only for smooth systems such as fans or centrifugal pumps driven by an electric or hydraulic motor operating under 10 hours per day. More severe conditions, such as shock loading, internal combustion engine drives, or longer hours of operation, require the application of a service factor. Reference 14 lists several such factors based on field experience with specific types of equipment. For problems in this book, factors from Table 9–1 can be used.